

Doomscrolling: TikTok and News Consumption*

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Abstract

How does TikTok use affect news consumption and downstream political outcomes? Leveraging detailed device-level mobile browsing and app use data and a shock to early adoption of TikTok as a novel, plausibly exogenous source of variation, I estimate the causal effect of TikTok use on news consumption, political knowledge, attitudes and behavior in the United States. I find that increased TikTok use substantially reduces news consumption related to major news events such as elections, national disasters and major political developments. TikTok use reduces news consumption along both the extensive and intensive margins. It also reduces the length of individuals' news consumption sessions, suggesting that reduced attention spans contribute to the effect. Survey evidence shows that the effect is not limited to mobile devices and has downstream consequences: individuals in high-TikTok-use media markets have lower traditional news consumption, political knowledge and turnout, and higher partisan polarization in presidential approval after, but not before, TikTok's proliferation.

Keywords: TikTok, Short-Video Platform, Social Media, News Consumption, Attention

JEL codes: L82, L86, D83, D72, D91, O33.

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1 Introduction

News consumption is central to democratic accountability ([Besley & Prat, 2006](#); [Snyder Jr & Strömberg, 2010](#); [Strömberg, 2015](#)), and it increasingly takes place on smartphones ([Newman et al., 2025](#)). TikTok is a rapidly growing competitor for smartphone users’ attention. The share of U.S. adults using TikTok rose from 21% in 2021 to 37% in 2025, while Facebook (69% to 71%) and Instagram (40% to 50%) saw modest or no growth ([Pew Research Center, 2025](#)). TikTok also stands out for the amount of time its users spend on the platform. Globally, the average TikTok user spends 97 minutes a day on the platform, compared with 73 minutes for the average Instagram user and 67 for the average Facebook user ([DataReportal, 2025](#)).

Unlike earlier platforms, which organized content around users’ online social networks, TikTok curates a continuous feed of short videos almost entirely from algorithmic “interest signals,” with little active selection by users ([Baptista, 2025](#)). Journalists, regulators, and researchers have raised concerns that this design encourages overuse ([The Washington Post, 2025](#); [Maarouf et al., 2025](#); [Bao, 2026](#); [European Commission, 2026](#)), crowds out more beneficial activities ([Agarwal & Ananthakrishnan, 2024](#); [Harvard Technology Review, 2025](#)), and erodes the capacity for sustained attention ([Jargon, 2022](#); [Nguyen et al., 2025](#); [Newport, 2026](#)). Time spent on TikTok necessarily displaces other activities, including other uses of mobile devices, as well as activities off the device ([Aridor, 2025](#)). Whether news consumption is among the activities displaced, and whether this matters for political knowledge and attitudes, is an empirical question.

This paper addresses the question using detailed device-level mobile browsing and app use data from 2019–2025 as well as survey data. I find that TikTok use substantially decreases engagement with news during weeks of major news events. Using survey data, I show that this effect extends beyond mobile devices to traditional news media. Further, TikTok use decreases political knowledge and turnout, and increases partisan polarization in approval of the president.

TikTok use is likely endogenous to mobile browsing behavior and news consumption. On one hand, people with little interest in news may be more likely to use TikTok because news is a small minority of content ([Aridor et al., 2025](#)). On the other hand, a habit of heavy mobile device use may increase both TikTok use and mobile news consumption. To address this potential endogeneity, I develop a novel instrument for TikTok use based on early marketing of TikTok’s predecessor, Musical.ly. Although TikTok launched internationally in September 2017, its U.S. adoption accelerated in 2018, after ByteDance, its founding company, acquired the U.S.-based platform Musical.ly and merged the two apps. Musical.ly’s users and content migrated automatically to TikTok, creating a large predetermined user base ([Dredge, 2018](#); [TikTok, 2018](#)). Musical.ly, an early short-form video platform launched in 2014, experienced a sharp increase in popularity in 2015–2016

when particular pop artists used the platform heavily to promote their music ([Robehmed, 2017](#); [Music Ally, 2017](#)).

Prior work on other social media platforms shows that early shocks to adoption can generate persistent geographic differences in use, through network effects and path dependence ([Enikolopov et al., 2020, 2023](#); [Mei & Wu, 2023](#); [Müller & Schwarz, 2023](#); [Müller et al., 2023](#); [Fujiwara et al., 2024](#)). Promotion by pop artists on Musical.ly incentivized fans to join the platform and, through network effects, contributed to broader adoption of the successor platform TikTok within their communities. The identification strategy exploits geographic variation in interest in these artists to isolate variation in TikTok use stemming from this early promotion, in an instrumental variable framework.

I construct the instrument from interest in nine pop artists who promoted their music on Musical.ly during the platform’s 2015–2016 rise. Interest in those artists in 2015, measured by Google search intensity for their names across 205 media markets covering the continental United States, predicts greater search for the Musical.ly platform after 2015, but not for Vine or Dubsmash, contemporaneous short-form video platforms ([Hu, 2018](#); [McCammond, 2017](#)). It also predicts both a higher likelihood of using TikTok and increased time on the platform throughout the following decade, from 2019 through 2025. Interest in the Musical.ly artists does not predict total time spent across other social media platforms, but predicts less time on Instagram and WhatsApp, suggesting some substitution away from those platforms toward TikTok.

I use this quasi-exogenous source of variation in TikTok use to estimate its causal effect on news consumption, political knowledge, attitudes and behavior. I measure TikTok use and mobile news consumption using device-level data from Comscore ([Comscore, Inc., 2019–2025](#)), which passively records browsing and app usage among 183,785 opt-in panelists observed between May 2019 and November 2025. Over this period, mobile news consumption varies widely, with substantial medium-term swings that often coincide with broader changes in mobile use rather than particular news events. However, news consumption also spikes sharply in weeks when major newsworthy events occur. Focusing on these short-run spikes separates changes in news consumption around major news events from broader shifts in mobile use. I therefore focus on those weeks in a device-week panel setting. I document 40 news event weeks between 2019 and 2025, defined as weeks in which national news consumption exceeds an eight-week rolling average by more than one standard deviation. They map to well-known newsworthy events such as election weeks, major political developments, natural disasters and mass shootings.

In the device-week panel, TikTok use substantially weakens the response of news consumption to major news events, when news is most salient. In the cross section, by contrast, TikTok use has no detectable effect on news consumption averaged over the weeks each individual is observed. The panel estimate implies that an additional hour of weekly TikTok use decreases news consumption during news event weeks by about 10%,

or 9 minutes per week compared to a mean of 91 minutes. The average panelist spends about 132 minutes per week on TikTok, so moving from no use to the average implies a decrease of roughly 22%, or about 20 minutes. The estimates are robust to alternative definitions of news event weeks.

A causal interpretation requires an exclusion restriction assumption that Musical.ly artist interest affects news consumption only through TikTok use. One potential threat to identification is that the interest is correlated with unobserved media-market determinants of news consumption. Another is that the artists' political engagement affects their fans' news consumption. I provide several pieces of evidence in support of the exclusion restriction. After accounting for a few key media-market characteristics, media markets with high and low interest in Musical.ly artists are balanced on pre-2018 voting outcomes and on demographic and socio-economic characteristics. At the level of the artists, I collect lists of comparison artists who are similar to the Musical.ly artists but did not promote their music on the platform, and control for 2015 interest in them. Both the demographics of their fans and their political endorsements from 2012–2020 are statistically indistinguishable from those of the Musical.ly artists. Interest in these artists predicts neither TikTok use nor news consumption, so unobserved characteristics correlated with artist fandom are unlikely to drive the results. Because the two groups have similar political profiles, this extends to the artists' politics.

As an additional check, I control for interest in the same Musical.ly artists measured in 2014 and 2016, before and after the platform's initial rise in popularity. Interest in these years predicts neither TikTok use nor news consumption, and the estimates are unchanged. The estimates are also unchanged when controlling for interest in Vine and Dubsmash, which does not correlate consistently with news consumption. These results reinforce confidence in the interpretation of the main estimates. They are unlikely to reflect an effect of artist fandom through channels other than TikTok use, or unobserved confounders related to artist fandom or early interest in short-form video platforms.

Comparing the OLS estimates to the IV estimates is informative about selection into TikTok use. OLS coefficients are small and positive, as selection on mobile use would predict: heavier mobile users spend more time on TikTok, and their mobile news consumption responds more strongly to newsworthy events. Consistent with this, conditioning on total mobile use in the weeks before an event renders the naive association negative, but leaves the instrumented estimate unaffected. The positive association also disappears when news consumption is expressed as a share of mobile time.

The negative effect of TikTok use on the news consumption response to major news events operates on several margins. TikTok use reduces both the probability of consuming any news during an event week and the amount consumed by those who do. It also shortens individual visits to news sites and apps. An additional weekly hour on TikTok decreases the longest news session of the week by about one minute against a mean of 21,

and shortens the 75th and 95th percentiles of an individual’s own session lengths, while the median does not move. The effect is therefore concentrated in individuals’ longer news sessions, consistent with a reduction in the capacity to sustain attention.

To examine whether these effects extend beyond mobile devices, I link TikTok use at the media-market level to Cooperative Election Study respondents from 2010 to 2024 ([Kuriwaki, 2025](#); [Cooperative Election Study, 2010–2024](#)). For each device in the Comscore panel, I residualize TikTok use on device-level characteristics. I then compute average adjusted TikTok use in each media market, and instrument this measure with 2015 interest in the Musically artists interacted with an indicator for years after 2018. After 2018, an additional weekly hour of TikTok use in a market decreases the probability of having watched television news, read a newspaper, or listened to radio news in the past day by 5.5 percentage points against a mean of 0.81.

TikTok use also affects downstream political outcomes. An additional weekly hour of average market-level TikTok use decreases how closely respondents report following government and public affairs by 0.22 points on a four-point scale, and decreases the number of correct answers to eight questions about the partisan composition of government by 0.63 answers compared to a mean of 5. TikTok use also decreases turnout substantially: an hour increase in average market-level TikTok use causes a 5 percentage point decrease in the likelihood of turnout in general elections, from an average of 54 percent. Finally, while TikTok use does not affect self-reported ideology or strength of partisanship, it polarizes presidential approval along party lines: an additional weekly hour increases approval of the president among copartisans, and disapproval of the president among out-partisans, by 0.22 points on average on a four-point scale, from a mean of 3.5. TikTok use also leads partisan survey respondents to view the opposing party as more extreme. Importantly, event-study estimates show that respondents in markets with high and low predicted TikTok use did not follow differential trends on these measures before 2018, supporting a causal interpretation.

Users may encounter news on TikTok itself, potentially offsetting the decline in news consumption measured off the platform. Content consumed within applications is not observed in the Comscore data, so this cannot be tested directly. I argue that such offsetting is unlikely. First, the negative effects of TikTok use on political knowledge suggest that news encountered on TikTok is not a perfect substitute for off-platform news as a source of political information. Second, [Aridor et al. \(2025\)](#) observe on-screen content for a panel of smartphone users between September 1 and November 4, 2024, and find that about 0.6% of time on TikTok is spent on election-related content. Assuming this share is constant across weeks, an additional weekly hour of TikTok use includes about 22 seconds of election-related content while displacing 9 minutes of news consumption off the platform. Even under the extreme assumption that all election-related content observed over the roughly nine-week sample period were consumed during the week of

the election, election-related content would account for 5.6% of TikTok time during that week. A 5.6% share corresponds to about 3.3 minutes per hour, against the 9-minute decline in news consumption off the platform. News encountered on TikTok is therefore unlikely to compensate for the decline in consumption off the platform.

This paper contributes to the literature on the societal consequences of social media (e.g., [Falck et al., 2014](#); [Allcott et al., 2020](#); [Guriev et al., 2021](#); [Zhuravskaya et al., 2020](#); [Aridor et al., 2024](#); [Enikolopov et al., 2024](#)) by providing novel evidence on the political effects of TikTok use. While this literature has so far concentrated on Facebook and Twitter ([Aridor et al., 2024](#)), the closest work on short-form video studies well-being and time use. [Maarouf et al. \(2025\)](#) compare TikTok adopters to Instagram and Facebook adopters before the latter introduced short-form video, and find that TikTok adoption increases phone use over nine weeks but not nighttime use. [Agarwal & Ananthakrishnan \(2024\)](#) compare early TikTok adopters to users who adopt several weeks later, tracking outcomes for six weeks after adoption. They find that adoption increases screen time and, among the heaviest users, delays bedtime and shortens inferred sleep. [Gao et al. \(2025\)](#) randomly incentivize Douyin users to browse while logged out for four weeks, attenuating the platform’s ability to personalize recommendations. Douyin use falls during the intervention, and login frequency remains lower over the following four weeks. [T. Chen et al. \(2025\)](#) find that higher city-level Douyin exposure worsens mental health, instrumenting exposure with pre-2016 4G infrastructure interacted with Douyin’s 2016 launch. This paper differs from this work in both outcome and identification. It estimates the effect of TikTok on news consumption, political knowledge and attitudes, using persistent geographic variation in TikTok use in the United States, whereas previous work used short-run or rollout-based variation.

This paper also contributes to the literature on the demand for news (e.g., [Gentzkow & Shapiro, 2011](#); [Chopra et al., 2022](#); [Capozza et al., 2022](#); [Aridor et al., 2025](#)) and on how social media shapes news consumption (e.g., [Bakshy et al., 2015](#); [Flaxman et al., 2016](#); [Allcott & Gentzkow, 2017](#); [Levy, 2021](#); [Nyhan et al., 2023](#); [Braghieri et al., 2026](#)). The available evidence on the latter comes from established platforms whose feeds were mostly populated by text, images and links, rather than a continuous stream of short videos. This paper is the first to study a short-form video platform in this literature, asking whether TikTok use reduces the quantity of news consumed, and whether that decline registers in political knowledge, attitudes and behavior. It finds that TikTok has qualitatively different effects from those of the platforms studied so far.

On knowledge, most experiments on established platforms find that access to these platforms increases factual knowledge about recent news and politics, whether the experiments deactivate Facebook or Instagram ([Allcott et al., 2020](#); [Mosquera et al., 2020](#); [Asimovic et al., 2021, 2023](#); [Arceneaux et al., 2024](#); [Allcott et al., 2024](#)) or alter the content delivered by their feeds ([Guess et al., 2023a,b](#)). [Allcott et al. \(2020\)](#) and [Mosquera et](#)

al. (2020) find that when access is removed people consume less news rather than turning to other sources, and conclude that Facebook is an important source of news. This paper finds that TikTok has the opposite effect to the platforms studied so far, reducing news consumption and lowering political knowledge.

On attitudes, the same experiments generally find little effect on affective polarization, measured by feelings toward the Democratic and Republican parties, despite reducing news consumption (Allcott et al., 2020; Guess et al., 2023b; Allcott et al., 2024). Switching from algorithmic to reverse-chronological feeds likewise leaves affective polarization unchanged (Guess et al., 2023a; Gauthier et al., 2026), although on X, switching to the algorithmic feed shifted political opinion toward conservative positions (Gauthier et al., 2026). Changing feed content to increase counter-attitudinal news or reduce political out-group hostility, however, lowers affective polarization (Levy, 2021; Piccardi et al., 2025; Stray et al., 2026). Evidence on issue polarization is mixed: Allcott et al. (2020) find greater polarization on current political issues alongside greater news exposure from Facebook access, while Allcott et al. (2024) find no effect of Facebook or Instagram deactivation. Melnikov (2025) finds that 3G access widened ideological polarization, measured by ideological self-placement, policy views, and voting behavior, alongside partisan divergence in news diets. The Cooperative Election Study does not measure affective polarization. Instead, I examine partisan alignment in presidential approval, perceived party ideology, and own ideology. Thus, while Facebook deactivation lowers news exposure with mixed effects on issue polarization and 3G widened partisan divergence in news diets alongside ideological polarization, TikTok produces a less-informed public that approves of the president more along partisan lines and views the opposing party as more extreme, without shifting its own ideology or partisanship.

The decline in news consumption and political knowledge speaks to a broader pattern in how new information technologies shape how informed the public is about politics and public affairs. The literature suggests that emerging technologies have drawn attention away from media that, at the time, carried more news. The introduction of television in the United States reduced newspaper and radio consumption and lowered political knowledge (Gentzkow, 2006). The spread of broadband shifted time toward online entertainment and depressed turnout, crowding out television and national newspapers in Germany (Falck et al., 2014) and local news in the United Kingdom (Gavazza et al., 2019). TikTok displaces news in the same way, but the decline is concentrated in the longest news sessions, the ones that demand sustained attention. This suggests that TikTok shapes not only the time available for news but the capacity to engage with it.

The rest of the paper is organized as follows. Section 2 describes the data and the evolution of TikTok use and news consumption over the sample period. Section 3 discusses the empirical strategy. Section 4 presents the main results. Section 5 concludes.

2 Data and background

This paper combines data on mobile device use, survey responses on traditional news consumption, political knowledge, political behavior and attitudes, and various auxiliary data sources. In this section, I describe the data sources and main variables, and document the evolution of TikTok use and news consumption over the sample period. The mobile device data span May 2019 to November 2025, and the survey data cover every wave from 2010 through 2024.

2.1 Data sources

Mobile device use. To measure device-level mobile browsing and app usage, this paper uses data from the media analytics company Comscore ([Comscore, Inc., 2019–2025](#)). Comscore recruits an opt-in panel of users aged 18 and older, who give informed consent and are compensated for participation. An app installed on their device passively records browsing and app usage ([Comscore, Inc., 2025](#)). The data include all websites and applications visited on iOS and Android devices, with precise timestamps and measures of time spent. Devices are geo-located to 205 designated market areas (DMAs) across the United States and linked to basic demographic and socio-economic characteristics, listed in Table A1. I refer to DMAs as media markets and to each device as an individual hereafter. In total, 183,785 individuals are observed at any point between May 2019 and November 2025. Figures A1–A4 compare the panel to the 2020 Decennial Census and American Community Survey ([US Census Bureau, 2023, 2022](#)). The panel matches the census closely on gender, on the Hispanic share of the population, and across most of the age and income distributions. Panelists over 60 are under-represented (13% vs. 28%), offset by an over-representation of those aged 31–50. Households below \$25K are over-represented (37% vs. 18%) and those above \$100K under-represented (11% vs. 31%). Single-person households are over-represented (22% vs. 11%), as are Black panelists (22% vs. 13%).

TikTok use is measured as the number of minutes or hours per week an individual spends in the TikTok application. Mobile news consumption is measured as weekly minutes spent browsing news websites or interacting with a news application. Appendix Tables A2 and A3 present descriptive statistics on TikTok use, use of other social media platforms, and mobile news consumption.

Traditional news consumption, political knowledge and attitudes. To examine the effects of TikTok use beyond mobile use, this paper uses the Cooperative Election Study, a large national survey of U.S. adults ([Cooperative Election Study, 2010–2024](#); [Kuriwaki, 2025](#)). Respondents are assigned to the media market containing their reported county of residence. This paper uses every CES wave from 2010 through 2024.

The flagship waves take place during midterm and presidential election years, and include between 53,640 and 63,272 respondents per wave, while the off-year waves are smaller (14,008 to 25,160 respondents). Across all waves between 2010 and 2024, the sample contains 597,687 respondent-year observations matched to a media market. This paper studies three families of outcomes.

Traditional news consumption. I measure traditional news consumption using a binary indicator equal to one if the respondent reports having watched TV news, read a newspaper or listened to radio news in the past 24 hours. These questions were not fielded in 2012, 2015 or 2017.

Attention to and knowledge about politics. Respondents are asked how often they follow what is going on in government and public affairs. I measure attention to politics on a four-point scale using responses to this question. The primary outcome is the number of correct answers to eight factual questions: the party affiliations of the respondent’s governor, two U.S. senators, and U.S. representative, and the majority parties in the U.S. House, U.S. Senate, state senate, and state lower chamber.

Political participation and polarization. For political participation, I use voter-file validated turnout in the general election, available in the eight federal-election years between 2010 and 2024. For political attitudes, I use a four-point job-approval item for the sitting president. I construct presidential approval polarization by signing this item by partisan alignment, so that higher values indicate approval among copartisans and disapproval among supporters of the other party.

Details on the questionnaire, coverage, variable construction, and summary statistics are reported in Appendix A.2 and Appendix Tables A4–A6.

Pop artist lists. Pop artists who promoted their music through Musical.ly during the platform’s 2015–2016 rise are identified from three kinds of contemporaneous material: a report for music-industry professionals documenting artists’ promotional campaigns on Musical.ly (Music Ally, 2017); music-press coverage of Musical.ly’s role in pop promotion (Spanos, 2017); and compilations of the viral Musical.ly challenges built around their songs (Zaira Ann, 2017; Maraki, 2015).¹

Comparison groups of artists are selected based on the Billboard’s 2015 year-end list of the 100 biggest U.S. hit songs (Billboard, 2015), the Teen Choice Award nominations

¹See a compilation of Musical.ly users participating in the “Watch Me (Whip/Nae Nae)” challenge, dancing to Silentó’s 2015 hit single of the same name popularized on the platform (<https://www.youtube.com/watch?v=BJFUrq07ybQ>), and a compilation of users participating in the #GetUglyDabChallenge, started by Jason Derulo in December 2015 to promote his single “Get Ugly” (<https://www.youtube.com/watch?v=vnYAbMH4Bik>). The Music Ally report and the Rolling Stone article are preserved on the Internet Archive at <https://web.archive.org/web/20251226164433/https://musically.com/wp-content/uploads/2017/06/Sandbox182-835969.pdf> and <https://web.archive.org/web/20190331104609/https://www.rollingstone.com/music/music-features/how-teen-craze-musical-ly-got-too-big-for-pop-to-ignore-108941/>.

for 2014 and 2015 (McGahan, 2014; PopCrush Staff, 2015), artist genre tags in the MusicBrainz database (MusicBrainz, 2026) and the Forbes Celebrity 100 (Forbes, 2014).² Section 3.1 lists the artists and describes how these sources are used to select comparison artists.

Pop artist characteristics. Artist characteristics used to compare the groups come from three sources. I code age, gender, place of birth, and stated race or ethnicity from each artist’s English-language Wikipedia biography (Wikipedia, 2026a). I code political endorsements from Wikipedia lists of endorsements in the 2012, 2016 and 2020 presidential campaigns (Wikipedia, 2026b). Finally, I use YouGov Ratings to measure the share of U.S. adults with a positive or negative opinion of each artist, overall and by generation and gender (YouGov, 2026). All pages were retrieved on May 16, 2026; Appendix Tables B1 and B2 list the articles and, for Wikipedia pages, the identifiers for the exact versions used.

Google search interest. Geographic interest in pop artists is measured using Google Trends data on relative search interest across U.S. media markets (Google, 2025–2026). For each artist, I collect relative interest in 2015, the year Musical.ly’s popularity surged, and in 2014 and 2016. I also collect yearly media-market search interest and monthly national search interest for Musical.ly, Vine, Dubsmash and TikTok. Appendix Tables B3 and B4 list every query with its Google Trends topic identifier and download date.

News pressure. News pressure is a TV-based measure of the intensity of newsworthy events. As in Eisensee & Strömberg (2007) and Durante & Zhuravskaya (2018), I compute news pressure as the number of minutes the NBC, ABC and CBS evening newscasts devote to their top three stories, averaged across the three networks each day and then across the days of the week. The data on broadcast programming come from the Vanderbilt Television News Archive (Vanderbilt Television News Archive, 2026), available until April 2025.

Media-market characteristics. Demographic characteristics come from the Census Bureau’s Vintage 2015 county population estimates (US Census Bureau, 2016b) and include total population and the shares of the population by age, sex, and race and ethnicity. Socio-economic characteristics come from the 2011–2015 five-year estimates of the Amer-

²The Billboard chart and the two Teen Choice Awards lists are preserved on the Internet Archive at <https://web.archive.org/web/20260413101228/https://www.billboard.com/charts/year-end/2015/hot-100-songs/>, <https://web.archive.org/web/20260315022616/https://popcrush.com/2014-teen-choice-awards-winners-list/> and <https://web.archive.org/web/20260218175439/https://popcrush.com/2015-teen-choice-awards-winners-list-see-the-awards/>. The 2014 Forbes Celebrity 100 is preserved at <https://web.archive.org/web/20140703073400/https://www.forbes.com/celebrities/>

ican Community Survey, retrieved through Social Explorer ([US Census Bureau, 2016a](#)). They are the shares of adults by educational attainment, log median household income, the poverty and unemployment rates, and the Gini index. Presidential vote shares for 2000–2016, used in the balance tests, come from the county returns of the MIT Election Data and Science Lab ([MIT Election Data and Science Lab, 2018](#)). Table A7 presents summary statistics for media market characteristics.

2.2 TikTok use and news consumption over time

TikTok use

Figure 1 shows that TikTok’s user base grew drastically over the sample period. The share of individuals that used the app in a given week rose from about 5% in 2019, before the pandemic, to about 12.5% in early 2020, coinciding with the first major COVID-19 lockdown orders in the U.S., and then climbed steadily to about 40% by the end of 2025. Among individuals with non-zero use in a given week, median weekly use has remained relatively stable at about one to two hours throughout the period, while the average rose from about 6 hours per week in 2019–2020 to above 10 hours, before declining and stabilizing at roughly 8 hours per week in 2022–2025.

[Figure 1 about here]

Figure A5 shows TikTok use from 2019 through 2025 by age groups. For those under 26, TikTok use grew from about 10% to 60% by the end of 2025. Even for those over the age of 65, TikTok use grew from essentially no use in 2019 to about 20% by the end of 2025. Figure A6 compares TikTok’s user base to that of other major platforms. YouTube and Facebook were used by about 70% of individuals per week throughout. Instagram use increased from about 30% to 40%, and Twitter/X use decreased from roughly 20% to 15%.

Mobile news consumption

Over the sample period, mobile news consumption drifts with pronounced medium-term swings, as Figure 2 shows. Before the 2020 pandemic, the average panelist spent about 80–100 minutes per week consuming news. During the 2020 pandemic, when lockdowns sharply raised overall mobile device use (see Figure A7), mobile news consumption increased sharply and was particularly volatile. Throughout 2021 news consumption declined in waves from about 120 weekly minutes to 80 weekly minutes, after which weekly consumption has drifted between 70 and 90 minutes per week.

[Figure 2 about here]

Within these swings, however, news consumption spikes sharply around major newsworthy events. The lower panel of Figure 2 shows deviations in news consumption relative to an 8-week rolling average, along with the news event weeks identified from those deviations. I define a week as a *news event week* if national news consumption exceeds its surrounding 8-week rolling average by more than one standard deviation. This yields 40 news event weeks between 2019 and 2025, which line up with well-known episodes: election weeks, mass shootings, natural disasters and major political news such as impeachment proceedings and national protest movements.

News consumption beyond mobile devices

The mobile use panel measures mobile news consumption from 2019 onward. To examine longer-run trends beyond mobile devices, I use survey measures of television, newspaper, and radio news consumption available from 2010 onward. The share reporting any of these forms of news consumption declined substantially between 2010 and 2024, as Appendix Figure A8 shows. The share was about 88% until 2014, fell to 82% by 2018, and to 73% by 2024. Appendix Figures A9–A11 show that this decline extends across television, newspapers, and radio. The share reporting television news consumption declined from about 73% to 57%, newspaper readership from about 57% to 29%, and radio news consumption from about 42% to 28%.

3 Empirical strategy

This section describes the empirical strategy. I start by discussing the empirical challenge and the construction of the instrument. Next, I describe the specifications estimated on mobile use data and discuss identification. Finally, I describe how device-level TikTok use is linked to survey respondents, the specifications estimated to uncover TikTok’s effects beyond mobile devices, and additional identification checks using survey data.

3.1 Empirical challenge and the source of variation

The effects of interest are those of TikTok use on mobile news consumption, on traditional news consumption, and on political knowledge, participation and attitudes. An ordinary least squares estimate of the effect of TikTok use is unlikely to recover the causal effect. Across individuals, those with little interest in news may use TikTok more heavily, biasing the estimate downward. Conversely, individuals who generally spend substantial time on mobile devices, including consuming news, may be more likely to use TikTok, biasing the estimate upward. Across media markets, those where TikTok use became high may differ from other markets in ways that independently shape news consumption, political knowledge and political attitudes. I therefore isolate variation in TikTok use

stemming from early marketing on TikTok’s predecessor, Musical.ly. In particular, I use geographic variation in 2015 fandom of artists who promoted their music on Musical.ly to predict TikTok use between 2019 and 2025, in an instrumental variable framework.

Musical.ly artists and selection of comparison artists

I identify nine pop artists who promoted their music through Musical.ly during the platform’s 2015–2016 rise. I refer to them as Musical.ly artists hereafter. They are: Lady Gaga, Katy Perry, Ariana Grande, Meghan Trainor, Selena Gomez, Silentó, Jason Derulo, Charli XCX, and Anne-Marie.

A natural concern is that interest in the Musical.ly artists proxies for a specific taste in music, which could correlate with unobserved determinants of news consumption, political knowledge, and attitudes. To separate interest in the Musical.ly artists from broader musical taste, I construct two groups of comparable pop artists who did not advertise on Musical.ly, and control for interest in them throughout.

Some of the Musical.ly artists were, by 2015, established superstars. In particular, Lady Gaga and Katy Perry both appear on the 2014 “Forbes Celebrity 100” list of the world’s most powerful celebrities ([Forbes, 2014](#)). To absorb general interest in mainstream pop stars, I construct a group of *superstar* placebo artists. This group consists of musicians on the Forbes Celebrity 100 who did not advertise on Musical.ly, and have a MusicBrainz genre classification of pop, hip-hop, R&B, or dance ([MusicBrainz, 2026](#)).

Most of the Musical.ly artists, however, were young, up-and-coming artists who used the platform as an alternative channel to promote their music and go viral, and who appealed disproportionately to the teenagers who populated Musical.ly at the time. I build a second comparison group of *up-and-coming* placebo artists out of rising pop artists with recent chart success and demonstrated teen appeal. I select artists that meet the following criteria:

1. *Hit single in 2015*: lead artist on the Billboard year-end list of 100 biggest U.S. hits ([Billboard, 2015](#)).
2. *Teen appeal*: Teen Choice Awards music-category nominee in 2014 or 2015, voted by teenagers aged 13–19 ([McGahan, 2014](#); [PopCrush Staff, 2015](#)).
3. *Career recency*: first Billboard year-end top 100 hit in 2010 or later, screening out long-tenured artists who happened to have a 2015 hit.
4. *Pop(ular) music*: MusicBrainz genre classification of pop, hip-hop, R&B, or dance ([MusicBrainz, 2026](#)).
5. *Not a superstar*: not in the Forbes Celebrity 100 ([Forbes, 2014](#)).

Applying these same criteria to the Musical.ly artists themselves confirms that they fall almost entirely into the two groups. Lady Gaga and Katy Perry qualify as *superstars*, and Ariana Grande, Meghan Trainor, Selena Gomez, and Silentó meet all five *up-and-coming* criteria. The remaining three nearly do: Jason Derulo first charted in 2009 and so misses

the career-recency cutoff by one year; Charli XCX appears on the 2015 Year-End Hot 100 only as a featured artist on “Fancy,” failing the lead-artist requirement; and Anne-Marie’s first Billboard hit single came in 2016, just outside the window. The comparison groups of artists are therefore built from the same observable profile as the Musical.ly artists, differing only in that they did not advertise on the platform. Table B5 lists every artist in each of the three groups.

Geographic variation in artist fandom

To measure geographic variation in fandom of the Musical.ly artists in 2015, I use relative search interest in each artist across U.S. media markets from Google Trends. To combine each artist’s relative search intensity into a single measure of interest, I standardize each artist’s search intensity across media markets and average across artists:

$$\text{Musical.ly Celebrity index}_m = \frac{1}{A} \sum_a \frac{\text{Google search}_{am} - \mu_a}{\sigma_a}, \quad (1)$$

where $\text{Google search}_{am}$ is relative Google search interest in Musical.ly artist a in media market m in 2015, μ_a and σ_a are the mean and standard deviation of $\text{Google search}_{am}$ across media markets, and A is the number of Musical.ly artists. Standardizing search for each artist before averaging keeps more-searched artists from dominating the index, so that the index reflects the relative geographic spread of interest across the Musical.ly artists rather than the popularity of any one of them. Superstar and up-and-coming celebrity indices are constructed analogously: for each superstar placebo artist, I standardize their relative Google search interest, and average across the group, creating a superstar index. I create an up-and-coming celebrity index in the same way.

[Figure 3 about here]

After accounting for four key media-market characteristics—the shares of women, Hispanic residents, residents in poverty, and high-school graduates—media markets with above- and below-median interest in the Musical.ly artists are balanced across a broad range of pre-2018 voting outcomes and demographic, socio-economic, and panel characteristics, as Table B6 reports. These key characteristics are controlled for throughout. Figure 3 presents the geographic variation in the Musical.ly Celebrity index that remains after accounting for those characteristics. Appendix Figure B1 presents the raw index, and Figure B2 reports the geographic variation in interest in up-and-coming and superstar artists.

3.2 Specifications using mobile use data

Cross-section

The causal relationship of interest is how time spent on TikTok affects time spent con-

suming news. In its simplest form, the relationship can be represented as:

$$News\ Consumption_{im} = \beta_1 TikTok_{im} + \beta_2 Up-and-coming_{im} + \beta_3 Superstar_{im} + \mathbf{X}'_{im}\boldsymbol{\beta} + \epsilon_i \quad (2)$$

where $News\ Consumption_{im}$ is the logarithm of average weekly minutes of news consumption for individual i in media market m , averaged across the weeks individual i is observed. $TikTok_{im}$ is average weekly hours of TikTok use, $Up-and-coming_{im}$ and $Superstar_{im}$ are the placebo indices in individual i 's media market, and $\mathbf{X}'_{im}$ is a vector of individual- and media-market-level controls. ϵ_i is the error term.

Estimated by ordinary least squares, β_1 is unlikely to recover a causal effect, for the reasons set out in Section 3.1. I therefore instrument $TikTok_{im}$ with the Musical.ly Celebrity index. The first stage is

$$TikTok_{im} = \phi_1 Musical.ly\ Celebs_{im} + \phi_2 Up-and-coming_{im} + \phi_3 Superstar_{im} + \mathbf{X}'_{im}\boldsymbol{\pi} + \eta_i \quad (3)$$

where $Musical.ly\ Celebs_{im}$ is the Musical.ly Celebrity index in individual i 's media market m , and other notation is as before. As the instrument varies by media market, standard errors are clustered by media market.

News event weeks

The medium-term swings in news consumption documented in Section 2.2 are largely unrelated to particular news stories and inject substantial noise into its level, whereas news consumption rises sharply and predictably around major news events. The empirical strategy therefore exploits these news-event-week spikes in place of the noisy level of news consumption: I estimate how pre-event TikTok use shapes the news-consumption response to these events, in a weekly panel. The specification is

$$News\ Consumption_{imt} = \beta_1 TikTok_{imt} \times Event_t + \beta_2 Up-and-coming_{im} \times Event_t + \beta_3 Superstar_{im} \times Event_t + \mathbf{X}'_{imt}\boldsymbol{\beta} + \alpha_i + \gamma_t + \epsilon_{it} \quad (4)$$

where $News\ Consumption_{imt}$ is weekly minutes of news consumption for individual i in media market m in week t ; $TikTok_{imt}$ is average weekly hours of TikTok use for individual i in the four weeks before week t ; $Event_t$ is an indicator equal to one if a week t is a news event week; α_i and γ_t are individual and week fixed effects; and the remaining notation is as above. This model is estimated on individual-week observations observed in the four weeks before and after each news event week. The coefficient of interest, β_1 , captures the differential effect of pre-event TikTok use on news consumption during news event weeks.

As in the cross-section, $TikTok_{imt} \times Event_t$ is instrumented, here with the interaction

of the Musical.ly Celebrity index and the event indicator. The first stage is

$$\begin{aligned} TikTok_{imt} \times Event_t = & \phi_1 Musical.ly\ Celebs_{im} \times Event_t + \phi_2 Up-and-coming_{im} \times Event_t \\ & + \phi_3 Superstar_{im} \times Event_t + \mathbf{X}'_{imt} \boldsymbol{\pi} + \lambda_i + \rho_t + \eta_{it} \end{aligned} \quad (5)$$

where λ_i and ρ_t are individual and week fixed effects. The controls are time-invariant and enter interacted with $Event_t$. Standard errors are clustered by media market.

Identification

A causal interpretation of β_1 in equations (2) and (4) rests on the exclusion restriction: conditional on the controls and the placebo indices, interest in the Musical.ly artists affects event-week news consumption only through its effect on TikTok use. One potential threat to identification is that interest in the artists reflects general mobile-use habits or a latent interest in news. A second potential threat is that the political engagement of the Musical.ly artists shapes how their fans engage with news and politics. I provide evidence against both, at the level of media markets and at the level of the artists.

At the media-market level, the balance tests in Section 3.1 speak to the first concern. Table B6 shows that, conditional on the four key characteristics included throughout, media markets with above- and below-median interest in the Musical.ly artists are statistically indistinguishable across a broad range of pre-2018 voting, demographic, socio-economic, and panel characteristics. The instrument is therefore unlikely to capture media-market traits that independently drive how a market’s news consumption responds to events.

At the artist level, I address both concerns using the comparison artists described in Section 3.1. Tables B7 and B8 compare the groups of artists, pooling comparison artists and splitting into superstars and up-and-coming artists respectively. The groups are statistically indistinguishable in their presidential-candidate endorsements from 2012 to 2020, both in how often they endorse and in whom they endorse. The Musical.ly artists are somewhat younger and more likely to be women. Unlike political engagement, which could affect fans directly, artist demographics matter only through the audiences they attract. In YouGov public-opinion data (YouGov, 2026), the two groups are equally popular overall, and equally popular within each age and gender group, indicating their audiences have similar demographic profiles.

Because both stages include interest in these artists, β_1 is identified only from the component of Musical.ly artist interest that is orthogonal to interest in comparable artists who did not advertise on the platform. A general taste for pop music therefore cannot drive the estimate. The placebo indices also serve as placebo treatments: if Musical.ly artist interest were to affect event-week news consumption through any channel other than

its effect on TikTok use, one would expect interest in similar comparison artists to do the same. The results show that they predict neither TikTok use nor news consumption, alleviating these concerns.

Two further sets of placebo treatments extend this logic. The first is interest in the Musical.ly artists themselves in 2014 and 2016, before and after the platform’s initial rise in popularity. The second is interest in contemporaneous early short-form video platforms, Vine and Dubsmash. Interest in the Musical.ly artists in the adjacent years predicts neither TikTok use nor news consumption. Similarly, interest in Vine and Dubsmash shows no consistent association with news consumption. Controlling for either set leaves the estimates unchanged.

The 2SLS estimate of the effect of TikTok use on news consumption, β_1 , is a local average treatment effect for individuals whose TikTok use responds to the instrument. Interest in the Musical.ly artists increased early adoption of Musical.ly, both directly among their fans and through network effects, creating geographic differences in the platform’s user base that persisted through the merger with TikTok. The estimate therefore applies to individuals whose TikTok use increased as a result of these differences in local adoption.

3.3 Effects beyond mobile devices

To explore effects of TikTok use beyond mobile use, I aggregate TikTok use to the media-market level and link to survey respondents from those media markets. To account for the demographic composition of Comscore panelists between media markets, I residualize each individual’s TikTok use on the individual-level characteristics listed in Table A1. I then average these residuals within each media market. The resulting measure is therefore the average media-market-level TikTok use adjusted for the composition of the Comscore panel. In this subsection the unit of observation is a survey respondent and t indexes survey years rather than weeks.

I compare respondents in markets where TikTok use became high and those in markets where it did not, before and after the platform’s arrival in the United States:

$$Y_{imt} = \beta_1 TikTok_{im} \times Post_t + \beta_2 Up-and-coming_{im} \times Post_t + \beta_3 Superstar_{im} \times Post_t + \mathbf{X}'_{imt}\boldsymbol{\beta} + \alpha_m + \gamma_t + \epsilon_{imt} \quad (6)$$

where Y_{imt} is the outcome of respondent i in media market m in survey year t , $TikTok_{im}$ is average adjusted weekly hours of TikTok use in respondent i ’s media market m , and $Post_t$ is an indicator for survey years after 2018, the year in which Musical.ly was consolidated into TikTok and the app became widely available in the United States. $Up-and-coming_{im}$ and $Superstar_{im}$ are the placebo indices in respondent i ’s market m ; α_m and γ_t are media-market and survey-year fixed effects, and $\mathbf{X}'_{imt}$ are individual- and pre-2018 media-market

controls, the latter interacted with $Post_t$. Observations are weighted to represent the adult U.S. population, and standard errors are clustered by media market.

As before, I instrument TikTok use with interest in Musically artists in 2015. As the index does not vary with time and is absorbed by α_m , I instrument $TikTok_{im} \times Post_t$ with the interaction of the Musically Celebrity index and $Post_t$. The first stage is

$$TikTok_{im} \times Post_t = \phi_1 Musical.ly\ Celebs_{im} \times Post_t + \phi_2 Up-and-coming_{im} \times Post_t + \phi_3 Superstar_{im} \times Post_t + \mathbf{X}'_{imt} \boldsymbol{\pi} + \lambda_m + \rho_t + \eta_{imt} \quad (7)$$

where λ_m and ρ_t are media-market and survey-year fixed effects and the remaining notation is as above. The coefficient of interest, β_1 , is the effect of an additional hour of average weekly TikTok use in a respondent's media market on that respondent's outcome, identified from the differential change after 2018 between markets whose residents were more and less interested in the Musically artists in 2015.

I also estimate an event-study version, in which the survey years are grouped into two-year periods and each period is allowed its own coefficient:

$$Y_{imt} = \sum_{k \neq k_0} \beta_k TikTok_{imk} \times \mathbf{1}[t \in k] + \sum_{k \neq k_0} (\delta_k Up-and-coming_{im} + \zeta_k Superstar_{im}) \times \mathbf{1}[t \in k] + \mathbf{X}'_{imt} \boldsymbol{\beta} + \alpha_m + \gamma_t + \epsilon_{imt} \quad (8)$$

where k indexes the periods, $\mathbf{1}[t \in k]$ is an indicator for survey year t falling in period k , and k_0 denotes the omitted period, 2016–2017, against which all others are measured. The market controls enter interacted with each period rather than with $Post_t$. $TikTok_{imk}$ represents average adjusted weekly hours of TikTok use in respondent i 's market m in period k . Each $TikTok_{imk} \times \mathbf{1}[t \in k]$ term is instrumented by the corresponding interaction of the Musically Celebrity index with the period indicator, so the model is exactly identified period by period.

Data on TikTok use are not available before May 2019, and 2019 data cover relatively few individuals. Respondents in the first post-period, 2018–2019, are therefore assigned their media market's average TikTok use in 2019–2020. Because TikTok use is instrumented, the measurement error induced by this assignment does not bias the 2SLS estimates, provided it is uncorrelated with the instrument (Angrist & Krueger, 2001). The pre-periods are assigned the market's average TikTok use over the full sample, so that the pre-period coefficients estimate whether markets that later had high TikTok use were already moving differently before the platform arrived.

As before, interest in up-and-coming and superstar artists provides placebo treatments to verify that effects are not driven by unobserved confounders. Further, the event-study estimates support the assumption that interest in the Musically artists affects survey

responses only through its effect on TikTok use. Markets that would later adopt TikTok heavily because of Musical.ly artist fandom were not already diverging from other markets before the platform arrived.

The survey-based analysis relates media-market-level TikTok use to individual-level outcomes. Therefore, it cannot separate the effect of an individual’s own TikTok use from that of TikTok use in that individual’s local environment. Many of the outcomes studied are shaped in part by horizontal transmission from others. Such spillovers are important for political information and political attitudes (Fafchamps & Vicente, 2013; Arias et al., 2019; Enríquez et al., 2024; Egorov et al., 2026). Spillovers from TikTok use within a media market are therefore a relevant quantity for assessing the platform’s aggregate consequences.

4 Results

4.1 The first stage

The instrumental strategy relies on 2015 interest in the Musical.ly artists capturing local adoption of Musical.ly and, through the 2018 merger, subsequent adoption of TikTok. Figure 4 provides evidence in support of the first link: higher interest in the Musical.ly artists predicts greater Google search interest in Musical.ly itself, with the relationship emerging as the platform became more popular over 2015–2018 and fading after the platform ceased operating. The placebo indices show no such pattern.

The relationship is specific to Musical.ly rather than to short-form video platforms in general. Appendix Figures C1 and C2 repeat the exercise with Google searches for Vine and Dubsmash, contemporaneous short-form video platforms (Hu, 2018; McCammond, 2017). They show that interest in the Musical.ly artists does not predict greater search for either of these two platforms after 2015.

[Figure 4 about here]

Table 1 reports the cross-sectional first stage, regressing average weekly TikTok minutes on the Musical.ly Celebrity index, the Superstar index, and the Up-and-coming celebrity index. The Musical.ly Celebrity index strongly and robustly predicts TikTok use. A unit increase in the index predicts about 28 to 37 additional weekly minutes across specifications, with an estimate of 37 minutes in the most stringent specification. A one-standard-deviation increase in the index, corresponding to an increase of 0.4, predicts about 15 additional weekly minutes. Across the nine specifications the cluster-robust first-stage F ranges from 14.7 to 24.5; with a single excluded instrument this is the effective F-statistic of Montiel Olea & Pflueger (2013). It clears the 20% worst-case-bias

threshold (15.1) in most specifications and reaches 24.51 in the most saturated specification, past the 10% threshold of 23.1.

[Table 1 about here]

In contrast, the superstar index does not predict TikTok use in any specification. The up-and-coming index is never positively associated with use, and the negative association is insignificant once the full set of controls is included. This suggests that the positive association between Musical.ly Celebrity interest in 2015 and TikTok use in 2019–2025 is driven by the artists’ advertising on the platform, not by unobservable characteristics related to general taste in artists with a similar profile.

Figure 5 visualizes this first stage with binned scatter plots. Panel (a) plots residualized average weekly TikTok minutes against the residualized Musical.ly Celebrity index, partialing out the media-market balance and individual-level controls of column 5 of Table 1; panel (b) partials out the full set of controls in column 9. The figure shows a clear positive relationship under both control sets. Figure C3 in the Appendix shows that the placebo indices have no such relationship with TikTok use.

[Figure 5 about here]

The first stage is robust across various measures of TikTok use, and the instrument affects use across the distribution. Table C1 shows that interest in Musical.ly celebrities predicts both higher median weekly minutes of TikTok use and a larger share of observed days with any recorded TikTok use. Table C2 shows that the instrument increases the probability of any use by 2.8 percentage points, and the probability that an individual’s weekly TikTok minutes exceed the twentieth, fortieth, sixtieth and eightieth percentiles among users. The Musical.ly Celebrity index therefore affects both whether individuals use TikTok and how much, and does so across the distribution rather than at one end of it. Finally, Appendix Figure C4 shows that the Musical.ly Celebrity index predicts increased TikTok use across all age groups, including for those over the age of 65. The effect is largest among young panelists aged 18–25 and those aged 26–35, as one would expect.

The Musical.ly Celebrity index predicts TikTok use specifically, not social media use in general. Appendix Table C3 and Figure C5 show how the three indices correlate with use of other social media platforms. The Musical.ly index predicts less time on Instagram and on WhatsApp, by about 13 and 12 minutes per week against means of 74 and 24, consistent with some substitution away from those platforms toward TikTok. The remaining coefficients are small relative to outcome means, and the index does not predict total time across the thirteen platforms tested. The instrument thus changes the composition of non-TikTok social media use slightly while leaving overall social media use unaffected. The IV estimates therefore capture effects specific to TikTok rather than changes in social media use more generally.

4.2 TikTok and mobile news consumption

4.2.1 Cross-sectional results

I begin with the cross-section of individuals. Table 2 reports two-stage least squares estimates of equation (2) for four measures of mobile use and mobile news consumption, each averaged over the weeks an individual is observed. Column (1) shows that TikTok use increases overall mobile use, consistent with earlier work (Maarouf et al., 2025; Agarwal & Ananthakrishnan, 2024). An additional weekly hour of TikTok increases total mobile minutes by about 5.5%, significant at the 10% level.

With outcomes averaged across weeks in this way, however, the estimates show no detectable effect of TikTok use on news consumption. Column (2) shows no change in the probability that an individual consumes any news. Columns (3) and (4) show no detectable effect on the amount of news consumed, whether measured among individuals with positive news consumption or across all individuals. The Anderson–Rubin confidence sets, which are robust to weak identification, are wide: they rule out neither a substantial decline nor a substantial increase. Appendix Table C4 reports the same estimates with increasingly stringent controls, and the pattern is stable throughout.

[Table 2 about here]

Importantly, overall mobile news consumption is noisy over time (see Figure 2). Medium-term swings often coincide with broader changes in overall mobile use, while some weeks contain major newsworthy events and others do not. I therefore focus next on weeks in which major news events occur, when news consumption is particularly important.

4.2.2 News consumption during weeks of major news events

Table 3 reports estimates of equation (4) for three measures of the news response: whether an individual consumes any news during a news event week, how much news they consume conditional on non-zero consumption, and a single measure combining the two.³ Panel A shows the two-stage least squares estimates, Panel B the reduced form estimates and Panel C the ordinary least squares estimates.

The corresponding first stage is reported in Appendix Table C6. In the panel, the first stage cluster-robust F-statistic is about 16.0. Anderson–Rubin confidence sets, robust to

³Weekly news consumption is zero for many individual-week observations, so a logarithm is not defined. As J. Chen & Roth (2024) show, the average effect of a log-like transformation of an outcome with zeros is not unit-invariant: the units of the outcome implicitly determine how heavily the estimate weights the extensive margin. Following their recommendation to make that weighting explicit, weeks with less than one minute of news are coded as zero, and column (3) values the move from zero to one minute as a doubling. Appendix Table C5 values the same move at zero and gives comparable results.

weak instruments, are reported throughout. As in the cross-section, neither placebo index positively predicts TikTok use.

Panel B reports the reduced form. Interest in the Musical.ly artists predicts less news consumption during news event weeks on all three measures, while neither comparison index predicts any of the three measures. The p -values at the foot of the panel test this directly, and equality between the Musical.ly coefficient and the comparison coefficients is rejected jointly on every measure. Panel A instruments TikTok use with the Musical.ly Celebrity index. An additional weekly hour of TikTok use decreases the probability that an individual consumes any news during a news event week by 1.7 percentage points, a 2.5% decrease from the mean, and decreases news minutes by about 6% among individual-week observations with positive consumption. Combining the intensive and extensive margins in column (3), an additional hour of weekly TikTok use decreases news consumption during news event weeks by about 10%. Against a mean of 91 weekly minutes of news during news event weeks, this is a decline of roughly 9 minutes. The average individual spends 132 minutes per week on TikTok, so moving from no use to the average implies a decline of about 22%. The 95% Anderson–Rubin confidence set ranges from -0.210 to -0.041 and excludes zero. The comparison artist indices are, with one exception, not associated with news consumption.

[Table 3 about here]

Comparing the ordinary least squares estimates in Panel C to the two-stage least squares estimates in Panel A is informative about selection into TikTok use. Ordinary least squares estimates are small and positive for every measure, as selection on mobile use would predict: heavier mobile users spend more time on TikTok, and their news consumption responds more strongly to newsworthy events. Appendix Table C7 supports this interpretation. Conditioning on total mobile use in the weeks preceding a news event week renders the naive association negative and leaves the instrumented estimate unaffected. Both the naive association and the two-stage least squares estimates are also negative when news consumption is expressed as a share of mobile time.

Appendix Tables C8–C11 show that the estimates are robust to alternative definitions of news event weeks. Raising the threshold to 1.5 standard deviations above the rolling average, lowering it to 0.75, and defining news event week as a week more than 5% above the rolling average all leave the results close to the baseline. The estimates are also robust to dropping the news event weeks from March through May 2020, during the first wave of COVID-19 stay-at-home orders (Moreland, 2020). This period is characterized by high and volatile mobile use (Figure A7), and dropping it raises the first-stage F-statistic from 16.0 to 19.4 and the estimated effect on the intensive margin from -0.063 to -0.089 . In Table C11 news event weeks are defined as the weeks in the top decile of news pressure, a measure of newsworthy events based on TV news rather than mobile data. This is a

demanding test, as weeks of high news pressure do not always coincide with substantial increases in mobile news consumption (see Figure A12), reflecting differences in mobile news demand and TV news programming. TikTok use reduces the news response during high news-pressure weeks on the extensive margin, while an effect is not detected on the intensive margin.

4.2.3 Alternative placebo instruments

Identification relies on the assumption that, conditional on controls, interest in Musical.ly artists in 2015 affects news consumption only through its effect on TikTok use. This section probes the plausibility of the exclusion restriction further by incorporating alternative “placebo instruments,” that is, variables likely to capture potential confounders correlated with the 2015 Musical.ly Celebrity index. I consider four additional placebo instruments, grouped into two pairs. The first pair is interest in the nine Musical.ly artists measured in 2014 and 2016, the years before and after Musical.ly’s initial surge in popularity (see Figure C6). The second pair is 2015 interest in Vine and Dubsmash, which captures general early enthusiasm for short-form video social media platforms. Figures 6 and 7 show the first and second stage including these pairs of placebo instruments, alone and in combination with each other. The underlying estimates are reported in Appendix Tables C12 and C13. The results are robust to the inclusion of the placebo instruments.

[Figure 6–7 about here]

Figure 6 shows that the Musical.ly Celebrity index from 2015 predicts TikTok use in every specification. The coefficient is stable around 0.5 and the cluster-robust F-statistic is 16 or higher in most specifications, above 10 in all specifications. Across six placebo instruments, none predict increased TikTok use. Consistent with Vine and Musical.ly competing for the same audience in 2015, leaving markets with greater interest in Vine fewer Musical.ly users to migrate to TikTok in 2018, interest in Vine correlates negatively with TikTok use.

The estimated causal effect of TikTok use on the news consumption response to major news events is robust to the inclusion of the additional placebo instruments. Figure 7 reports the second-stage estimates for the outcome combining the intensive and extensive margins. Across specifications, an additional hour of weekly TikTok use decreases news consumption during news event weeks by between 7% and 15%, significant at the 5% level throughout. The 27 coefficients on placebo instruments are close to zero and insignificant, with one exception: Vine interest predicts a smaller news response when no celebrity index is controlled for. With the celebrity indices included, the estimate is smaller and indistinguishable from zero.⁴ Appendix Figures C7–C9 repeat the exercise

⁴One of the 27 placebo coefficients is significant at the 5% level, or 4% of coefficients. At the 5% level, this is about what could be expected by chance.

for the extensive margin, the intensive margin, and the combined outcome valuing the extensive margin at zero, with the same pattern.

Appendix Tables C14–C17 report the reduced-form estimates from the same specifications. They also test explicitly whether the placebo coefficients differ from the one on the 2015 Musical.ly Celebrity index. In the specification including all six placebo instruments, equality is rejected on every outcome. For individual placebo instruments, equality is rejected for interest in Vine and Dubsmash throughout, and for interest in superstar and up-and-coming celebrities in 2015 in almost every case. Equality of coefficients between interest in the Musical.ly artists in 2014 and 2016 versus 2015 is rarely rejected: interest in these artists is persistent from year to year, making comparison between their coefficients too statistically demanding.⁵

Taken together, these results reinforce confidence in the causal interpretation of the main estimates. The estimated effect of TikTok use on news consumption is not driven by unobserved confounders related to interest in artists similar to the Musical.ly artists or general early interest in short-form video platforms. Further, interest in the same artists in adjacent years neither predicts increased TikTok use nor changes the estimated effect.

4.3 TikTok and sustained attention to news

The relative decline in news consumption during major, salient news events is characterized by a decrease in the length of individuals' longest news consumption sessions. In a typical week an individual visits news sites and applications several times, and Table 4 studies how TikTok use affects the length of those visits. It takes as outcomes the length of an individual's longest news consumption session in a given week, the mean length, the median, and the 75th and 95th percentiles. As before, the estimates compare news event weeks to other weeks, across individuals with varying TikTok use.

Panel A uses all 40 news event weeks. Column (1) shows that an hour of TikTok use decreases an individual's longest news consumption session in news event weeks by 0.62 minutes against a mean of 20.8 minutes, while average length is not affected. Panel B excludes the weeks during the first wave of COVID-19 stay-at-home orders (Moreland, 2020), as mobile use during this period reflected the unusual circumstances of confinement rather than ordinary behavior. Outside of the first COVID-19 lockdown period, the effect on the longest session grows to about one minute. An additional hour also significantly decreases the length of news sessions at the 75th and 95th percentiles by 0.21 and 0.38 minutes, respectively, at the 10% level. Appendix Figure C10 shows the effect on the longest session outside of the first lockdown period while controlling for the alternative placebo instruments. The effect is negative and significant at the 1% level throughout.

⁵A raw correlation coefficient between the Musical.ly Celebrity index from 2015 and 2014 is 0.838. Between the index from 2015 and 2016 the correlation is 0.637.

[Table 4 about here]

Overall, as a result of TikTok use, an individual’s average news session is unchanged, while longer sessions are shortened, with the largest reduction in the longest session of the week. TikTok use therefore reduces the propensity to stay with news for an extended period. Because longer news sessions require more sustained attention, this pattern is consistent with a reduced capacity to sustain attention as a mechanism contributing to TikTok’s effect on news consumption.

4.4 TikTok’s effect beyond mobile devices: traditional news consumption, political knowledge, behavior and attitudes

Table 5 studies the effect of aggregate media-market-level TikTok use on survey respondents in the Cooperative Election Study. The table reports 2SLS difference-in-differences estimates, comparing respondents within media markets before and after TikTok’s proliferation. Appendix Table C18 reports the corresponding ordinary least squares and reduced-form estimates. Figures 8–10 report event-study versions of the same specifications, documenting the dynamic effect of media-market-level TikTok use.

Column (1) of Table 5 reports the first stage. Interest in the Musically artists in 2015 predicts subsequent market-level TikTok use, with a cluster-robust F-statistic of about 27.5. Column (2) shows that the effect of TikTok on news consumption extends beyond mobile devices. An additional weekly hour of TikTok use in a media market decreases the probability of having watched television news, read a newspaper or listened to the radio in the past 24 hours by 5.5 percentage points, a 7% decrease from the mean of 0.81. Appendix Table C19 takes each medium separately. The same increase in TikTok use decreases newspaper readership by 8.9 percentage points compared to a mean of 0.40, decreases television news consumption by 4.1 points, with a mean of 0.65, and has no detectable effect on radio news.

[Table 5 about here]

Columns (3)–(4) of Table 5 show that TikTok use also affects political engagement and knowledge. An additional hour of TikTok use leads respondents to report following government and public affairs less closely, by 0.22 points on a four-point scale, and to answer 0.63 fewer of the eight knowledge questions correctly, against a mean of 4.97. Both estimates are significant at the 5% level. Appendix Table C20 shows that TikTok affects both knowledge about national politics and about state and district politics.

Columns (5)–(6) of Table 5 show that market-level TikTok use also affects political behavior and attitudes. An additional hour of market-level TikTok use leads to a 5 percentage point decrease in the likelihood of turnout among respondents, compared to a

mean of 54%. Column (6) of Table 5 shows that TikTok penetration polarizes presidential approval along party lines. The outcome is a partisan-aligned measure of presidential approval on a four-point scale, so that higher values indicate greater approval among copartisans of the president and greater disapproval among supporters of the other party. An additional hour of market-level TikTok use increases partisan-aligned approval by 0.22 points from an average of 3.5. Appendix Table C21 shows that the effect is robust to an alternative definition of partisanship, and is driven by increased approval of a sitting president among copartisans, while point estimates for approval among out-partisan are negative but insignificant. Appendix Table C22 reports effects on perceived party extremity, respondents' own ideological extremity, and strength of partisanship. The estimates show that TikTok use also increases perceived ideological extremity for the opposing party, while leaving perceptions of own party's ideology, distance between own ideology and the parties' ideologies, one's own ideological extremity, and strength of partisanship unaffected.

Event-study estimates

Figure 8 shows how the first-stage relationship evolves over time. The Musical.ly Celebrity index predicts TikTok use from the first survey period after the merger and strengthens as the platform diffuses, from 0.21 in 2018–2019 to 0.83 in 2024. The placebo indices generally do not correlate with TikTok use. The period-by-period F-statistics are weak in 2018–2019 and of moderate strength thereafter (10.5–18.2). The event-study 2SLS figures therefore report Anderson–Rubin confidence sets rather than conventional confidence intervals. The weak first stage in 2018–2019 is reflected in the width of its confidence sets in every outcome. Appendix Figure C11 shows first-stage coefficients for the pre-2018 periods, where respondents are assigned the market's average TikTok use from 2019–2025. These are the first stages behind the pre-2018 estimates in the event studies that follow.

[Figure 8–9 about here]

Figure 9 shows the dynamic effect of TikTok use on traditional news consumption, attention to public affairs, and political knowledge. Panel A shows that a negative effect appears in 2020 and remains significant at the 5% level from 2020 to 2023. Panel B shows that the negative effect on self reported attention to government and public affairs is detectable at the 5% level consistently from 2018 to 2024. Panel C shows a negative effect on factual political knowledge, detectable at the 5% level in 2020–2021 and 2024, and at the 10% level in 2022–2023.

[Figure 10 about here]

Figure 10 reports the dynamic effect of TikTok use on turnout and partisan polarization in presidential approval. Panel A shows that the effect on turnout is negative throughout the post-2018 period, significant at the 5% level in 2020 and 2022 and at the 10% level from 2018 to 2024. Panel B shows that the effect on partisan polarization in presidential approval grows over time, significant at the 5% level from 2022 to 2024.

Importantly, media markets where the instrument predicts higher TikTok use were not already diverging from other markets before the platform arrived. Across all five outcomes, the pre-period coefficients are close to zero and do not exhibit a systematic trend. The pre-period coefficients are not jointly significant for any outcome (p-values range from 0.19 to 0.75), and there are no significant linear trends across pre-periods (p-values range from 0.33 to 0.68), supporting a causal interpretation of the post-2018 estimates.

Appendix Figures C12 and C13 report the reduced-form coefficients on all three indices by survey period. The placebo indices attain significance in a few of the pooled specifications, but they do not reproduce the pattern of the instrument. In some cases the pooled significance comes from differences that are already present before 2018, as for the up-and-coming index on traditional news consumption. Where the placebo coefficients differ from zero after 2018, they generally carry the opposite sign to the Musical.ly coefficient. Unobserved characteristics correlated with interest in comparable artists are therefore unlikely to drive the 2SLS estimates.

4.5 Can news content on TikTok compensate for decline in news consumption?

Users may encounter news on TikTok itself, offsetting the decline in consumption measured off the platform. Content within applications is not observed in the Comscore data, so this cannot be tested directly. Two pieces of evidence suggest that such offsetting is limited.

First, the decline in political knowledge is the clearest evidence that TikTok’s content does not fully compensate for the information it displaces. Regardless of the news encountered on the platform, the survey estimates suggest that respondents in higher-TikTok-use media markets are less politically informed.

Second, direct content analysis points the same way. Aridor, Dekel, Jiménez Durán, Levy, & Song (2025) observe on-platform content for a panel of smartphone users over the 65 days (about 9.3 weeks) from September 1 to November 4, 2024, around the presidential election. They find that about 0.6% of time on TikTok is spent on election-related content, which is 22 seconds per hour. Assuming this share is constant over the period and approximates the share of news content on the platform, an additional weekly hour of TikTok use includes about 22 seconds of election news while displacing 9 minutes

of news consumption off the platform. Even under the extreme assumption that no election-related content appeared on TikTok in the first 8.3 weeks of their sample period and all of it appeared in the final week before the election, news on TikTok offsets only a fraction of the news it displaces. If weekly TikTok use is constant over the 65-day period, concentrating all election-related content in the final week implies that it would account for 5.6% of TikTok time in the final week before the election. An hour of TikTok use would include 3.3 minutes of election-related content, still well short of the 9-minute decline. Thus, news encountered on TikTok is unlikely to compensate for the decline in consumption off the platform.

5 Conclusion

Using a plausibly exogenous shock to early TikTok adoption, this paper estimates how TikTok use affects news consumption, political knowledge, attitudes and behavior. TikTok use increases total time on mobile devices by 5%. During weeks of major news events, TikTok leads to a relative decline in mobile news consumption, while it has no detectable effect on news consumption averaged over the weeks each individual is observed. An additional hour of weekly TikTok use decreases news consumption during news event weeks by about 10%, roughly nine minutes against a mean of 91.

TikTok use reduces both the probability of consuming any news during a news event week and the amount consumed by individuals who do consume news. It also shortens the sessions of news consumption, and the effect is concentrated in the longest sessions of the week, the ones that require sustained attention. This is consistent with a reduced capacity to sustain attention and engage deeply with news stories.

Survey evidence shows that the effect is not limited to mobile news, as respondents in media markets where the instrument predicts higher TikTok use are less likely to report watching television news or reading newspapers after TikTok’s proliferation. Furthermore, TikTok use has downstream political consequences. Higher media-market-level use reduces how closely respondents report following government and public affairs, lowers their political knowledge and turnout, and polarizes approval of the sitting president along party lines.

The decline in knowledge is itself the clearest evidence that information encountered on TikTok does not compensate for what is displaced, and direct content analysis points the same way. Because an informed public is a precondition for democratic accountability, the social consequences of TikTok’s proliferation extend beyond the time the platform absorbs.

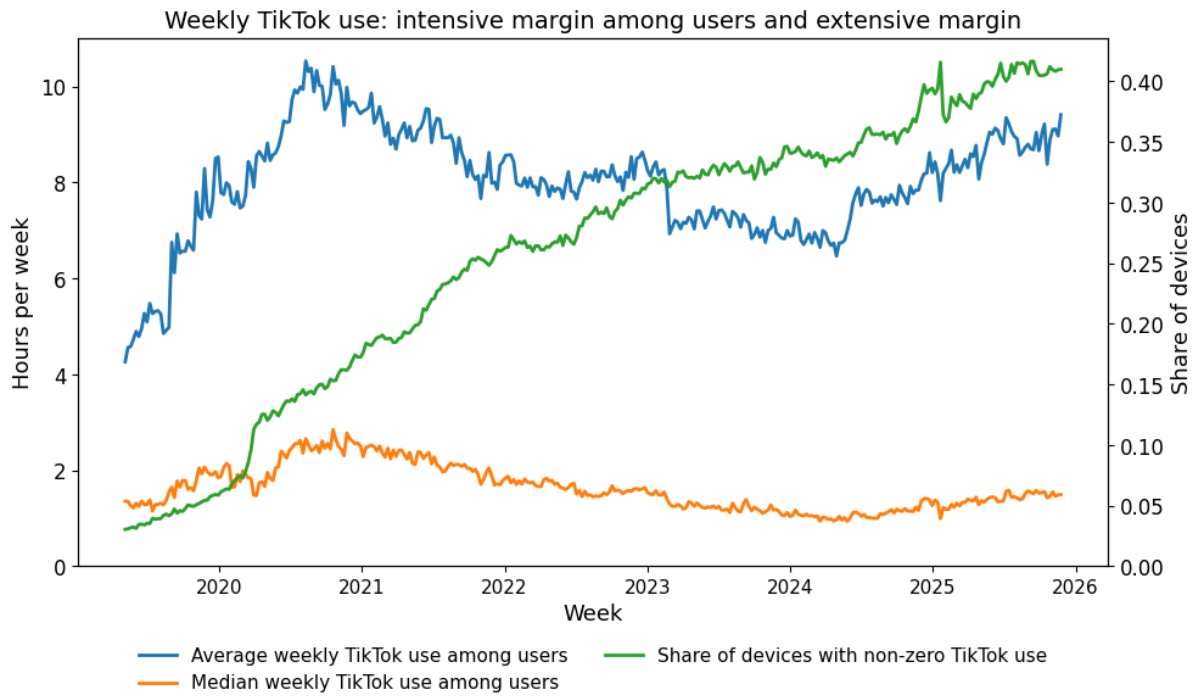


Figure 1: TikTok use from 2019 to 2025

Note: This figure plots average TikTok use among Comscore panelists from May 2019 to November 2025, per week. The green line shows the share of observed devices with non-zero TikTok use. The orange line shows the median hours spent on TikTok, among devices with non-zero TikTok use. The blue line shows the average hours spent on TikTok among devices with non-zero TikTok use.

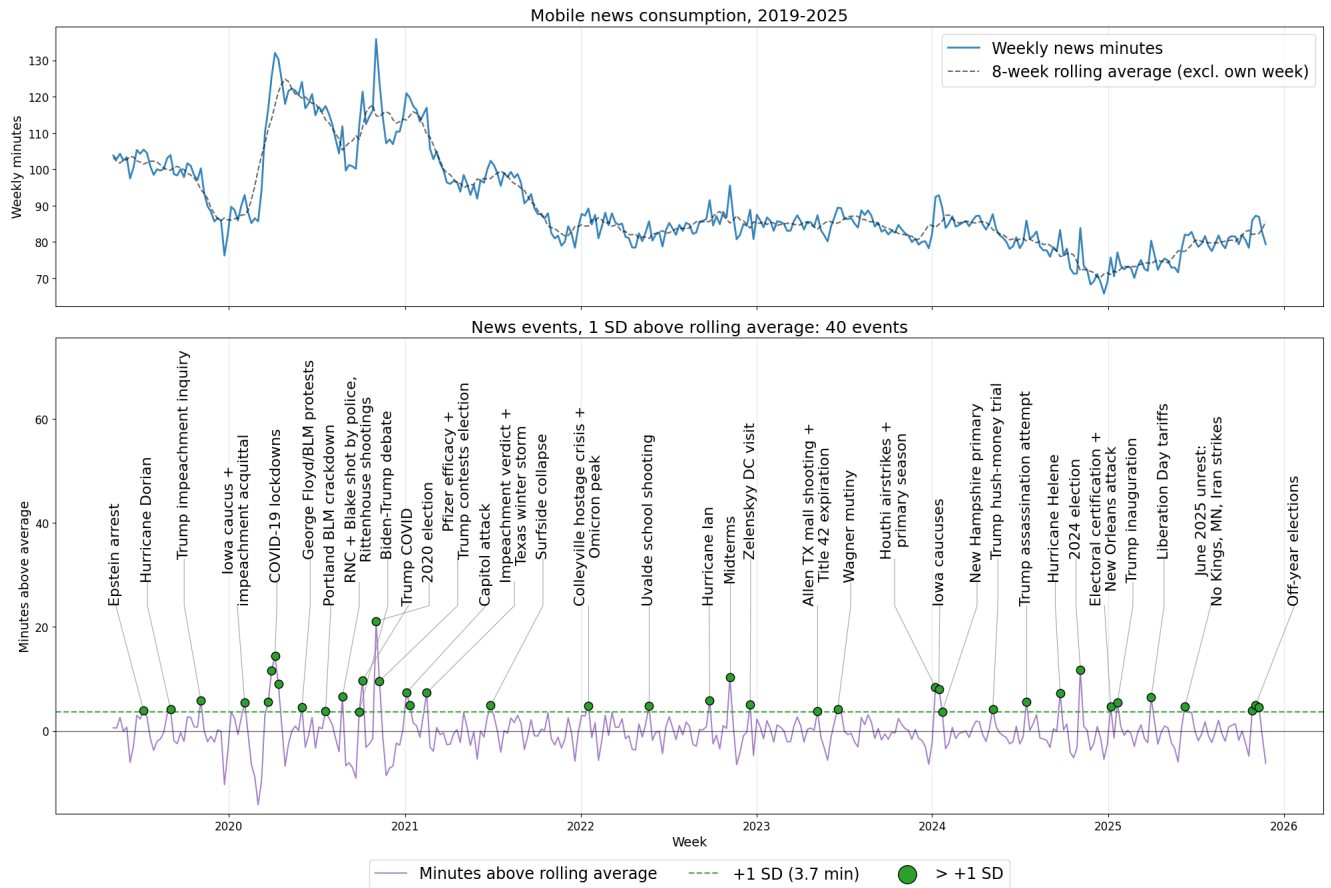


Figure 2: Mobile news consumption and news event weeks

Note: This figure plots weekly minutes of news consumption among Comscore panelists from May 2019 to November 2025, and news event weeks. In the top panel, the blue line shows average weekly minutes of news consumption and the dashed line shows an 8-week rolling average that excludes the current week. In the bottom panel, the purple line shows deviations from the rolling average, the green dashed line represents a one standard deviation excess of the rolling average. The green points denote the 40 news event weeks, defined as weeks in which national news consumption exceeds the rolling average by one standard deviation.

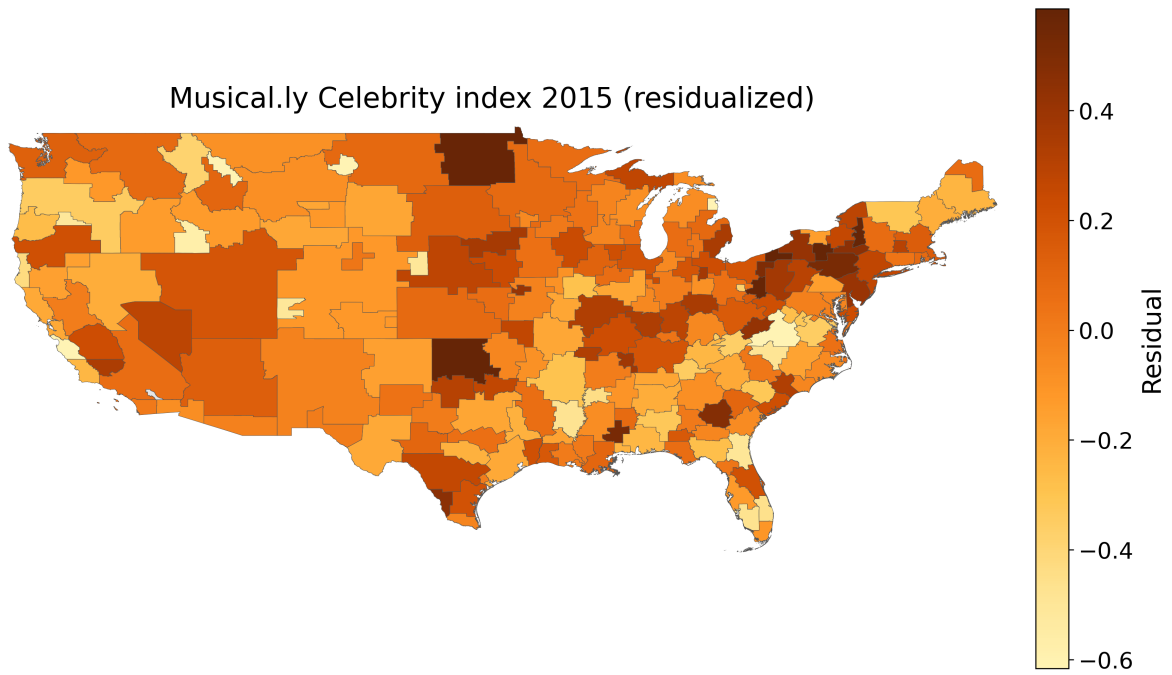


Figure 3: Geographic variation in the residualized Musical.ly Celebrity index

Note: This figure maps the Musical.ly Celebrity index across U.S. media markets (DMAs), constructed from 2015 Google Trends search interest in the Musical.ly artists and residualized on the media-market shares of women, Hispanic residents, residents in poverty, and high-school graduates—the four characteristics included as controls in every specification. Darker shading indicates higher relative interest after residualization. The raw (unresidualized) index is mapped in [Figure B1](#).

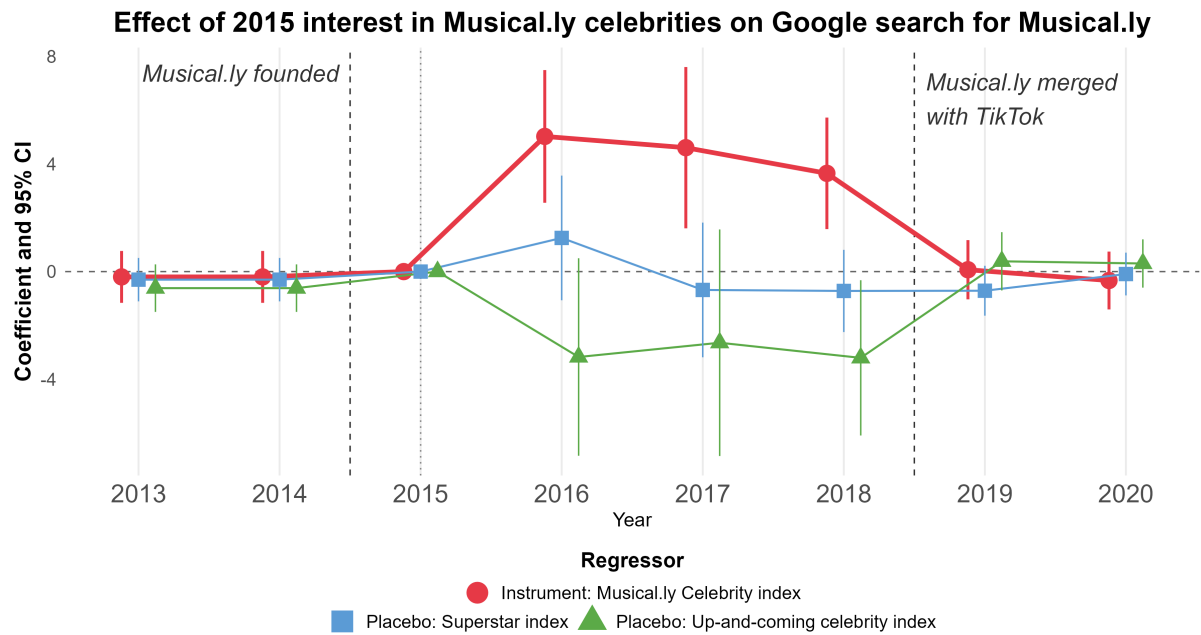


Figure 4: The Musical.ly Celebrity index predicts Google search interest in Musical.ly

Note: This figure plots point estimates and 95% confidence intervals from a regression of DMA-by-year Google search interest in Musical.ly on the three 2015 celebrity indices, each interacted with year dummies, with DMA and year fixed effects (205 DMAs). 2015 is the omitted reference year. Each year's DMA-level Google Trends series is normalized within year, so to make coefficients comparable across years each DMA's interest is rescaled by national annual search for Musical.ly (Figure C6). Dots are point estimates and bars are 95% confidence intervals, based on standard errors clustered by DMA. The Musical.ly Celebrity index is in red, the superstar and up-and-coming placebo indices in blue and green.

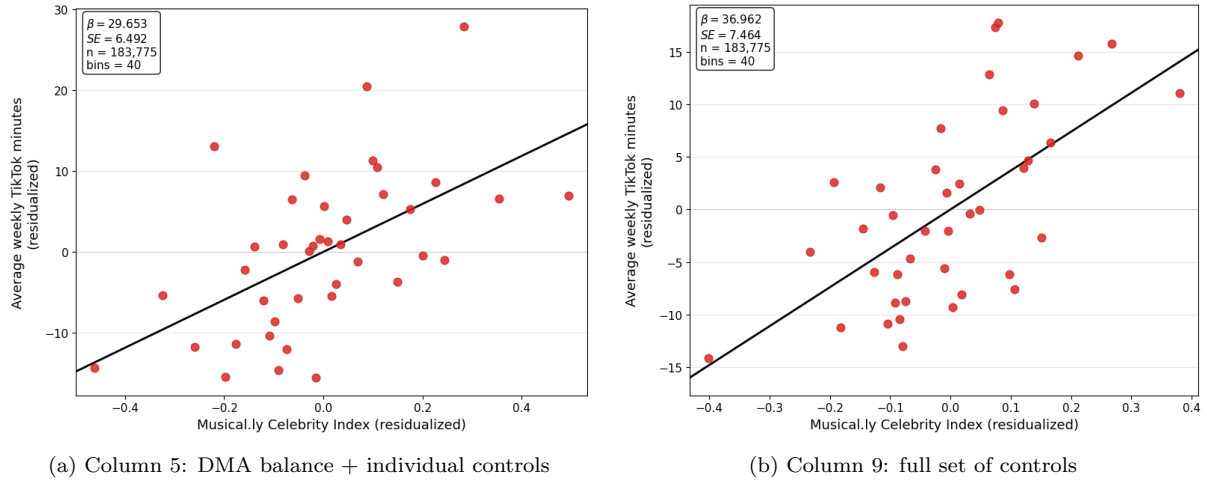


Figure 5: First stage among Comscore panelists: Binned residual scatter plots of average weekly TikTok minutes and the Musical.ly Celebrity index

Note: This figure presents binned scatter plots of the first stage, plotting residualized average weekly TikTok minutes against the residualized Musical.ly Celebrity index across device-years. Panel (a) residualizes both variables on the DMA balance controls and the individual-level controls (column 5 of Table 1); panel (b) on the full set of controls, which adds the DMA population, demographic and socio-economic controls (column 9). *Individual controls* are the device user's gender, age in 5-year bins, device type (iPhone, iPad, or Android), indicators for Black and for Hispanic users, household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, an indicator for children in the household, and the race and ethnicity indicators interacted with the age bins. *DMA controls* are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates; indicators for deciles of DMA total population; the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older; the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races; and the DMA shares of adults with at least a high-school diploma, a bachelor's degree, and a master's degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. The data are binned into 40 equal-interval bins on the x-axis and dots represent average values within each bin. The line is an OLS fit on the unbinned data; its slope reproduces the corresponding first-stage coefficient.

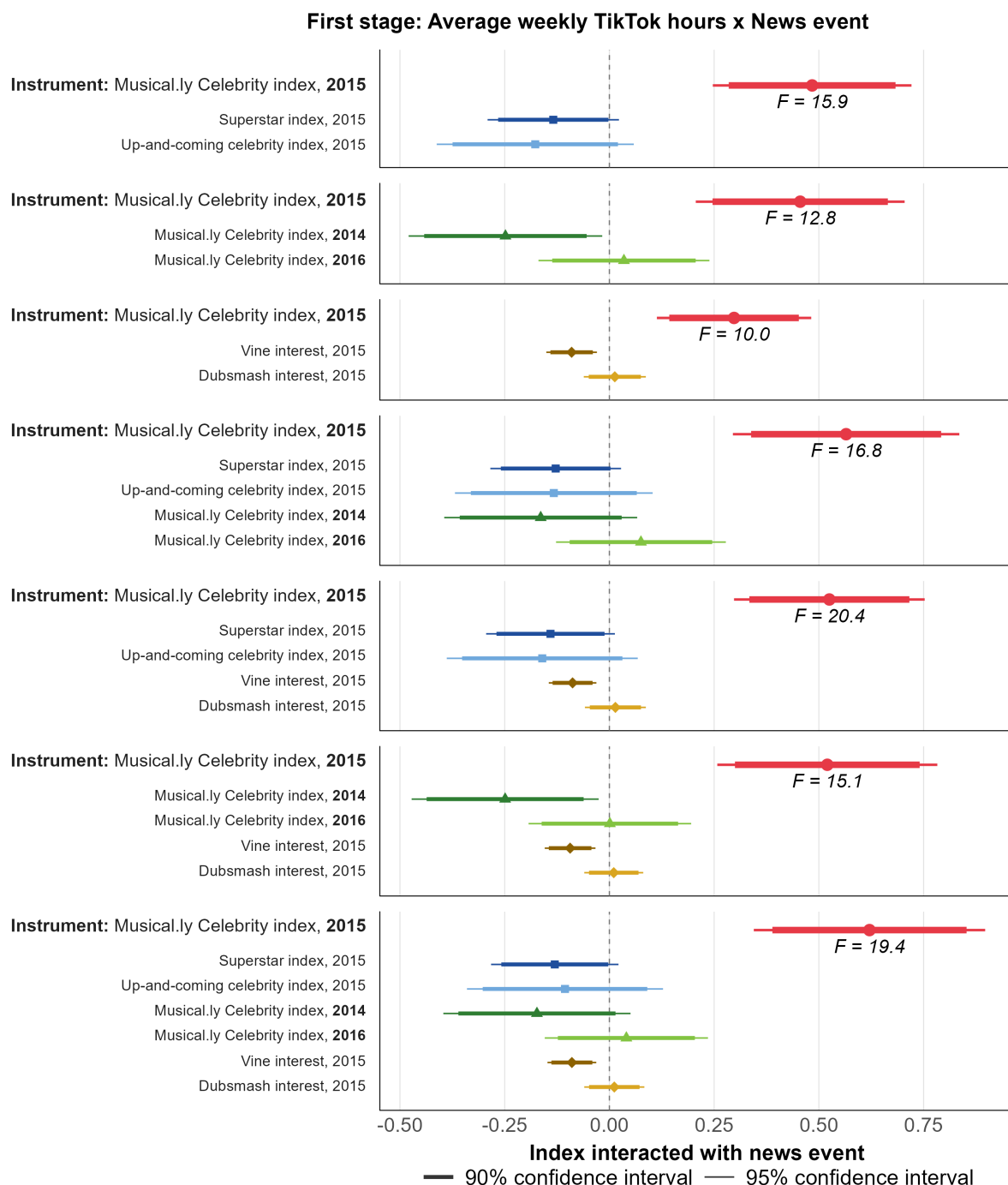


Figure 6: First stage among Comscore panelists robustness: Controlling for alternative placebo instruments

Note: This figure plots point estimates and confidence intervals from a device-week level regression. The dependent variable is average weekly TikTok hours in the four weeks before an event, interacted with the event indicator. The placebo indices enter in three pairs. The first pair consists of the 2015 superstar and up-and-coming placebo celebrity indices. The second holds the Musical.ly artists fixed but constructs their index from 2014 and from 2016 Google Trends data rather than 2015, capturing interest in the same artists in the years before and after they promoted their music on the platform; the 2014 index excludes Silentó, whose 2014 search interest is non-zero in only one media market. The third is 2015 Google search interest in Vine and in Dubsmash, the contemporaneous short-form video platforms (Hu, 2018; McCammond, 2017). Each column of the figure is a separate regression. The matrix below the plot marks with a filled circle which indices enter that column, and the cluster-robust first-stage F-statistic for that column is reported beneath it. All seven coefficients are index coefficients on the same dependent variable and are plotted on a common scale, with the Musical.ly Celebrity index in red. Thick bars are 90% and thin bars 95% confidence intervals. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user's gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor's and a master's degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA.

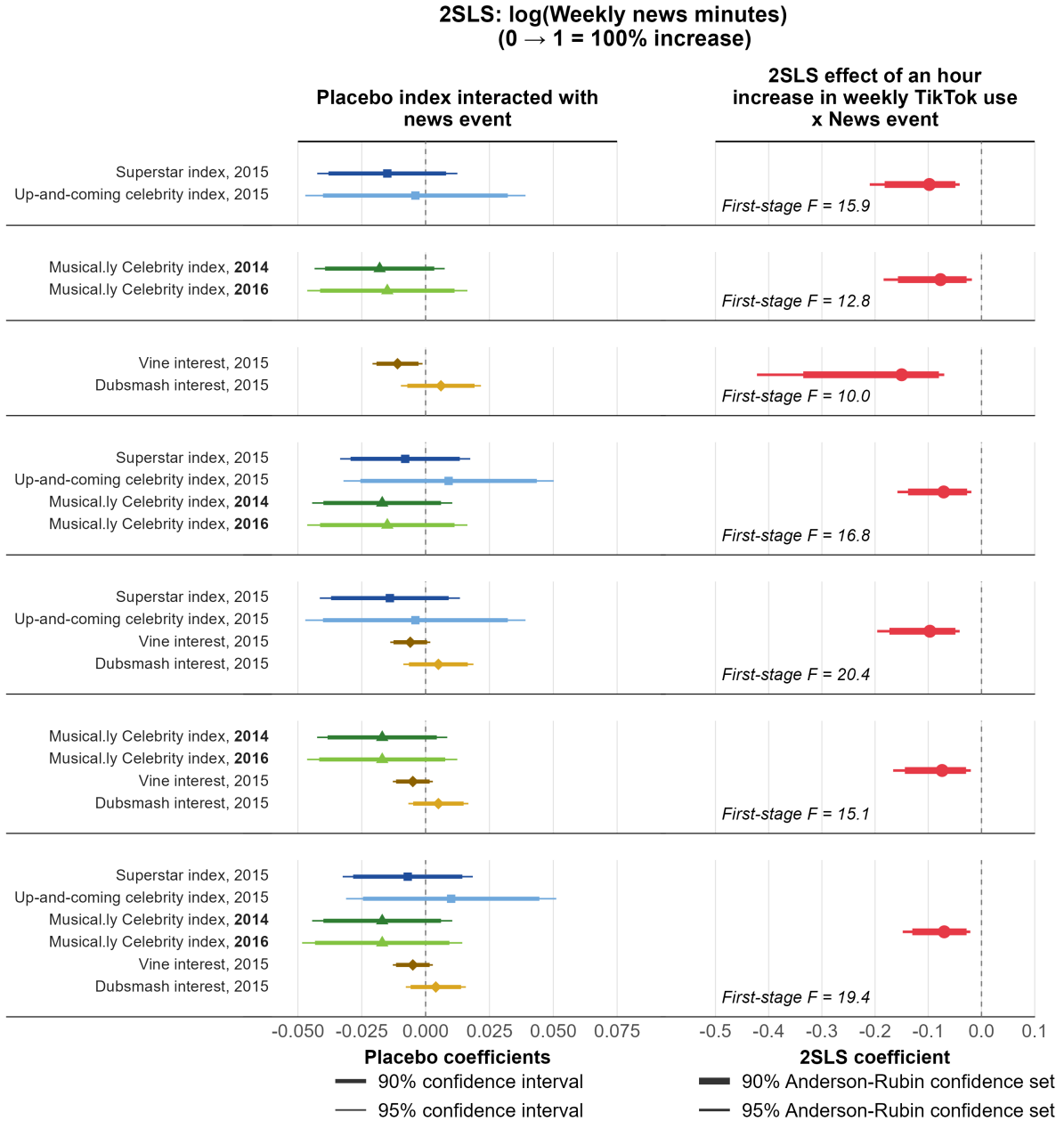


Figure 7: Second stage among Comscore panelists robustness: Controlling for alternative placebo instruments

Note: This figure presents device-week level 2SLS regressions. The dependent variable is the logarithm of weekly news minutes, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued as a doubling. Average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, is instrumented with the 2015 Musical.ly Celebrity index interacted with the same indicator. The placebo indices enter in three pairs. The first pair consists of the 2015 superstar and up-and-coming placebo celebrity indices. The second holds the Musical.ly artists fixed but constructs their index from 2014 and from 2016 Google Trends data rather than 2015, capturing interest in the same artists in the years before and after they promoted their music on the platform; the 2014 index excludes Silentó, whose 2014 search interest is non-zero in only one media market. The third is 2015 Google search interest in Vine and in Dubsmash, the contemporaneous short-form video platforms (Hu, 2018; McCammond, 2017). Each column is a separate regression. The matrix below the plot marks with a filled circle which indices enter that column, and the cluster-robust first-stage F-statistic for that column is reported beneath it. The upper panel plots the coefficient on TikTok use, with thick bars giving 90% and thin bars 95% Anderson–Rubin confidence sets, which are robust to weak identification. The lower panel plots the second-stage coefficients on the placebo indices, on their own scale, with thick bars giving 90% and thin bars 95% confidence intervals based on standard errors clustered by DMA. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA.

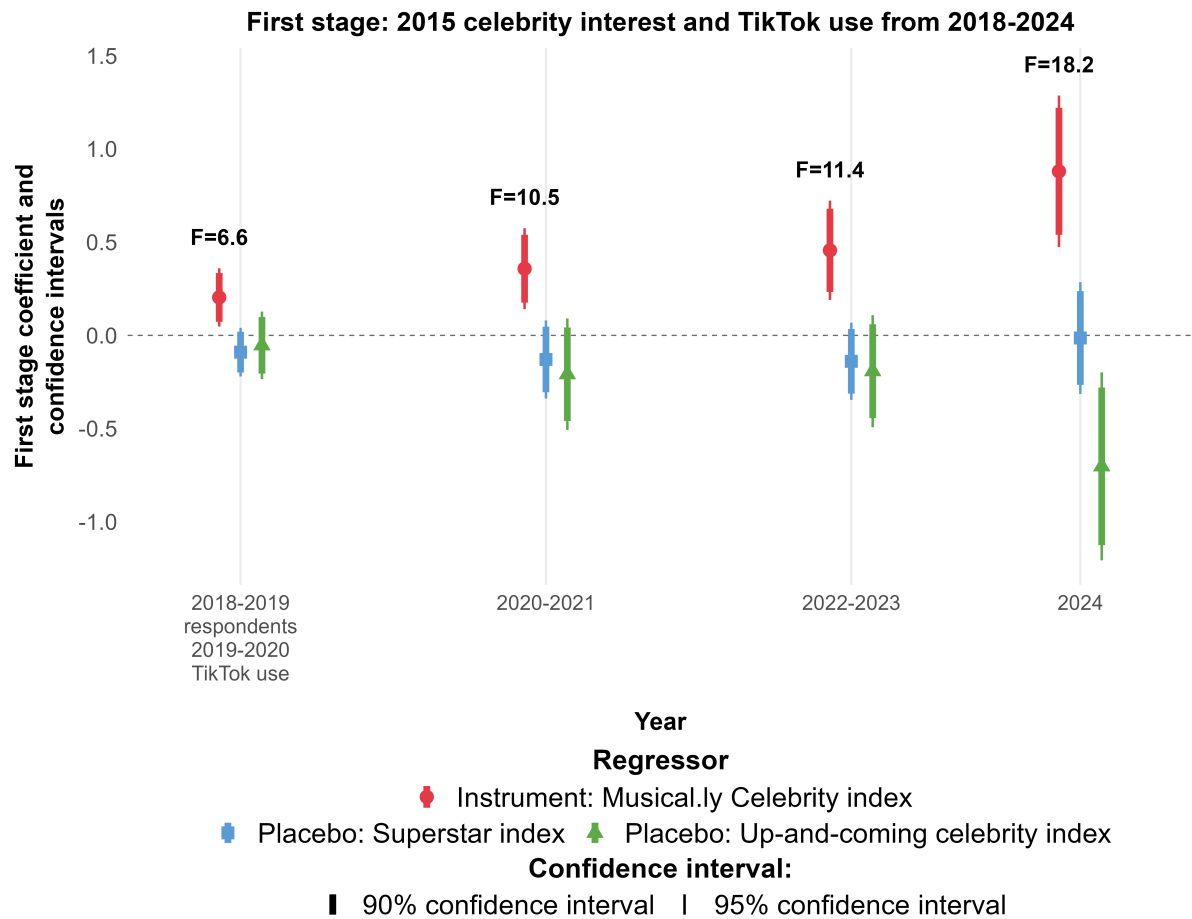


Figure 8: First stage among CES respondents: Musical.ly celebrity interest in 2015 and media-market TikTok use over time

Note: This figure reports survey respondent-level regressions. It plots the coefficients on the three 2015 celebrity indices from a regression of media-market-level average demographically adjusted weekly TikTok hours on each index interacted with survey-period indicators, with 2016–2017 as the omitted reference period. Only the post-2018 periods are shown; Appendix Figure C11 adds the pre-2018 periods, where respondents from 2010–2015 are assigned the market’s average TikTok use from 2019–2025. Cluster-robust first-stage F-statistics are reported above each period. Thick bars are 90% and thin bars 95% confidence intervals. The specification includes DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with the period indicators, and individual controls for age, gender, race and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA.

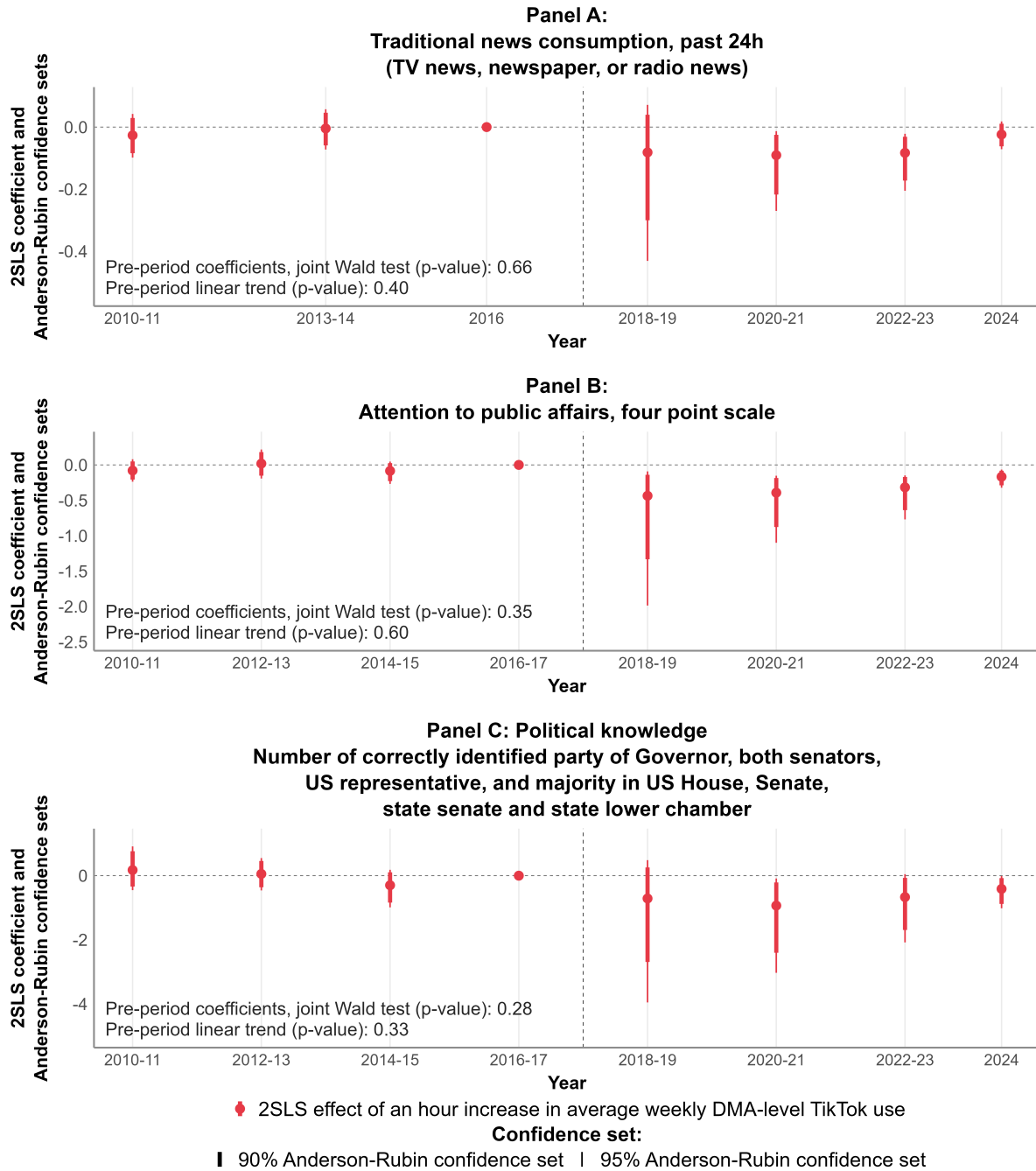


Figure 9: Second stage among CES respondents: The dynamic effect of TikTok use on news consumption and political information

Note: This figure plots survey respondent-level two-stage least squares estimates of the effect of an additional weekly hour of media-market-level TikTok use, estimated separately by survey period, with 2016–2017 as the omitted reference. In Panel A the outcome is an indicator for having watched television news, read a newspaper or listened to radio news in the past 24 hours. In Panel B the outcome is how closely the respondent reports following government and public affairs, on a four-point scale. In Panel C the outcome is the number of correct answers to eight factual questions: the party of the respondent’s governor, of both U.S. senators and of their U.S. representative, and the majority party in the U.S. House, the U.S. Senate, the respondent’s state senate and their state’s lower chamber. Because the traditional media items are not fielded in 2012, 2015 or 2017, Panel A pools survey years into 2010–2011, 2013–2014 and 2016 before the merger and uses the same post-2018 periods as the other panels. Thick bars are 90% and thin bars 95% Anderson–Rubin confidence sets, which are robust to weak instrument bias. Media-market-level average weekly TikTok hours, interacted with survey-period indicators, is instrumented with the 2015 Musical.ly Celebrity index interacted with the same indicators; the superstar and up-and-coming placebo indices enter as exogenous controls. The specification includes DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with the period indicators, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA.

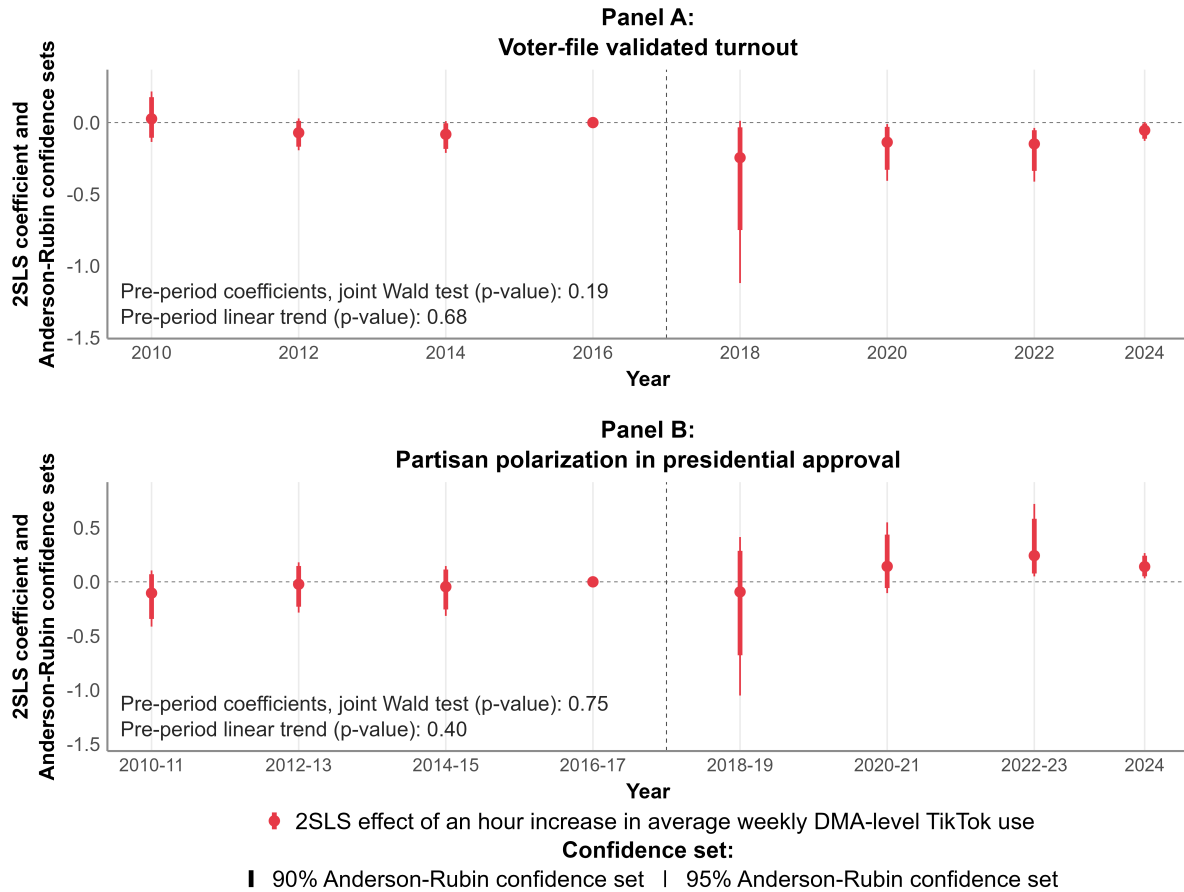


Figure 10: Second stage among CES respondents: The dynamic effect of TikTok use on turnout and partisan polarization in presidential approval

Note: This figure plots survey respondent-level two-stage least squares estimates of the effect of an additional weekly hour of media-market-level TikTok use, estimated separately by survey period, with 2016 as the omitted reference in Panel A and 2016–2017 in Panel B. In Panel A the outcome is voter-file validated turnout, which is available only in federal-election years. In Panel B the outcome is presidential approval signed by partisan alignment, among respondents who describe themselves as Democrats or Republicans. Thick bars are 90% and thin bars 95% Anderson–Rubin confidence sets, which are robust to weak instrument bias. Media-market-level average weekly TikTok hours, interacted with survey-period indicators, is instrumented with the 2015 Musical.ly Celebrity index interacted with the same indicators; the superstar and up-and-coming placebo indices enter as exogenous controls. The specification includes DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with the period indicators, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA.

Table 1: First stage: average weekly TikTok minutes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependent variable	Average weekly minutes on TikTok								
Musical.ly Celebrity index	31.302*** (8.161)	28.013*** (6.661)	28.558*** (6.557)	29.177*** (6.471)	29.653*** (6.493)	30.793*** (6.463)	29.129*** (6.623)	31.902*** (6.715)	36.962*** (7.466)
Superstar index	-2.510 (7.801)	-1.629 (6.102)	-3.364 (5.936)	-5.163 (5.806)	-5.359 (5.746)	-0.923 (5.682)	-1.121 (5.757)	-4.324 (6.230)	-5.794 (6.371)
Up-and-coming celebrity index	-6.071 (11.404)	-9.871 (8.469)	-12.076 (8.408)	-12.055 (8.204)	-11.532 (8.124)	-17.400** (8.292)	-15.249* (8.563)	-14.459* (8.420)	-14.483 (9.038)
Mean dependent variable	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14
Number of devices	183,775	183,775	183,775	183,775	183,775	183,775	183,775	183,775	183,775
Number of DMAs	205	205	205	205	205	205	205	205	205
R ²	0.00	0.09	0.10	0.10	0.10	0.10	0.10	0.10	0.10
Cluster-robust F-statistic	14.71	17.69	18.97	20.33	20.85	22.70	19.34	22.57	24.51
DMA controls for balance	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Gender, 5y age buckets, device type		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Race and ethnicity			Yes	Yes	Yes	Yes	Yes	Yes	Yes
HH income, size, children				Yes	Yes	Yes	Yes	Yes	Yes
Race and ethnicity specific age buckets					Yes	Yes	Yes	Yes	Yes
DMA population deciles						Yes	Yes	Yes	Yes
DMA age composition							Yes	Yes	Yes
DMA race composition								Yes	Yes
DMA socio-economic									Yes

Notes: The unit of observation is a device; each column regresses a device's average weekly TikTok minutes, averaged over the weeks the device is observed, on the three celebrity indices. Standard errors clustered by DMA. The DMA balance controls, included in every column, are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates. The remaining controls are added cumulatively across columns and are defined as follows. *Gender, 5y age buckets, device type* is the device user's gender, age in 5-year bins, and device type (iPhone, iPad, or Android). *Race and ethnicity* is an indicator for Black users and an indicator for Hispanic users. *HH income, size, children* is household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, and an indicator for children in the household. *Race and ethnicity specific age buckets* interacts the race and ethnicity indicators with the age bins. *DMA population deciles* is a set of indicators for deciles of DMA total population. *DMA age composition* is the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older. *DMA race composition* is the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races. *DMA socio-economic* is the DMA shares of adults with at least a high-school diploma, a bachelor's degree, and a master's degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Cross-sectional 2SLS: TikTok use, total mobile use, and mobile news consumption

	(1)	(2)	(3)	(4)
Dependent variable	log(Weekly total mobile minutes)	Any news consumption	log(Weekly news minutes)	
Sample	All devices	All devices	News > 0	All devices (0 → 1 = 100% increase)
Avg. weekly TikTok hours	0.055* (0.031)	0.006 (0.008)	0.088 (0.077)	0.084 (0.077)
90% AR CI	[0.001, 0.111]	[-0.007, 0.019]	[-0.041, 0.230]	[-0.046, 0.222]
95% AR CI	[-0.010, 0.123]	[-0.010, 0.022]	[-0.068, 0.265]	[-0.074, 0.254]
Superstar index	-0.021 (0.016)	-0.013*** (0.004)	-0.082** (0.033)	-0.110*** (0.036)
Up-and-coming celebrity index	-0.041* (0.023)	-0.000 (0.005)	0.049 (0.042)	0.047 (0.047)
Mean dependent variable	7.687	0.943	2.582	2.409
Number of devices	183,772	183,775	173,273	183,775
Number of DMAs	205	205	205	205
First-stage F-statistic	24.59	24.51	22.40	24.51
Individual controls	Yes	Yes	Yes	Yes
DMA controls	Yes	Yes	Yes	Yes

Notes: The unit of observation is a device, and each column reports the two-stage least squares estimate of equation (2). Average weekly TikTok hours is instrumented with the Musically Celebrity index; the superstar and up-and-coming placebo indices enter as exogenous controls and their coefficients are reported. The dependent variable in column (1) is the logarithm of average weekly total mobile minutes. In column (2) it is an indicator equal to one if the device consumes any news. In column (3) it is the logarithm of average weekly news minutes among devices with positive news consumption. In column (4) it is average weekly news minutes over all devices, transformed as $\log(x)$ for $x \geq 1$ and -1 for $x < 1$, so that devices with little or no news consumption are retained and the move from below one minute to one minute is valued as a doubling. All quantities are averaged over the weeks the device is observed. Anderson–Rubin confidence sets are robust to weak identification. All columns include the full set of controls. *Individual controls* are the device user’s gender, age in 5-year bins, device type (iPhone, iPad, or Android), indicators for Black and for Hispanic users, household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, an indicator for children in the household, and the race and ethnicity indicators interacted with the age bins. *DMA controls* are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates; indicators for deciles of DMA total population; the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older; the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races; and the DMA shares of adults with at least a high-school diploma, a bachelor’s degree, and a master’s degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. Table C4 reports the same estimates adding control blocks sequentially. Standard errors clustered by DMA.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: TikTok use and mobile news consumption during news event weeks

	(1)	(2)	(3)
Margin	Extensive	Intensive	Combined
Dependent variable	Any news consumption	log(Weekly news minutes) news > 0	log(Weekly news minutes) (0 → 1 = 100% increase)
Panel A: 2SLS			
Avg. weekly TikTok hours × News event	-0.017** (0.007)	-0.063** (0.029)	-0.098*** (0.036)
90% AR CI	[-0.033, -0.008]	[-0.129, -0.020]	[-0.182, -0.049]
95% AR CI	[-0.039, -0.006]	[-0.151, -0.011]	[-0.210, -0.041]
Superstar index × News event	0.0002 (0.0023)	-0.023** (0.011)	-0.015 (0.014)
Up-and-coming celebrity index × News event	-0.003 (0.004)	0.009 (0.017)	-0.004 (0.022)
Cluster-robust F-statistic	15.95	12.64	15.95
Panel B: Reduced form			
Musical.ly Celebrity index × News event	-0.008*** (0.002)	-0.030** (0.013)	-0.047*** (0.014)
Superstar index × News event	0.002 (0.002)	-0.014 (0.012)	-0.002 (0.013)
Up-and-coming celebrity index × News event	-0.0004 (0.0030)	0.020 (0.016)	0.013 (0.019)
R ²	0.480	0.575	0.649
Within-R ²	0.0000	0.0001	0.0001
<i>p-values for the equality of coefficients with the Musical.ly Celebrity index</i>			
Superstar index	0.002***	0.448	0.039**
Up-and-coming celebrity index	0.075*	0.019**	0.020**
Both indices jointly	0.008***	0.065*	0.022**
Panel C: OLS			
Avg. weekly TikTok hours × News event	0.001*** (0.000)	0.002*** (0.000)	0.004*** (0.000)
R ²	0.480	0.575	0.649
Within-R ²	0.0001	0.0001	0.0002
Mean dependent variable	0.68	3.38	1.99
Observations	5,660,705	3,852,220	5,660,705
Number of devices	176,321	160,833	176,321
Number of weeks	235	235	235
Number of DMAs	205	205	205
Week & device FEs	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes

Notes: The unit of observation is a device-week. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. In column (1) the dependent variable is an indicator for any news consumption in the week; in column (2) it is the logarithm of weekly news minutes among device-weeks with positive news consumption; and in column (3) it is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued as a doubling. Panel A reports two-stage least squares estimates of equation (4), instrumenting average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musical.ly Celebrity index interacted with the same indicator; the corresponding first stage is reported in Appendix Table C6. Anderson–Rubin confidence sets are robust to weak identification. Panel B reports the reduced form, in which the Musical.ly Celebrity index interacted with the event indicator enters the outcome equation directly in place of TikTok use. Panel C reports ordinary least squares estimates of equation (4). All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: 2SLS estimation of the effect of TikTok use on the length of news sessions during news event weeks

	(1)	(2)	(3)	(4)	(5)
News session within the week	Longest	Mean	Median	p75	p95
Dependent variable	News session length (minutes)				
Panel A: All news event weeks					
Avg. weekly TikTok hours $\times$ News event	-0.623** (0.304)	-0.019 (0.079)	0.058 (0.082)	-0.095 (0.115)	-0.134 (0.213)
90% AR CI	[-1.273, -0.133]	[-0.161, 0.132]	[-0.076, 0.230]	[-0.335, 0.095]	[-0.517, 0.273]
95% AR CI	[-1.477, -0.026]	[-0.197, 0.174]	[-0.106, 0.284]	[-0.408, 0.137]	[-0.614, 0.385]
Superstar index $\times$ News event	-0.252** (0.126)	-0.029 (0.027)	-0.010 (0.026)	-0.002 (0.040)	-0.151** (0.077)
Up-and-coming celebrity index $\times$ News event	0.061 (0.168)	-0.020 (0.035)	-0.017 (0.037)	-0.061 (0.052)	0.060 (0.109)
Mean dependent variable	20.78	5.40	3.60	6.68	14.26
Cluster-robust F-statistic	12.64	12.64	12.64	12.64	12.64
Panel B: Excluding the first wave of COVID-19 lockdowns					
Avg. weekly TikTok hours $\times$ News event	-0.998*** (0.337)	-0.097 (0.079)	0.006 (0.074)	-0.207* (0.126)	-0.382* (0.221)
90% AR CI	[-1.846, -0.547]	[-0.263, 0.030]	[-0.128, 0.144]	[-0.498, -0.025]	[-0.855, -0.036]
95% AR CI	[-2.151, -0.473]	[-0.313, 0.057]	[-0.162, 0.181]	[-0.595, 0.009]	[-1.002, 0.036]
Superstar index $\times$ News event	-0.382** (0.149)	-0.053* (0.029)	-0.018 (0.027)	-0.025 (0.043)	-0.253*** (0.086)
Up-and-coming celebrity index $\times$ News event	0.133 (0.206)	0.007 (0.037)	0.001 (0.036)	-0.029 (0.059)	0.146 (0.112)
Mean dependent variable	20.62	5.42	3.63	6.69	14.22
Cluster-robust F-statistic	13.81	13.81	13.81	13.81	13.81
Week & device FEs	Yes	Yes	Yes	Yes	Yes
Controls $\times$ News event	Yes	Yes	Yes	Yes	Yes

Notes: The unit of observation is a device-week, and the sample is device-weeks with positive news consumption, for which session length is defined. Panel A covers 3,852,220 device-weeks across 160,833 devices and 235 weeks; Panel B covers 3,453,415 device-weeks across 158,277 devices and 220 weeks. Both panels cover 205 media markets. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. A news session is an uninterrupted visit to a news site or application; the dependent variable in each column is the length of the stated session within the device-week, in minutes. Panel A uses all 40 news event weeks. Panel B drops the news event weeks falling inside the period of state and territorial stay-at-home orders, which ran from March through May 2020 (Moreland, 2020), together with 1 June 2020, whose ± 4 -week control window reaches back into that period, leaving 35 news event weeks. Each column reports two-stage least squares estimates instrumenting average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musical.ly Celebrity index interacted with the same indicator. Anderson–Rubin confidence sets are robust to weak identification. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: TikTok use, traditional news consumption, and political outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
Specification	First stage	2SLS				
			Political information		Political behavior and attitudes	
Dependent variable	Avg. weekly TikTok hours $\times$ Post	Traditional news, past 24h	Attention to public affairs	Political knowledge	Validated turnout	Partisan polarization in presidential approval
Avg. weekly TikTok hours $\times$ Post		-0.055*** (0.019)	-0.224*** (0.058)	-0.633** (0.260)	-0.053* (0.030)	0.223*** (0.076)
95% AR CI		[-0.105, -0.023]	[-0.386, -0.132]	[-1.310, -0.184]	[-0.123, 0.003]	[0.100, 0.432]
Musical.ly Celebrity index $\times$ Post	0.499*** (0.095)					
Superstar index $\times$ Post	-0.091 (0.084)	0.036*** (0.007)	0.012 (0.026)	-0.115 (0.106)	0.012 (0.013)	-0.088*** (0.029)
Up-and-coming celebrity index $\times$ Post	-0.320*** (0.122)	-0.040*** (0.011)	-0.050 (0.042)	0.209 (0.164)	-0.046** (0.023)	0.132*** (0.041)
Cluster-robust F-statistic	27.46	28.44	28.14	27.46	30.13	27.73
Mean dependent variable	0.777	0.805	3.184	4.973	0.538	3.497
Number of respondents	574,436	512,216	580,025	574,436	463,193	368,331
Number of survey years	15	12	15	15	8	15
Number of DMAs	204	205	205	204	205	205
DMA & survey-year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
DMA balance controls $\times$ Post	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports survey respondent-level regressions. Survey years 2010–2024 are pooled and *Post* equals one after 2018, the year of the Musical.ly–TikTok merger. Column (1) reports the first stage, in which media-market-level average demographically adjusted weekly TikTok hours interacted with *Post* is regressed on the three 2015 celebrity indices interacted with *Post*; it is estimated on the political-knowledge sample. Columns (2)–(6) instrument that same term with the Musical.ly Celebrity index interacted with *Post*, with the placebo indices entering as exogenous controls. The cluster-robust F-statistic is computed on each column’s own sample. Anderson–Rubin confidence sets are robust to weak identification. The outcomes are: in column (2), an indicator for having watched television news, read a newspaper or listened to radio news in the past 24 hours; in column (3), how closely the respondent reports following government and public affairs, coded 1–4 with higher values indicating closer attention; in column (4), the number of correct answers to eight factual questions: the party of the respondent’s governor, of both U.S. senators and of their U.S. representative, and the majority party in the U.S. House, the U.S. Senate, the respondent’s state senate and their state’s lower chamber; in column (5), voter-file validated turnout in the general election, available only in federal-election years; and in column (6), the 1–4 presidential job-approval item recoded so that higher values indicate an evaluation more in line with the respondent’s own party: 4 where a copartisan strongly approves or an out-partisan strongly disapproves, down to 1 where a copartisan strongly disapproves or an out-partisan strongly approves. Party is taken from the seven-point party identification item, and column (6) is restricted to respondents who describe themselves as Democrats or Republicans; those who name no party but say they feel closer to one are excluded, as are pure independents. Appendix Table C21 reports the alternative in which the former are counted as partisans. All columns include DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with *Post*, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Appendix

A Data

Table A1: Control variables

Variables	Description	Source
Individual-level controls	Age in 5-year bins; gender; device type (iPhone, iPad, Android); household-income buckets; household-size indicators for 1, 2, 3, 4, and 5+; an indicator for children in the household; and indicators for Black and for Hispanic users	Comscore (Comscore, Inc., 2019–2025)
Demographic controls	Total population; shares of the population in age buckets 0–19, 20–39, 40–59 and 60 or older; share women; shares white, Black, Asian, Native American, Pacific Islander and of two or more races; share Hispanic	Census Bureau county population estimates, Vintage 2015, July 1, 2015 (US Census Bureau, 2016b)
Socio-economic controls	Shares of adults with at least a high-school diploma, a bachelor’s degree, and a master’s degree; log median household income; unemployment rate; poverty rate among those aged 18–64; Gini index	American Community Survey, 2011–2015 5-year estimates (US Census Bureau, 2016a)

Notes: County-level Census Bureau and ACS variables are aggregated to media markets, summing counts across counties and weighting median household income and the Gini index by county population. The presidential vote shares for 2000–2016 in Table B6 are aggregated from the county returns of [MIT Election Data and Science Lab \(2018\)](#) in the same way.

A.1 Comscore

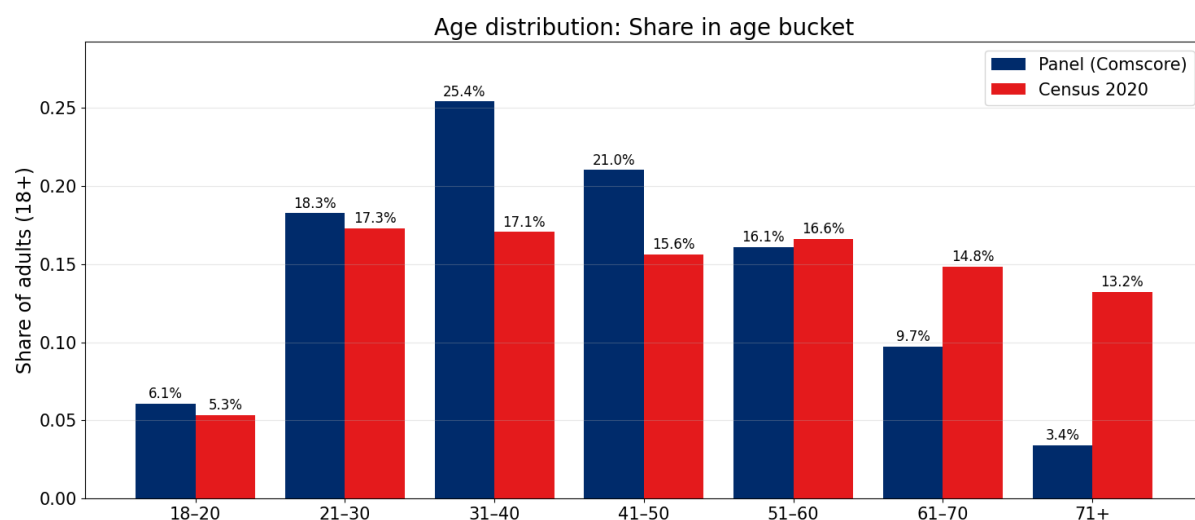


Figure A1: Comscore panel representativeness: Age distribution in Panel vs. Census

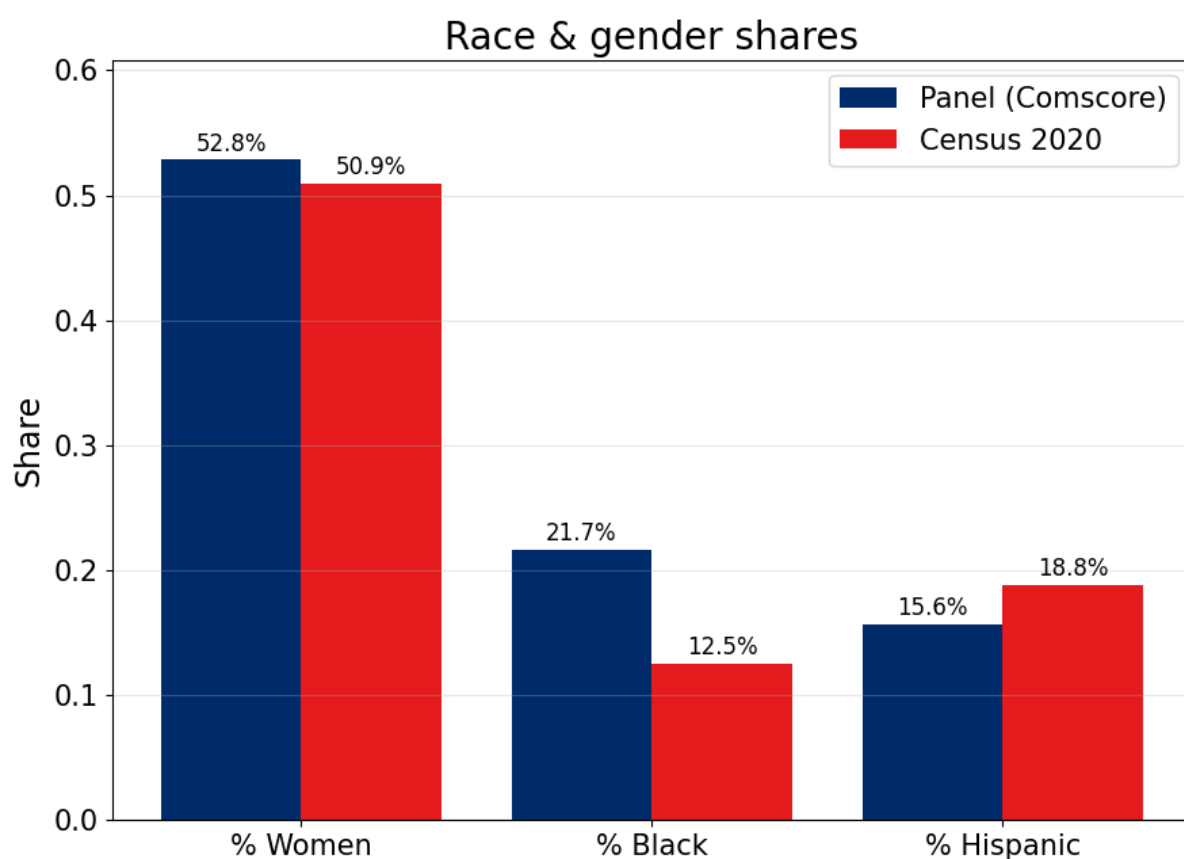


Figure A2: Comscore panel representativeness: Race and sex distribution in Panel vs. Census

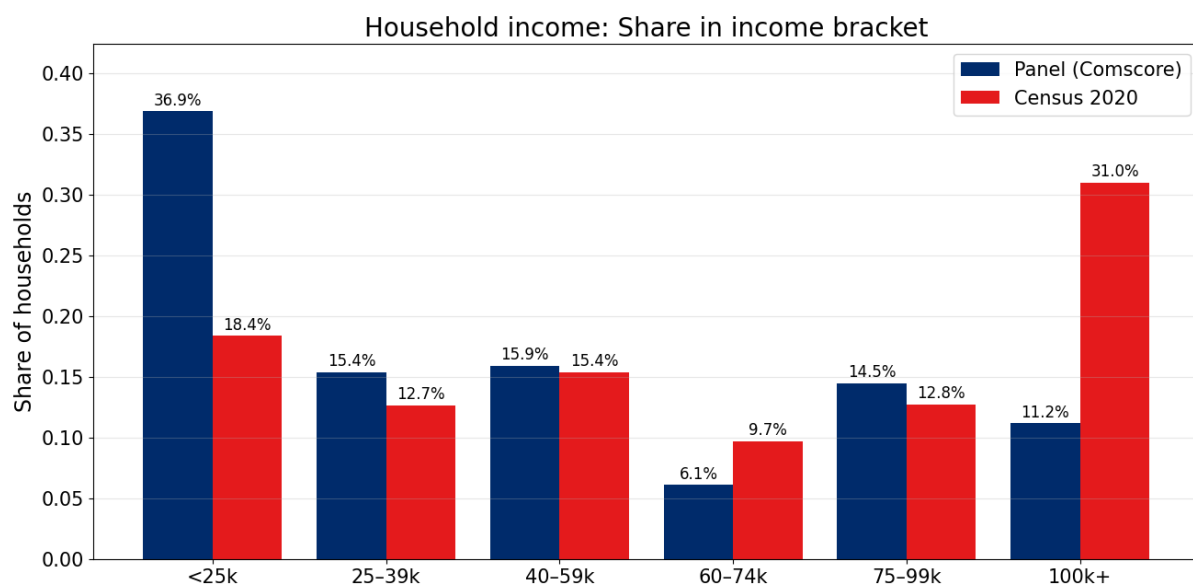


Figure A3: Comscore panel representativeness: Income distribution in Panel vs. Census

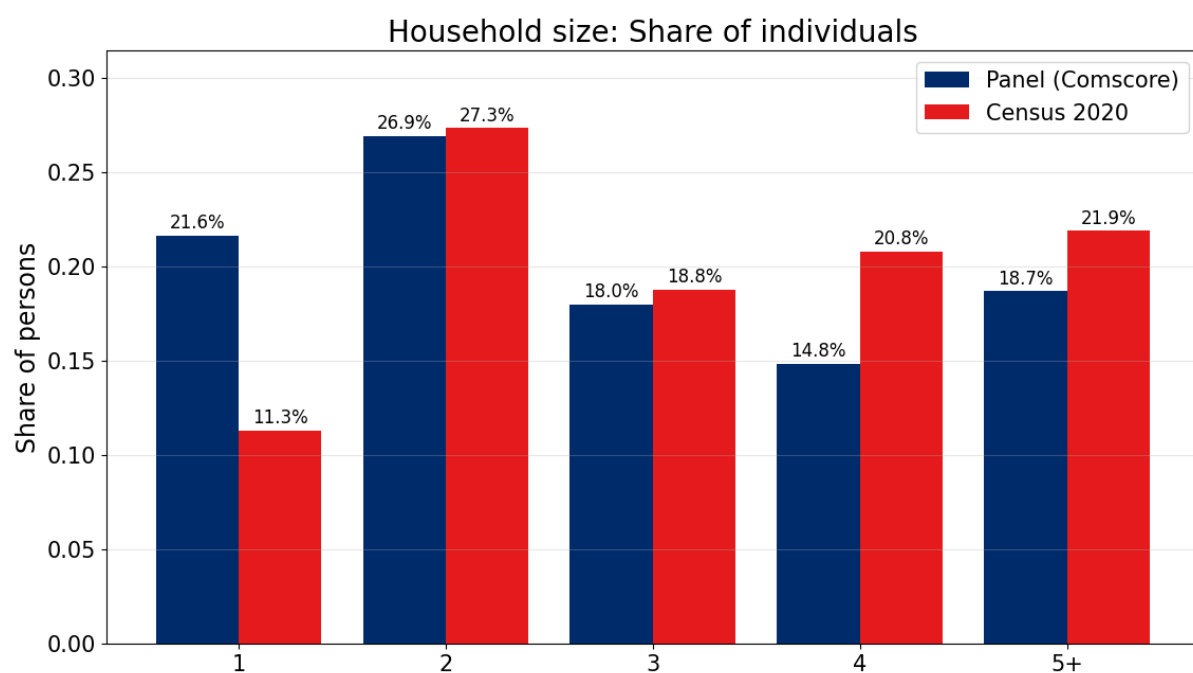


Figure A4: Comscore panel representativeness: Household size distribution in Panel vs. Census

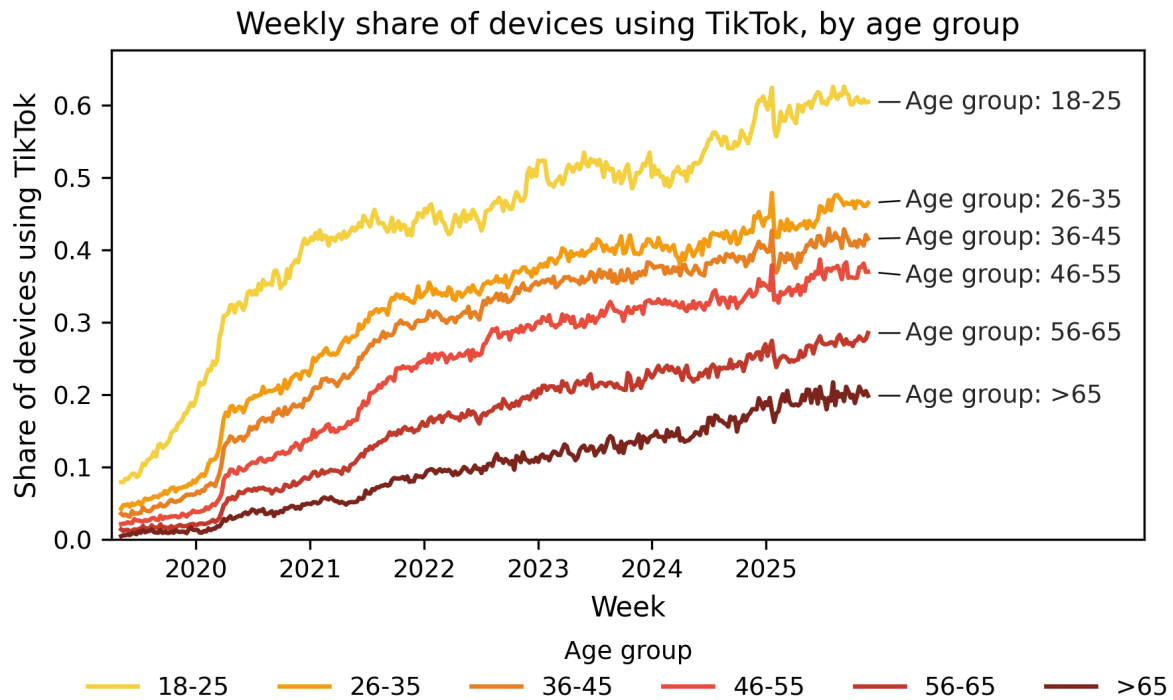


Figure A5: TikTok use by age

Note: This figure plots the weekly share of devices with non-zero TikTok use, by age group, among Comscore panelists from May 2019 to November 2025.

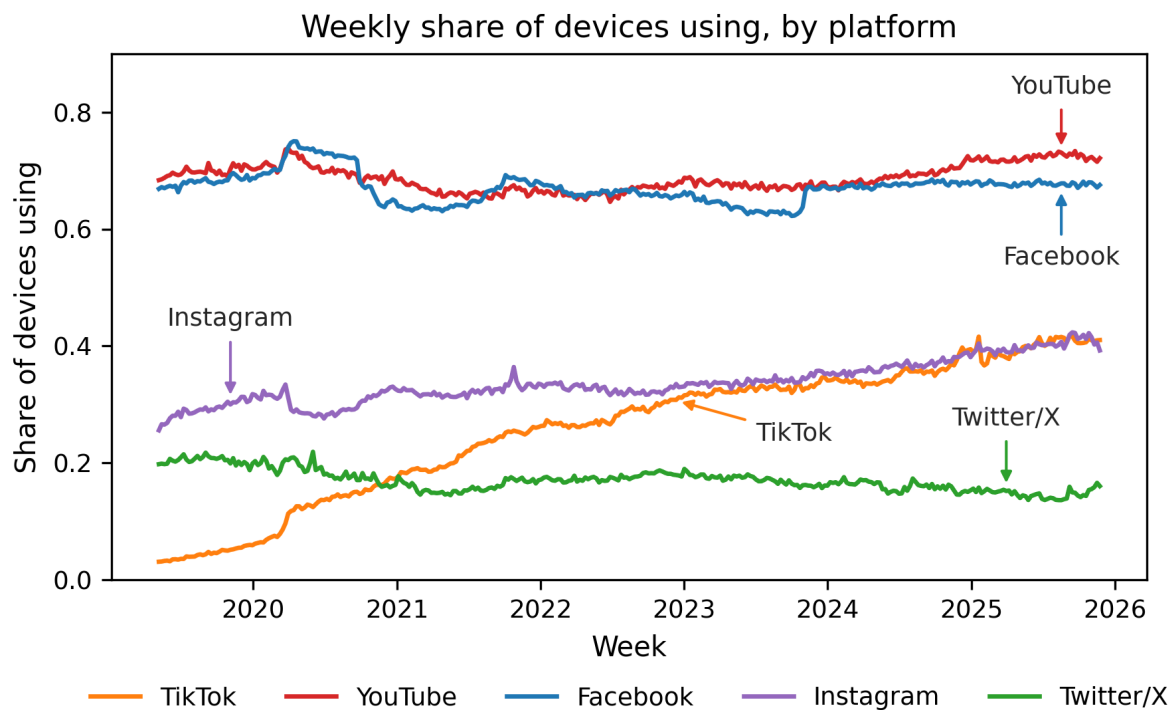


Figure A6: Social media use from 2019 to 2025

Note: This figure plots the weekly share of devices with non-zero use of TikTok, Youtube, Facebook, Instagram, and Twitter (now X), among Comscore panelists from May 2019 to November 2025.

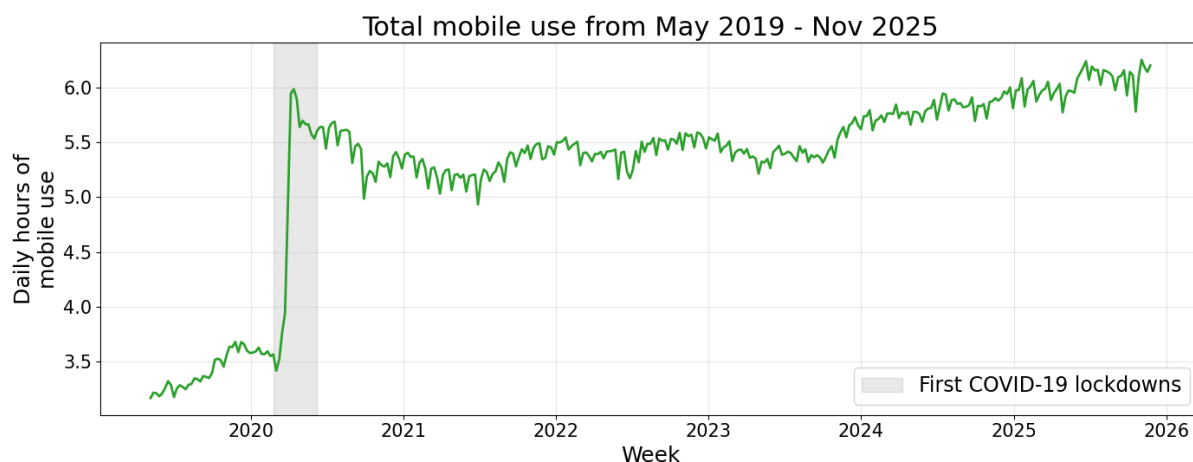


Figure A7: Total time spent on mobile browsing and app use from 2019 to 2025

Note: This figure plots the weekly average hours spent on mobile browsing and app use among Comscore panelists from May 2019 to November 2025.

Table A2: Device-level weekly minutes of platform use and browsing

Platform	% > 0	All devices				Users (> 0) only			
		Mean	p25	Median	p75	Mean	p25	Median	p75
TikTok	64.5	129.42	0.00	0.42	29.97	200.74	0.56	9.47	123.05
YouTube	97.3	212.47	16.30	76.25	256.26	218.37	19.18	81.61	264.98
Facebook	94.0	701.87	15.23	284.74	1073.25	746.77	32.64	349.80	1142.03
Instagram	76.3	72.72	0.01	1.07	22.09	95.27	0.47	4.58	45.38
Twitter/X	62.7	16.55	0.00	0.12	1.09	26.41	0.16	0.59	3.76
Snapchat	44.7	21.96	0.00	0.00	3.84	49.15	0.77	6.02	32.53
Reddit	52.8	8.17	0.00	0.02	0.59	15.47	0.14	0.51	2.22
Pinterest	58.4	8.21	0.00	0.06	1.15	14.04	0.15	0.70	5.18
LinkedIn	43.7	1.54	0.00	0.00	0.18	3.53	0.07	0.26	1.14
Discord	15.5	6.00	0.00	0.00	0.00	38.63	0.31	1.98	14.81

Social media excl. TikTok 99.1 1049.49 242.78 742.24 1551.19 1059.45 253.41 753.81 1561.54

Notes: The unit of observation is a device. For each device, weekly minutes of use are averaged within each year and then across years, weighting by the number of weeks the device is observed. The sample is the set of devices entering the column-(9) regression for total browsing (N devices = 183,785). For each platform, device-years with no recorded value are dropped before averaging. “% > 0” is the share of devices with a positive average. “Social media excl. TikTok” aggregates the nine non-TikTok platforms. All figures are weekly minutes.

Table A3: Weekly news consumption and total mobile use

Outcome	% > 0	All observations				Users (> 0) only			
		Mean	p25	Median	p75	Mean	p25	Median	p75
<i>Panel A. Across devices, averaged over the weeks each device is observed</i>									
News consumption (min/wk)	94.3	65.33	2.56	13.33	52.81	69.29	3.66	15.58	57.66
Total mobile use (hrs/wk)	100.0	35.57	17.38	33.45	50.76	35.57	17.38	33.45	50.76
<i>Panel B. Across device-weeks, in news event weeks only</i>									
News consumption (min/wk)	68.5	91.35	0.00	10.33	69.83	133.30	9.13	35.80	124.87
Total mobile use (hrs/wk)	100	37.00	11.02	31.92	56.83	37.08	11.13	32.01	56.89

Notes: News consumption is measured in weekly minutes and total mobile use in weekly hours. “% > 0” is the share of observations with a positive value. The two panels differ in their unit of observation. Panel A is a cross-section of devices: for each device, weekly values are averaged over the weeks the device is observed, so a device counts as positive if it consumes any news at any point. Panel B is at the device-week level and covers only the 40 news event weeks, so a device-week counts as positive if it consumes news in that week. The share with positive news is therefore mechanically higher in Panel A.

A.2 Cooperative Election Study

This appendix documents the survey items behind each outcome and the rules used to construct them. Respondents report their county of residence, which is standardized to a five-digit FIPS code and mapped to a media market. This gives 597,687 respondent-year observations from 3,006 counties in 205 media markets.

Political knowledge. The score sums eight binary items. Four ask the party of the respondent’s governor, of each of their two U.S. senators, and of their U.S. representative, with options *Never heard of person*, *Republican*, *Democrat*, *Other party / Independent* and *Not sure*. Four ask which party holds a majority in the U.S. House, the U.S. Senate, the respondent’s state senate and their state’s lower chamber, with options *Republicans*, *Democrats*, *Neither* and *Not sure*. Each answer is scored against the true officeholder or chamber majority in that state and year. Expressions of ignorance are scored zero rather than dropped, so the index measures correct knowledge rather than accuracy conditional on answering; items the respondent skipped are missing, and the score is missing unless all eight items are answered.

Traditional news consumption. A check-all grid asks which media the respondent used in the past 24 hours: social media, TV news, newspaper, radio news, or none of these. The indicator equals one if any of TV news, newspaper or radio news is selected.

Attention to public affairs. A single item asks how often the respondent follows what is going on in government and public affairs, with options *Most of the time*, *Some of the time*, *Only now and then*, *Hardly at all* and *Don’t know*. The scale is reversed so that higher values indicate more frequent attention, and “Don’t know” is treated as missing.

Validated turnout. The CES matches respondents to a national voter file and records the mode by which they voted in the general election. The indicator equals one

for respondents recorded as having voted. Validation is tied to the November general election and is therefore carried out only in midterm and presidential election years.

Presidential approval. Respondents rate the sitting president’s job performance as *Strongly approve*, *Somewhat approve*, *Somewhat disapprove* or *Strongly disapprove*, with *Not sure* treated as missing. Partisanship comes from the seven-point party identification item. Respondents who describe themselves as Democrats are classified as partisans; those who only lean toward a party and pure independents are excluded. Partisan polarization in approval rescores the item from 1 to 4 so that higher values indicate an evaluation more in line with the respondent’s own party: 4 for a copartisan of the president who strongly approves or an out-partisan who strongly disapproves, down to 1 for a copartisan who strongly disapproves or an out-partisan who strongly approves.

Table A4 reports which outcomes are available in which waves, and Table A5 the Dataverse release of each wave used. Table A6 presents summary statistics of main survey outcomes.

Table A4: CES outcome availability by wave, 2010–2024

Wave	Respondents	Knowledge (0–8)	Traditional media	News interest	Validated turnout	Partisan polarization
2010	54,316	Yes	Yes	Yes	Yes	Yes
2011	19,809	Yes	Yes	Yes		Yes
2012	53,640	Yes		Yes	Yes	Yes
2013	16,134	Yes	Yes	Yes		Yes
2014	55,274	Yes	Yes	Yes	Yes	Yes
2015	14,008	Yes		Yes		Yes
2016	63,272	Yes	Yes	Yes	Yes	Yes
2017	17,823	Yes		Yes		Yes
2018	58,886	Yes	Yes	Yes	Yes	Yes
2019	17,614	Yes	Yes	Yes		Yes
2020	59,862	Yes	Yes	Yes	Yes	Yes
2021	25,160	Yes	Yes	Yes		Yes
2022	58,911	Yes	Yes	Yes	Yes	Yes
2023	23,946	Yes	Yes	Yes		Yes
2024	59,032	Yes	Yes	Yes	Yes	Yes
Waves available		15	12	15	8	15

Table A5: Cooperative Election Study waves used

Wave	Authors (as on Dataverse)	Version	Released	DOI
2010	Ansolabehere	3	2012	10.7910/DVN/VKKRWA
2011	Ansolabehere	1	2013	10.7910/DVN/GLRXZ0
2012	Ansolabehere, Schaffner	9	2017	10.7910/DVN/HQEVPK
2013	Ansolabehere, Schaffner	1	2019	10.7910/DVN/KPP85M
2014	Schaffner, Ansolabehere	6	2024	10.7910/DVN/XFXJY
2015	Ansolabehere, Schaffner	2	2018	10.7910/DVN/SWMWX8
2016	Ansolabehere, Schaffner	5	2025	10.7910/DVN/GDF6Z0
2017	Schaffner, Ansolabehere	2	2019	10.7910/DVN/3STEZY
2018	Schaffner, Ansolabehere, Luks	6	2019	10.7910/DVN/ZSBZ7K
2019	Ansolabehere, Schaffner, Luks	1	2020	10.7910/DVN/WOT7O8
2020	Schaffner, Ansolabehere, Luks	4	2022	10.7910/DVN/E9N6PH
2021	Ansolabehere, Schaffner	1	2022	10.7910/DVN/OPQOCU
2022	Schaffner, Ansolabehere, Shih	4	2024	10.7910/DVN/PR4L8P
2023	Schaffner, Ansolabehere, Pope, Shih	2	2025	10.7910/DVN/JQJTCC
2024	Schaffner, Shih, Ansolabehere, Pope	9	2025	10.7910/DVN/X11EP6

Notes: Each row is one Common Content release of the Cooperative Election Study (CES; Cooperative Congressional Election Study, CCES, through 2019), archived on the Harvard Dataverse and distributed under a CC0 1.0 license. Dataverse titles are “CCES Common Content, *year*” (2010–2019) and “CES Common Content, *year*” (2020–2024); author order is as recorded on Dataverse. “Version” is the Dataverse version of the file used in the analysis, downloaded May 14–24, 2026; “Released” is the year that version was published. The 2014 and 2018 datasets were re-released as Version 7 in July 2026, after the analysis files were obtained. Harmonized demographics are taken from the Cumulative CES Common Content, Version 11, which covers 2006–2024 ([Kuriwaki, 2025](#)).

Table A6: Summary statistics: CES outcomes

Outcome	Range	Mean	By period		Respondents	Survey years
			Pre 2019	2019 and later		
<i>Panel A: News consumption, past 24 hours</i>						
Traditional news (any of the three)	0–1	0.805 (0.396)	0.860 (0.347)	0.744 (0.436)	512,216	12
Television news	0–1	0.645 (0.479)	0.710 (0.454)	0.574 (0.495)	512,216	12
Newspaper	0–1	0.404 (0.491)	0.482 (0.500)	0.319 (0.466)	512,216	12
Radio news	0–1	0.334 (0.472)	0.376 (0.484)	0.288 (0.453)	512,216	12
<i>Panel B: Political information</i>						
Political knowledge	0–8	4.973 (2.757)	4.918 (2.776)	5.053 (2.728)	574,436	15
Attention to public affairs	1–4	3.184 (0.967)	3.224 (0.941)	3.124 (1.001)	580,025	15
<i>Panel C: Political behavior and attitudes</i>						
Validated turnout	0–1	0.538 (0.499)	0.533 (0.499)	0.546 (0.498)	463,193	8
Partisan polarization in presidential approval	1–4	3.497 (0.792)	3.481 (0.805)	3.520 (0.772)	368,331	15

Notes: This table reports the range and mean for the outcomes from the Cooperative Election study used in the paper. Standard deviations are reported in parentheses below each mean. Each row is computed on the respondents of the corresponding regression column: the estimation sample (survey years 2010–2024, respondents with non-missing celebrity indices, 2015 DMA balance controls and demographics, and a positive survey weight) restricted to those non-missing on that outcome. Respondent counts and survey years differ across rows because the CES does not field the 24-hour media battery in 2012, 2015 or 2017, because validated turnout is available only in federal-election years, and because partisan polarization is only defined among those self-reporting as Democrat or Republican. Statistics by period are the means and standard deviations before and after 2018, the year of the Musical.ly–TikTok merger. Means and standard deviations are survey-weighted; respondent counts are unweighted.

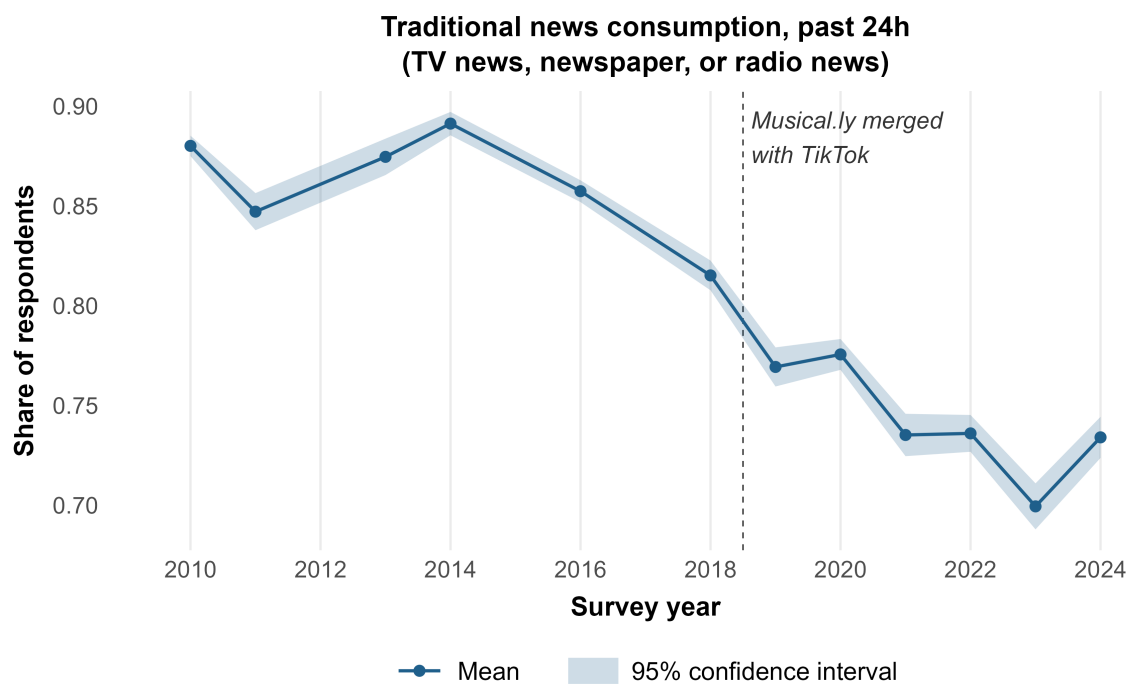


Figure A8: News consumption over time among CES respondents

Note: This figure plots yearly means and 95% confidence intervals of an indicator variable for having watched television news, read a newspaper or listened to radio news in the past 24 hours. Respondents are weighted to represent the adult U.S. population.

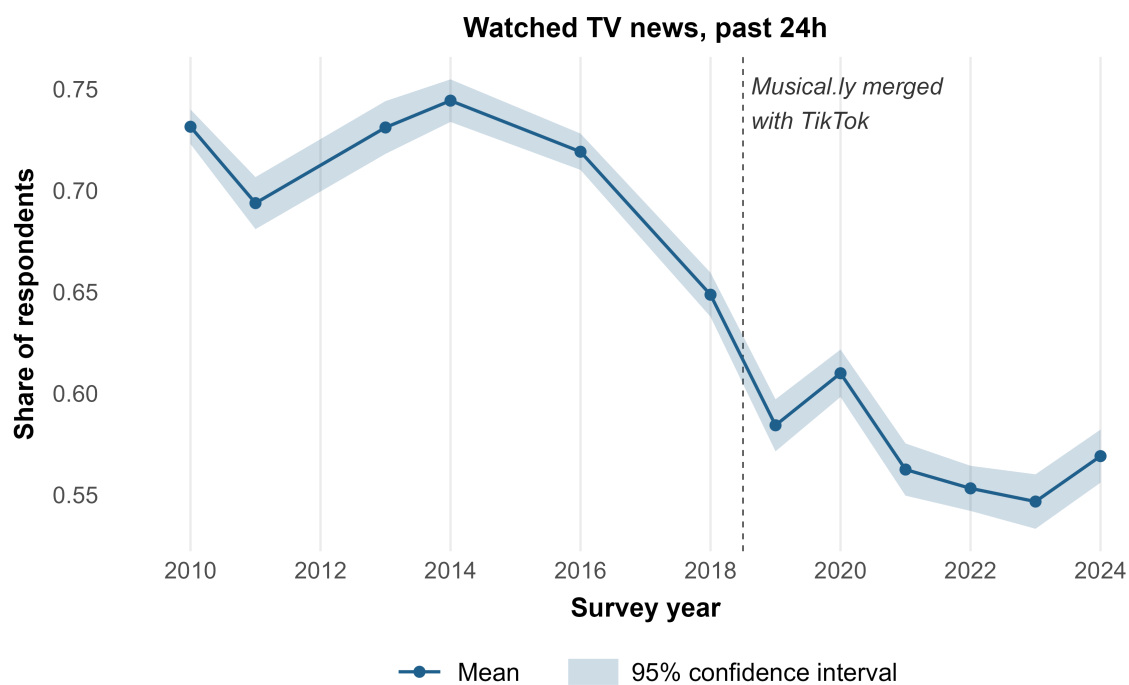


Figure A9: News consumption over time among CES respondents

Note: This figure plots yearly means and 95% confidence intervals of an indicator variable for having watched television news in the past 24 hours. Respondents are weighted to represent the adult U.S. population.

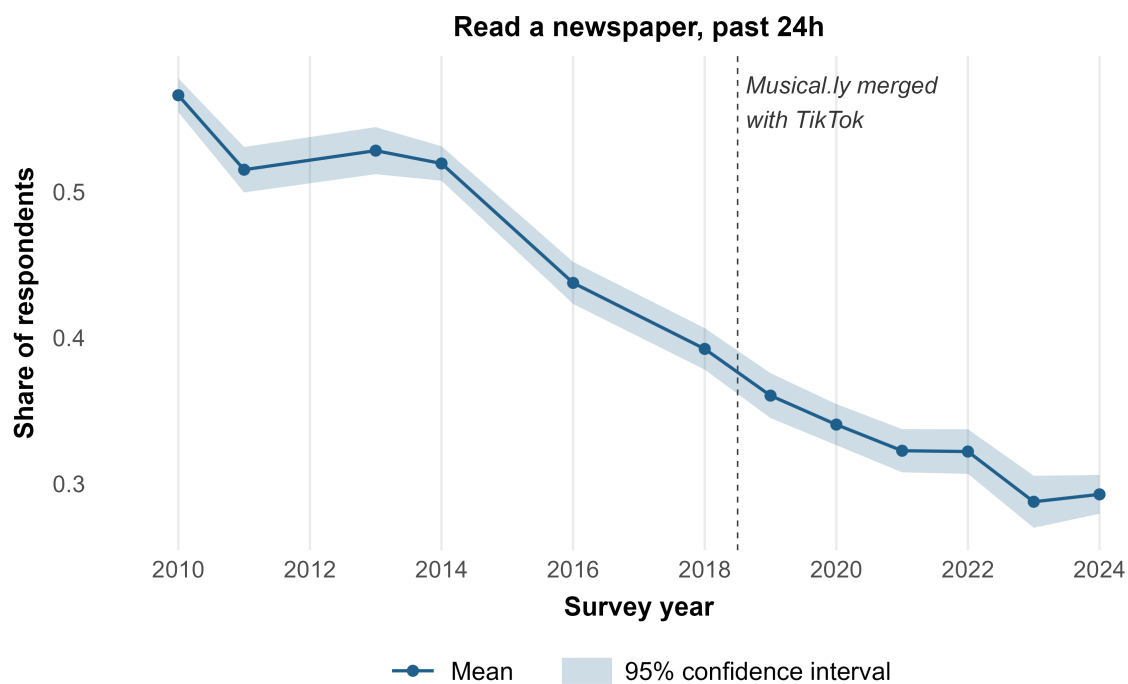


Figure A10: News consumption over time among CES respondents

Note: This figure plots yearly means and 95% confidence intervals of an indicator variable for having read a newspaper in the past 24 hours. Respondents are weighted to represent the adult U.S. population.

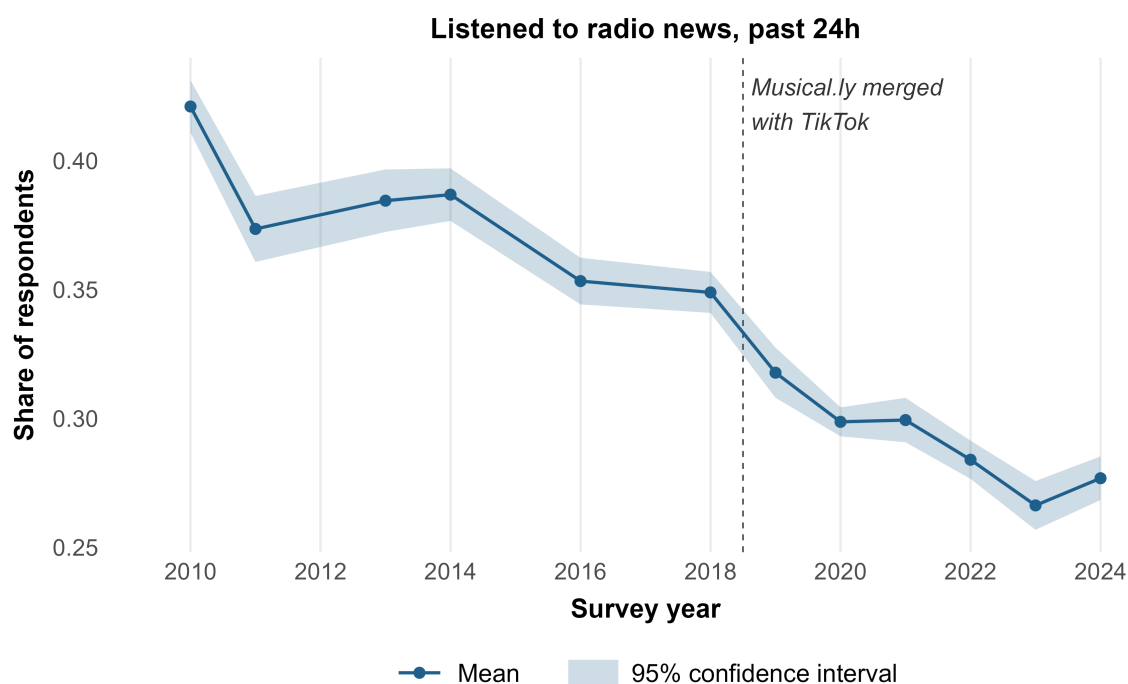


Figure A11: News consumption over time among CES respondents

Note: This figure plots yearly means and 95% confidence intervals of an indicator variable for having listened to radio news in the past 24 hours. Respondents are weighted to represent the adult U.S. population.

A.3 News pressure and media-market characteristics

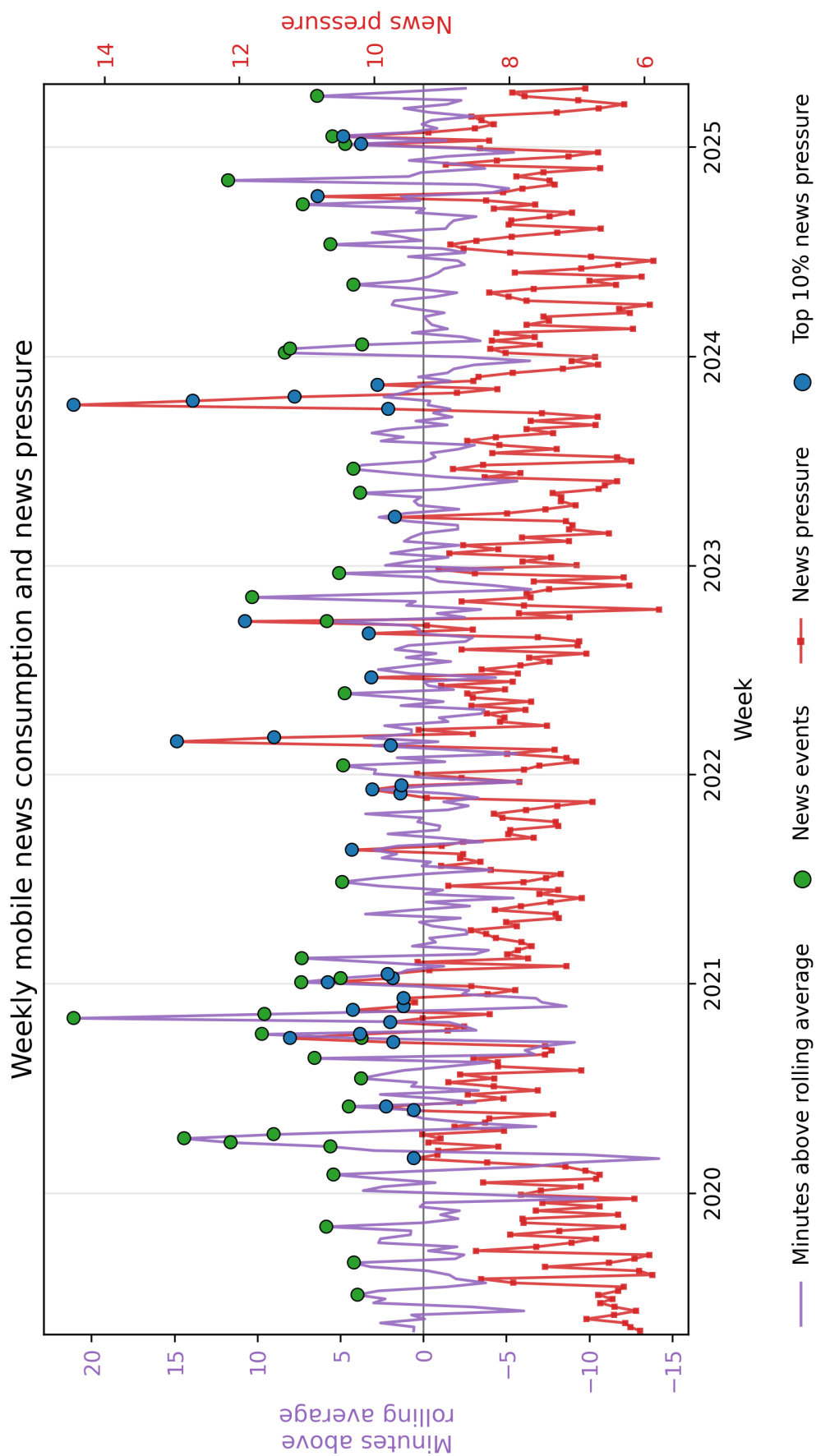


Figure A12: Mobile news consumption and news pressure, 2019–2025

Note: This figure overlays weekly minutes of news browsing among Comscore panelists with an independent, TV-based news-pressure measure: the minutes the NBC, ABC, and CBS evening newscasts devote to their top three stories each week (Eisenstein & Strömberg, 2007; Durante & Zhuravskaya, 2018). Green circles mark the 40 news event weeks, blue circles mark the top 10% news pressure weeks.

Table A7: Summary statistics: media-market characteristics

Variable	Mean	SD	Min	p25	Median	p75	Max
<i>Panel A. Demographic characteristics</i>							
log(population)	13.5593	1.1775	9.5450	12.8232	13.5520	14.3644	16.8832
Share women	0.5050	0.0087	0.4781	0.4998	0.5062	0.5106	0.5341
Share White	0.8205	0.1186	0.3486	0.7537	0.8587	0.9107	0.9756
Share Black	0.1151	0.1192	0.0040	0.0271	0.0684	0.1696	0.6342
Share Native American	0.0148	0.0217	0.0016	0.0041	0.0071	0.0148	0.1402
Share Asian	0.0260	0.0276	0.0044	0.0112	0.0181	0.0299	0.2619
Share Pacific Islander	0.0014	0.0016	0.0002	0.0005	0.0010	0.0015	0.0096
Share two or more races	0.0223	0.0090	0.0038	0.0170	0.0206	0.0244	0.0693
Share Hispanic	0.1241	0.1590	0.0099	0.0350	0.0606	0.1363	0.9512
Share age 0–19	0.2554	0.0248	0.1941	0.2417	0.2540	0.2671	0.3724
Share age 20–39	0.2645	0.0241	0.1879	0.2468	0.2655	0.2784	0.3713
Share age 40–59	0.2579	0.0157	0.2067	0.2482	0.2595	0.2691	0.2910
Share age 60+	0.2222	0.0336	0.1271	0.2015	0.2186	0.2422	0.3589
<i>Panel B. Socio-economic characteristics</i>							
Gini index	0.4527	0.0178	0.4009	0.4405	0.4514	0.4640	0.5073
Unemployment rate	0.0798	0.0210	0.0282	0.0656	0.0794	0.0930	0.1699
Poverty rate (18–64)	0.1609	0.0376	0.0817	0.1325	0.1561	0.1777	0.3060
log(median household income)	10.7891	0.1644	10.2615	10.6765	10.7738	10.8801	11.3963
Share HS graduates	0.8667	0.0487	0.6223	0.8407	0.8761	0.8992	0.9444
Share Bachelor’s	0.2505	0.0619	0.1423	0.1995	0.2453	0.2913	0.4480
Share Master’s	0.0915	0.0293	0.0460	0.0693	0.0871	0.1067	0.2138
<i>Panel C. Presidential vote shares</i>							
Republican vote share 2000	0.5325	0.0910	0.2967	0.4715	0.5377	0.5900	0.7650
Republican vote share 2004	0.5596	0.0959	0.2885	0.4887	0.5684	0.6235	0.7977
Republican vote share 2008	0.5141	0.1047	0.2388	0.4423	0.5147	0.5889	0.7715
Republican vote share 2012	0.5332	0.1140	0.2293	0.4533	0.5398	0.6179	0.7921
Republican vote share 2016	0.5459	0.1229	0.1858	0.4648	0.5496	0.6385	0.7906

Notes: This table reports summary statistics for the media-market characteristics used in the paper across the 205 media markets. Panel A reports the demographic characteristics, taken from the Census Bureau’s Vintage 2015 county population estimates ([US Census Bureau, 2016b](#)); Panel B reports the socio-economic characteristics, taken from the 2011–2015 five-year estimates of the American Community Survey ([US Census Bureau, 2016a](#)); and Panel C reports the Republican share of the two-party presidential vote in each election from 2000 to 2016, aggregated from the county returns of the MIT Election Data and Science Lab ([MIT Election Data and Science Lab, 2018](#)). County-level variables are aggregated to media markets, summing counts across counties and weighting median household income and the Gini index by county population. Table [A1](#) lists the same variables with their sources.

B The instrument

Table B1: Artist-level sources: Wikipedia biographies and YouGov Ratings pages

Artist	Group	Wikipedia article	Version	YouGov slug
Selena Gomez	Musical.ly	Selena_Gomez	1354375718	Selena_Gomez
Ariana Grande	Musical.ly	Ariana_Grande	1353518029	Ariana_Grande
Anne-Marie	Musical.ly	Anne-Marie_(singer)	1348512184	Anne_Marie
Charli XCX	Musical.ly	Charli_XCX	1353886864	Charli_XCX
Katy Perry	Musical.ly	Katy_Perry	1354354611	Katy_Perry
Meghan Trainor	Musical.ly	Meghan_Trainor	1351264491	Meghan_Trainor
Lady Gaga	Musical.ly	Lady_Gaga	1354333258	Lady_Gaga
Jason Derulo	Musical.ly	Jason_Derulo	1353967829	Jason_Derulo
Silentó	Musical.ly	Silentó	1351808406	Silento
Avicii	Superstar	Avicii	1353885321	Avicii
Beyoncé	Superstar	Beyoncé	1353689617	Beyonce
Bruno Mars	Superstar	Bruno_Mars	1353676847	Bruno_Mars
Calvin Harris	Superstar	Calvin_Harris	1354472809	Calvin_Harris
Dr. Dre	Superstar	Dr._Dre	1354188205	Dr_Dre
Jay-Z	Superstar	Jay-Z	1354200088	Jay_Z
Justin Bieber	Superstar	Justin_Bieber	1354276602	Justin_Bieber
Kanye West	Superstar	Kanye_West	1353713655	Kanye_West
Rihanna	Superstar	Rihanna	1354398877	Rihanna
Miley Cyrus	Superstar	Miley_Cyrus	1354076137	Miley_Cyrus
One Direction	Superstar	One_Direction	1352030394	One_Direction
Pharrell Williams	Superstar	Pharrell_Williams	1354220389	Pharrell_Williams
Taylor Swift	Superstar	Taylor_Swift	1354357535	Taylor_Swift
Andy Grammer	Up-and-coming	Andy_Grammer	1353580118	Andy_Grammer
Charlie Puth	Up-and-coming	Charlie_Puth	1353298738	Charlie_Puth
David Guetta	Up-and-coming	David_Guetta	1352739430	David_Guetta
Demi Lovato	Up-and-coming	Demi_Lovato	1349605570	Demi_Lovato
DJ Snake	Up-and-coming	DJ_Snake	1352592515	DJ_Snake
Ed Sheeran	Up-and-coming	Ed_Sheeran	1354258804	Ed_Sheeran
Ellie Goulding	Up-and-coming	Ellie_Goulding	1353637731	Ellie_Goulding
Fifth Harmony	Up-and-coming	Fifth_Harmony	1351880757	Fifth_Harmony
Jessie J	Up-and-coming	Jessie_J	1352472100	Jessie_J
Mark Ronson	Up-and-coming	Mark_Ronson	1353361633	Mark_Ronson
Nick Jonas	Up-and-coming	Nick_Jonas	1354111840	Nick_Jonas
Nicki Minaj	Up-and-coming	Nicki_Minaj	1354470259	Nicki_Minaj
OMI	Up-and-coming	Omi_(singer)	1346353794	OMI
Rachel Platten	Up-and-coming	Rachel_Platten	1354431401	Rachel_Platten
Sam Smith	Up-and-coming	Sam_Smith	1353517309	Sam_Smith
Skrillex	Up-and-coming	Skrillex	1353785071	Skrillex
The Weeknd	Up-and-coming	The_Weeknd	1354228468	The_Weeknd
Tove Lo	Up-and-coming	Tove_Lo	1354421236	Tove_Lo
Walk the Moon	Up-and-coming	Walk_the_Moon	1345634784	Walk_the_Moon

Notes: One row per artist in Table B5. “Wikipedia article” is the title of the English Wikipedia article, so that <https://en.wikipedia.org/wiki/<article>> is the page. “Version” is Wikipedia’s number for the version of the article retrieved on May 16, 2026, from which age, gender, place of birth and any stated race or ethnicity were coded (Wikipedia, 2026a); Wikipedia keeps every past version, and <https://en.wikipedia.org/w/index.php?title=<article>&oldid=<version>> displays that exact text regardless of later edits. “YouGov slug” completes <https://yougov.com/en-us/topics/public-figure/<slug>>, the YouGov Ratings page retrieved on May 16, 2026, from which fame and popularity overall and by age cohort and gender were taken (YouGov, 2026). The retrieved HTML of every page is included in the replication package.

Table B2: Wikipedia lists of presidential-campaign endorsements used to code artist endorsements

Election	Candidate	Wikipedia article	Version
2012	Barack Obama (D)	List_of_Barack_Obama_2012_presidential_campaign_endorsements	1353411077
2012	Mitt Romney (R)	List_of_Mitt_Romney_2012_presidential_campaign_endorsements	1354239605
2016	Hillary Clinton (D)	List_of_Hillary_Clinton_2016_presidential_campaign_celebrity_endorsements	1354206534
2016	Hillary Clinton (D)	List_of_Hillary_Clinton_2016_presidential_campaign_non-political_endorsements	1354206544
2016	Donald Trump (R)	List_of_Donald_Trump_2016_presidential_campaign_endorsements	1354206296
2016	Bernie Sanders (D, primary)	List_of_Bernie_Sanders_2016_presidential_campaign_endorsements	1352797771
2020	Joe Biden (D)	List_of_Joe_Biden_2020_presidential_campaign_endorsements	1353299165
2020	Donald Trump (R)	List_of_Donald_Trump_2020_presidential_campaign_non-political_endorsements	1351280802
2020	Bernie Sanders (D, primary)	List_of_Bernie_Sanders_2020_presidential_campaign_endorsements	1352798002

Notes: The English Wikipedia list articles searched for the 41 artists' names ([Wikipedia, 2026b](#)), retrieved on May 16, 2026, one row per article. <https://en.wikipedia.org/wiki/<article>> is the page. "Version" is Wikipedia's number for the version of the article retrieved; Wikipedia keeps every past version, and <https://en.wikipedia.org/w/index.php?title=<article>&oldid=<version>> displays that exact text regardless of later edits. Where Wikipedia has split a campaign's endorsements across several articles, the articles covering musicians are listed.

Table B3: Google Trends queries: artist topics

Topic	Topic identifier	Series	Downloaded
<i>Panel A. Musical.ly artists (2015 instrument; 2014 and 2016 placebo indices)</i>			
Selena Gomez	/m/0gs6vr	DMA, 2014, 2015, 2016	2026-08-18 / 2026-02-04 / 2025-05-02
Ariana Grande	/m/09gkdy4	DMA, 2014, 2015, 2016	2026-08-18 / 2026-02-04 / 2025-05-02
Anne-Marie	/g/11c52019vx	DMA, 2014, 2015, 2016	2026-08-18 / 2026-02-04 / 2025-05-02
Charli XCX	/m/0j4214h	DMA, 2014, 2015, 2016	2026-08-18 / 2026-02-04 / 2025-05-02
Katy Perry	/m/03y82t6	DMA, 2014, 2015, 2016	2026-08-18 / 2026-02-04 / 2025-05-02
Meghan Trainor	/m/0j8djs0	DMA, 2014, 2015, 2016	2026-08-18 / 2026-02-04 / 2025-05-02
Lady Gaga	/m/0478_m	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-20 / 2026-08-18
Jason Derulo	/m/04fzp51	DMA, 2014, 2015, 2016	2026-08-18 / 2026-02-04 / 2025-05-02
Silentó	/m/0134_6cg	DMA, 2014, 2015, 2016	2026-08-18 / 2025-05-02 / 2025-05-02
Watch Me (Whip/Nae Nae)	/g/11btxcrlz9	DMA, 2014, 2015, 2016	2026-08-18 / 2025-05-02 / 2025-05-02
<i>Panel B. Superstar placebo artists</i>			
Avicii	/m/0bs3d41	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Beyoncé	/m/01mpq7s	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Bruno Mars	/m/0bs1g5r	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Calvin Harris	/m/0288c4g	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Dr. Dre	/m/01vvydl	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Jay-Z	/m/01vw20h	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Justin Bieber	/m/06w2sn5	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Kanye West	/m/021840	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Rihanna	/m/06mt91	DMA, 2014, 2015, 2016	2026-08-18 / 2026-02-04 / 2025-05-02
Miley Cyrus	/m/0bdxs5	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-14
One Direction	/m/0fqqn timer	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Pharrell Williams	/m/04mn81	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
Taylor Swift	/m/0d1567	DMA, 2014, 2015, 2016	2025-05-02 / 2025-05-02 / 2025-05-02
<i>Panel C. Up-and-coming placebo artists</i>			
Andy Grammer	/m/0g5rt4p	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Charlie Puth	/m/0131js50	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
David Guetta	/m/0840vq	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Demi Lovato	/m/02wb6yq	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
DJ Snake	/m/01w65jh	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Ed Sheeran	/m/0g9sr1k	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Ellie Goulding	/m/080gct2	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Fifth Harmony	/m/0wzv3cm	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Jessie J	/m/0ds7zj1	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Mark Ronson	/m/0892sx	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Nick Jonas	/m/04f7c55	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Nicki Minaj	/m/047sxrj	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
OMI	/m/0134wpwt	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Rachel Platten	/m/0h54vcq	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Sam Smith	/m/0t_ctj1	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Skrillex	/m/0462ph1	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
The Weeknd	/m/0gjd4c	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Tove Lo	/m/0wt63mj	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18
Walk the Moon	/m/0gdtrc1	DMA, 2014, 2015, 2016	2026-08-18 / 2026-03-19 / 2026-08-18

Notes: Each row is one Google Trends topic ([Google, 2025–2026](#)), one per artist in Table B5 plus Silentó’s single “Watch Me (Whip/Nae Nae)”, which enters the Musical.ly index alongside the artist. Each series is a Google Trends *topic*, Google’s entry for the artist or platform, which pools searches about it regardless of spelling or language, rather than a search term. “Topic identifier” is the code Google Trends assigns to the topic; it appears in the address bar when the topic is selected (`q=/m/09gkdy4` for Ariana Grande, for example) and re-selects the same topic when entered as the query, so that each series can be reproduced. “DMA” series are the “Interest by subregion” export at the designated-market-area level. Google Trends values are normalized within each query, so re-downloading a query can return slightly different values; the files used are included in the replication package. Download dates (year-month-day) are listed in the order of the years in the “Series” column.

Table B4: Google Trends queries: short-form video platforms

Topic	Topic identifier	Series	Period	Downloaded
Musical.ly	/g/11bzq3s9f4	DMA, yearly	2013–2020	2026-02-02
Musical.ly	/g/11bzq3s9f4	National, monthly	Jan 2013–Dec 2020	2026-06-10
Vine	/m/0q417mg	DMA, yearly	2013–2020	2026-07-02
Vine	/m/0q417mg	National, monthly	Jan 2013–Dec 2020	2026-07-02
Dubsmash	/m/012d30vw	DMA, yearly	2014–2020	2026-07-02
Dubsmash	/m/012d30vw	National, monthly	Jan 2013–Dec 2020	2026-07-02
TikTok	/g/11f555cn81	National, monthly	Aug 2017–Sep 2019	2026-09-02

Notes: Google Trends series for the platforms ([Google, 2025–2026](#)). “DMA, yearly” is one “Interest by subregion” export per calendar year in the period; “National, monthly” is the “Interest over time” export for the United States over the period. Each series is a Google Trends *topic*, Google’s entry for the artist or platform, which pools searches about it regardless of spelling or language, rather than a search term. “Topic identifier” is the code Google Trends assigns to the topic; it appears in the address bar when the topic is selected (q=/m/09gkdy4 for Ariana Grande, for example) and re-selects the same topic when entered as the query, so that each series can be reproduced. “DMA” series are the “Interest by subregion” export at the designated-market-area level. Google Trends values are normalized within each query, so re-downloading a query can return slightly different values; the files used are included in the replication package.

Table B5: Musical.ly artists and placebo comparison artists

Musical.ly	Superstar placebo	Up-and-coming placebo	
Lady Gaga	Avicii	Andy Grammer	Mark Ronson
Katy Perry	Beyoncé	Charlie Puth	Nick Jonas
Ariana Grande	Bruno Mars	David Guetta	Nicki Minaj
Meghan Trainor	Calvin Harris	Demi Lovato	OMI
Selena Gomez	Dr. Dre	DJ Snake	Rachel Platten
Silentó	Jay-Z	Ed Sheeran	Sam Smith
Jason Derulo	Justin Bieber	Ellie Goulding	Skrillex
Charli XCX	Kanye West	Fifth Harmony	The Weeknd
Anne-Marie	Miley Cyrus	Jessie J	Tove Lo
	One Direction	Walk the Moon	
	Pharrell Williams		
	Rihanna		
	Taylor Swift		

Notes: Musical.ly artists are pop artists that promoted their music through Musical.ly in 2015–2016, identified from trade- and music-press coverage of marketing on the platform and from compilations of viral Musical.ly challenges ([Music Ally, 2017](#); [Spanos, 2017](#); [Zaira Ann, 2017](#); [Maraki, 2015](#)). Superstar placebo artists are musicians in the 2014 Forbes Celebrity 100 ([Forbes, 2014](#)) who meet the popular-music criterion, a MusicBrainz genre classification of pop, hip-hop, R&B or dance ([MusicBrainz, 2026](#)). Up-and-coming placebo artists are rising pop artists matched to the Musical.ly artists’ 2015 profile, selected from the Billboard Year-End Hot 100 for 2015 ([Billboard, 2015](#)), Teen Choice Award nominations for 2014 and 2015 ([McGahan, 2014](#); [PopCrush Staff, 2015](#)) and MusicBrainz genre tags ([MusicBrainz, 2026](#)) (criteria in Section 3.1).

Table B6: Balance of the Musical.ly Celebrity index: residualized median split

	Above median	Below median	<i>p</i> -value	<i>q</i> -value
<i>Panel A. Media-market (DMA) characteristics</i>				
log(population)	13.6512 (0.1122)	13.4683 (0.1201)	0.2668	0.6350
Share women	0.5045 (0.0008)	0.5056 (0.0009)	0.3469	0.6623
Share White	0.8332 (0.0100)	0.8080 (0.0131)	0.1287	0.6350
Share Black	0.1050 (0.0096)	0.1251 (0.0136)	0.2270	0.6350
Share Native American	0.0131 (0.0018)	0.0164 (0.0025)	0.2846	0.6350
Share Asian	0.0255 (0.0023)	0.0265 (0.0031)	0.8064	0.9408
Share Pacific Islander	0.0013 (0.0002)	0.0015 (0.0002)	0.3364	0.6623
Share Hispanic	0.1089 (0.0150)	0.1391 (0.0163)	0.1742	0.6350
Share age 0–19	0.2550 (0.0024)	0.2557 (0.0025)	0.8502	0.9651
Share age 20–39	0.2651 (0.0025)	0.2639 (0.0023)	0.7232	0.8679
Share age 40–59	0.2585 (0.0016)	0.2573 (0.0015)	0.5989	0.8674
Share age 60+	0.2214 (0.0031)	0.2231 (0.0035)	0.7190	0.8679
Gini index	0.4514 (0.0016)	0.4541 (0.0019)	0.2813	0.6350
Unemployment rate	0.0782 (0.0019)	0.0815 (0.0023)	0.2543	0.6350
Poverty rate (18–64)	0.1591 (0.0034)	0.1626 (0.0040)	0.5090	0.7917
log(median household income)	10.7896 (0.0148)	10.7886 (0.0176)	0.9658	0.9658
Share HS graduates	0.8715 (0.0045)	0.8619 (0.0051)	0.1579	0.6350
Share Bachelor’s	0.2446 (0.0058)	0.2564 (0.0064)	0.1744	0.6350
Share Master’s	0.0893 (0.0027)	0.0936 (0.0031)	0.2946	0.6350
Republican vote share 2000	0.5243 (0.0086)	0.5407 (0.0094)	0.1991	0.6350
Republican vote share 2004	0.5549 (0.0091)	0.5642 (0.0099)	0.4880	0.7917
Republican vote share 2008	0.5135 (0.0096)	0.5147 (0.0110)	0.9363	0.9658
Republican vote share 2012	0.5326 (0.0107)	0.5337 (0.0119)	0.9422	0.9658
Republican vote share 2016	0.5561 (0.0111)	0.5358 (0.0130)	0.2389	0.6350
<i>Panel B. Comscore panel characteristics, 2019–2025</i>				
Average observed days per device	298.19 (4.13)	300.70 (5.14)	0.7041	0.8679
Mean age	42.09 (0.12)	42.39 (0.14)	0.0990*	0.6350
log(age)	3.7394 (0.0028)	3.7464 (0.0033)	0.1069	0.6350
Share women	0.5464 (0.0046)	0.5439 (0.0052)	0.7144	0.8679
Average household income (midpoint)	43.49 (0.82)	42.66 (0.94)	0.5066	0.7917
Average household size	2.8227 (0.0119)	2.8010 (0.0130)	0.2202	0.6350
Children present	0.4373 (0.0033)	0.4280 (0.0047)	0.1045	0.6350
Share Black	0.1629 (0.0112)	0.1704 (0.0146)	0.6866	0.8679
Share Hispanic	0.1293 (0.0124)	0.1434 (0.0138)	0.4483	0.7845
Share iPhone users	0.1436 (0.0030)	0.1458 (0.0040)	0.6479	0.8679
Share iPad users	0.0582 (0.0021)	0.0609 (0.0026)	0.4272	0.7801

Notes: Media markets are split at the median of the Musical.ly Celebrity index after residualizing the index on the shares of women, Hispanic residents, residents in poverty, and high-school graduates. Cells report group means with standard errors in parentheses. *p*-values are from two-sided tests of equality of means; *q*-values are false-discovery-rate adjusted. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

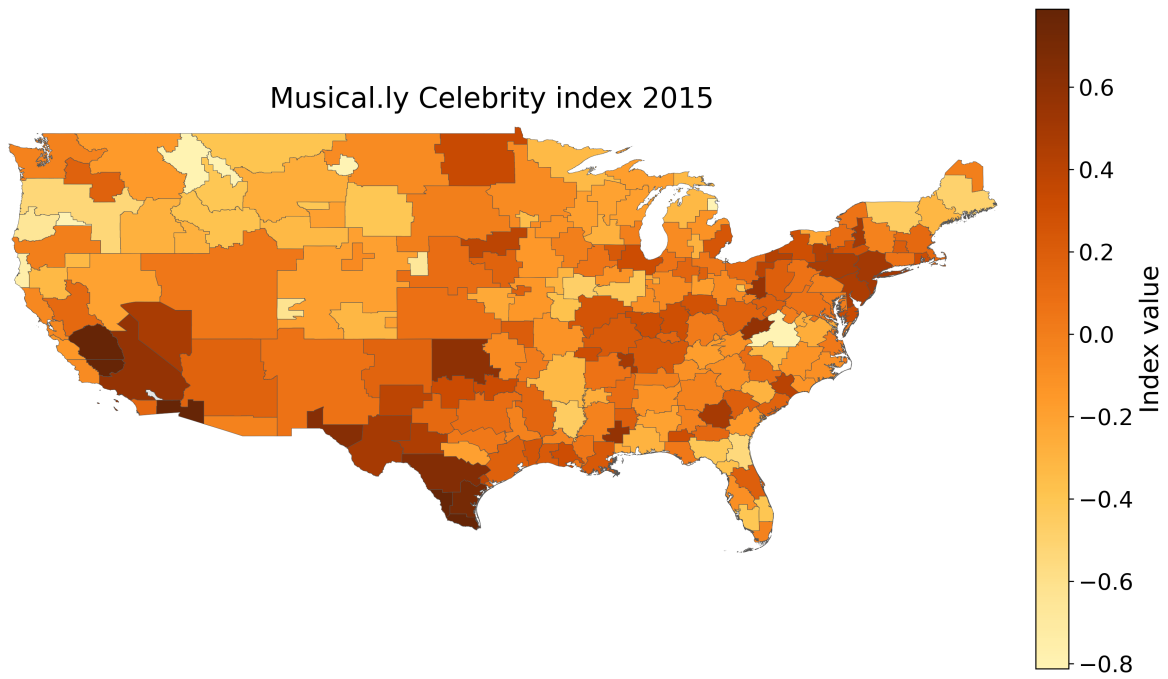


Figure B1: Geographic variation in the raw Musical.ly Celebrity index

Note: This figure maps the raw (unresidualized) Musical.ly Celebrity index across U.S. media markets (DMAs), constructed from 2015 Google Trends search interest in the Musical.ly artists. Darker shading indicates higher relative interest. The residualized index used throughout the paper is mapped in [Figure 3](#).

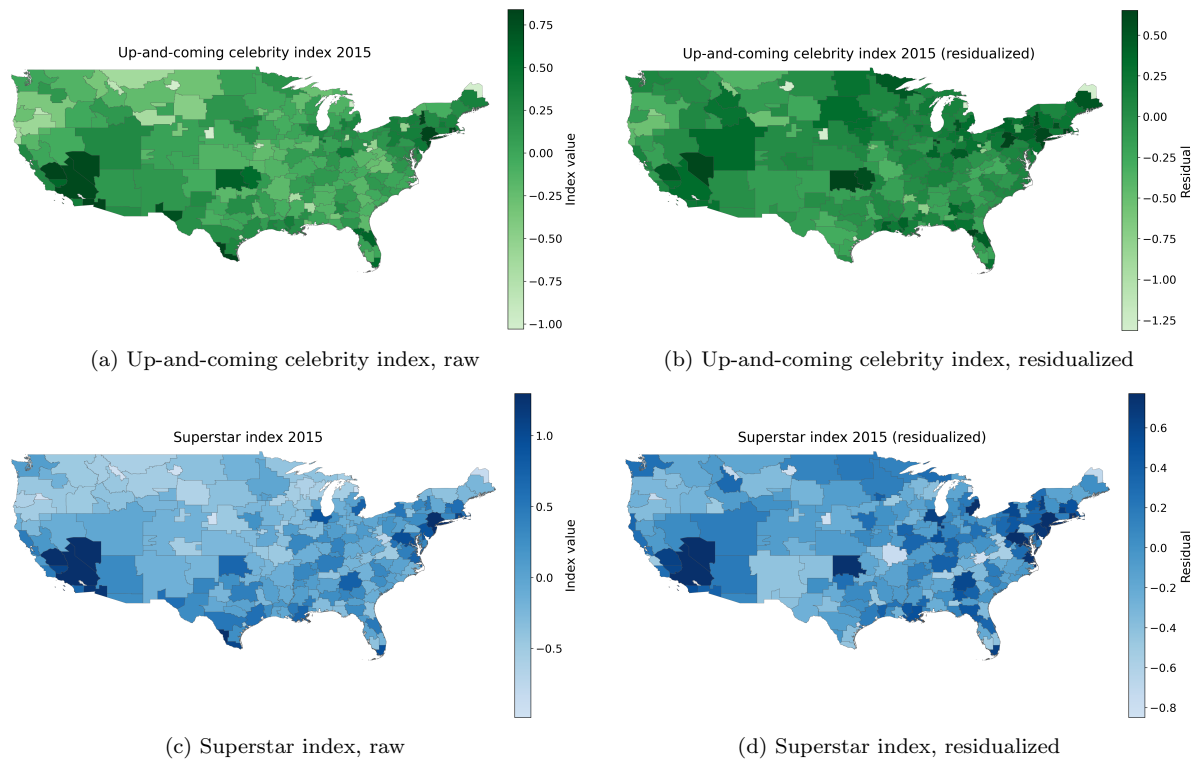


Figure B2: Geographic variation in the placebo indices, raw and residualized

Note: This figure maps the up-and-coming (top) and superstar (bottom) placebo indices across U.S. media markets (DMAs), constructed from 2015 Google Trends search interest in the respective placebo artists. The left column shows the raw indices; the right column shows the indices residualized on the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates. Darker shading indicates higher relative interest.

Table B7: Balance of Musical.ly artists versus placebo artists

	Musical.ly	Placebo	<i>p</i> -value
<i>Panel A. Presidential-candidate endorsements, 2012–2020</i>			
Any endorsement	0.44 (0.18)	0.38 (0.09)	0.729
Any Democratic endorsement	0.44 (0.18)	0.38 (0.09)	0.729
Any Republican endorsement	0.00 (0.00)	0.03 (0.03)	0.325
# general-election endorsements	0.78 (0.32)	0.56 (0.16)	0.560
Any endorsement in 2012	0.33 (0.17)	0.16 (0.07)	0.345
Any endorsement in 2016	0.33 (0.17)	0.31 (0.08)	0.913
Any endorsement in 2020	0.11 (0.11)	0.09 (0.05)	0.890
Endorsed Obama 2012	0.33 (0.17)	0.16 (0.07)	0.345
Endorsed Clinton 2016	0.33 (0.17)	0.28 (0.08)	0.783
Endorsed Sanders 2016 (primary)	0.00 (0.00)	0.03 (0.03)	0.325
Endorsed Trump 2016	0.00 (0.00)	0.03 (0.03)	0.325
Endorsed Sanders 2020 (primary)	0.11 (0.11)	0.06 (0.04)	0.692
Endorsed Trump 2020	0.00 (0.00)	0.03 (0.03)	0.325
<i>Panel B. Artist demographics</i>			
log(age in 2015)	3.15 (0.06)	3.38 (0.04)	0.006***
Male (share)	0.22 (0.15)	0.56 (0.09)	0.067*
US-born (share)	0.78 (0.15)	0.53 (0.09)	0.173
Race/ethnicity stated on Wikipedia	0.22 (0.15)	0.19 (0.07)	0.835
<i>Panel C. Artist popularity, 2026 (YouGov)</i>			
% positive opinion	36.89 (5.50)	33.59 (2.33)	0.592
% negative opinion	13.11 (1.98)	13.75 (2.18)	0.829
Popularity skew, millennials – boomers	12.44 (2.71)	14.84 (1.68)	0.464
Popularity skew, women – men	4.00 (2.21)	3.62 (1.22)	0.884
<i>Panel D. Fan demographics: % with positive opinion among (YouGov, 2026)</i>			
Millennials	40.67 (5.02)	37.84 (2.18)	0.616
Gen X	33.00 (6.33)	28.84 (2.61)	0.556
Baby boomers	28.22 (6.98)	23.00 (2.88)	0.504
Men	35.11 (5.03)	31.75 (2.11)	0.551
Women	39.11 (6.11)	35.38 (2.66)	0.586
<i>N</i> artists	9	32	

Notes: Cells report group means with standard errors in parentheses, comparing the nine Musical.ly artists to the 32 placebo artists (13 superstar, 19 up-and-coming). Endorsements are coded from Wikipedia candidate-endorsement lists; artist demographics from English-language Wikipedia biographies; popularity and fan demographics from YouGov public-figure ratings (2026). *p*-values are from two-sided tests of equality of means. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table B8: Balance of Musically artists versus superstar and up-and-coming placebo artists

	Musically	Superstar	Up-and-coming	<i>p</i> (vs. Super.)	<i>p</i> (vs. Upc.)
<i>Panel A. Presidential-candidate endorsements, 2012–2020</i>					
Any endorsement	0.44 (0.18)	0.46 (0.14)	0.32 (0.11)	0.941	0.544
Any Democratic endorsement	0.44 (0.18)	0.46 (0.14)	0.32 (0.11)	0.941	0.544
Any Republican endorsement	0.00 (0.00)	0.08 (0.08)	0.00 (0.00)	0.337	–
# general-election endorsements	0.78 (0.32)	0.92 (0.33)	0.32 (0.11)	0.756	0.207
Any endorsement in 2012	0.33 (0.17)	0.31 (0.13)	0.05 (0.05)	0.906	0.141
Any endorsement in 2016	0.33 (0.17)	0.46 (0.14)	0.21 (0.10)	0.568	0.534
Any endorsement in 2020	0.11 (0.11)	0.15 (0.10)	0.05 (0.05)	0.782	0.643
Endorsed Obama 2012	0.33 (0.17)	0.31 (0.13)	0.05 (0.05)	0.906	0.141
Endorsed Clinton 2016	0.33 (0.17)	0.38 (0.14)	0.21 (0.10)	0.817	0.534
Endorsed Sanders 2016 (primary)	0.00 (0.00)	0.08 (0.08)	0.00 (0.00)	0.337	–
Endorsed Trump 2016	0.00 (0.00)	0.08 (0.08)	0.00 (0.00)	0.337	–
Endorsed Sanders 2020 (primary)	0.11 (0.11)	0.08 (0.08)	0.05 (0.05)	0.804	0.643
Endorsed Trump 2020	0.00 (0.00)	0.08 (0.08)	0.00 (0.00)	0.337	–
<i>Panel B. Artist demographics</i>					
log(age in 2015)	3.15 (0.06)	3.44 (0.08)	3.34 (0.05)	0.010**	0.026**
Male (share)	0.22 (0.15)	0.62 (0.14)	0.53 (0.12)	0.068*	0.124
US-born (share)	0.78 (0.15)	0.62 (0.14)	0.47 (0.12)	0.434	0.124
Race/ethnicity stated on Wikipedia	0.22 (0.15)	0.15 (0.10)	0.21 (0.10)	0.709	0.948
<i>Panel C. Artist popularity, 2026 (YouGov)</i>					
% positive opinion	36.89 (5.50)	41.46 (3.20)	28.21 (2.67)	0.485	0.181
% negative opinion	13.11 (1.98)	20.85 (3.98)	8.89 (1.81)	0.100*	0.131
Popularity skew, millennials – boomers	12.44 (2.71)	13.85 (3.33)	15.53 (1.76)	0.747	0.356
Popularity skew, women – men	4.00 (2.21)	3.00 (1.84)	4.05 (1.65)	0.732	0.985
<i>Panel D. Fan demographics: % with positive opinion among (YouGov, 2026)</i>					
Millennials	40.67 (5.02)	45.54 (2.56)	32.58 (2.65)	0.404	0.178
Gen X	33.00 (6.33)	37.85 (3.80)	22.68 (2.83)	0.523	0.164
Baby boomers	28.22 (6.98)	31.69 (4.56)	17.05 (3.13)	0.683	0.171
Men	35.11 (5.03)	40.00 (2.94)	26.11 (2.16)	0.416	0.128
Women	39.11 (6.11)	43.00 (3.67)	30.16 (3.26)	0.594	0.219
<i>N</i> artists	9	13	19		

Notes: Cells report group means with standard errors in parentheses, comparing the nine Musically artists to the 13 superstar and 19 up-and-coming placebo artists. The last two columns report *p*-values from two-sided tests of equality of means against the superstar and up-and-coming groups, respectively; “–” indicates an undefined test (both groups identically zero). Endorsements are coded from Wikipedia candidate-endorsement lists; artist demographics from English-language Wikipedia biographies; popularity and fan demographics from YouGov public-figure ratings (2026). **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

C Additional results

C.1 The first stage

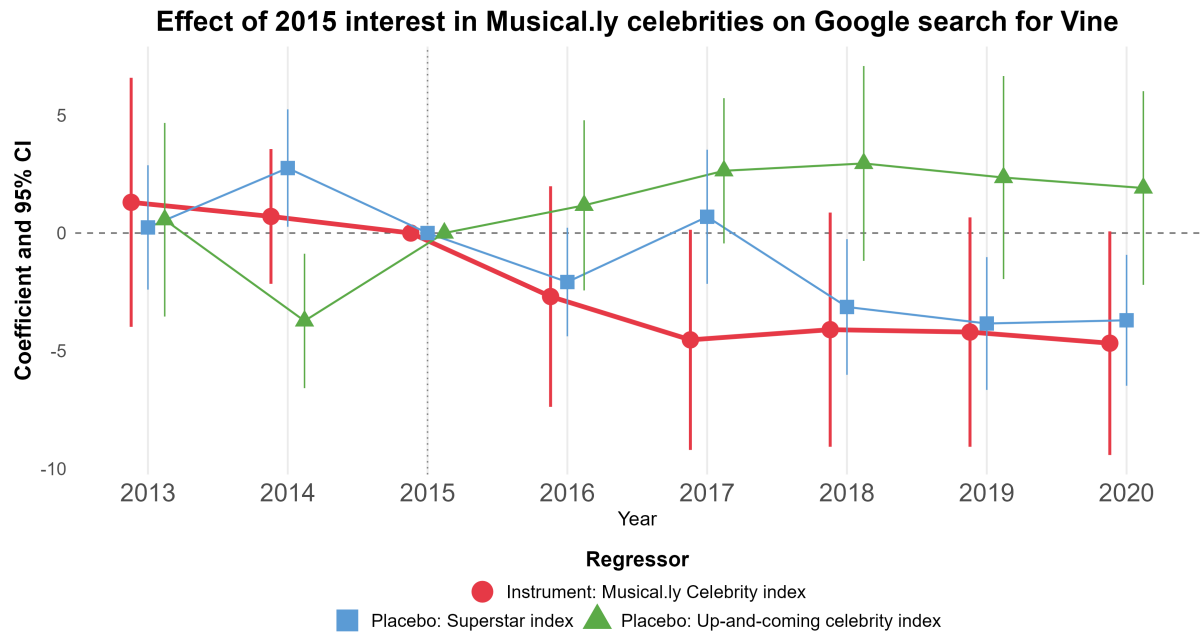


Figure C1: The Musical.ly Celebrity index and Google search interest in Vine

Note: This figure plots point estimates and 95% confidence intervals from a regression of DMA-by-year Google search interest in Vine on the three 2015 celebrity indices, each interacted with year dummies, with DMA and year fixed effects (205 DMAs). 2015 is the omitted reference year. Each year's DMA-level Google Trends series is normalized within year, so to make coefficients comparable across years each DMA's interest is rescaled by national annual search for Vine. Dots are point estimates and bars are 95% confidence intervals, based on standard errors clustered by DMA. The Musical.ly Celebrity index is in red, the superstar and up-and-coming placebo indices in blue and green.

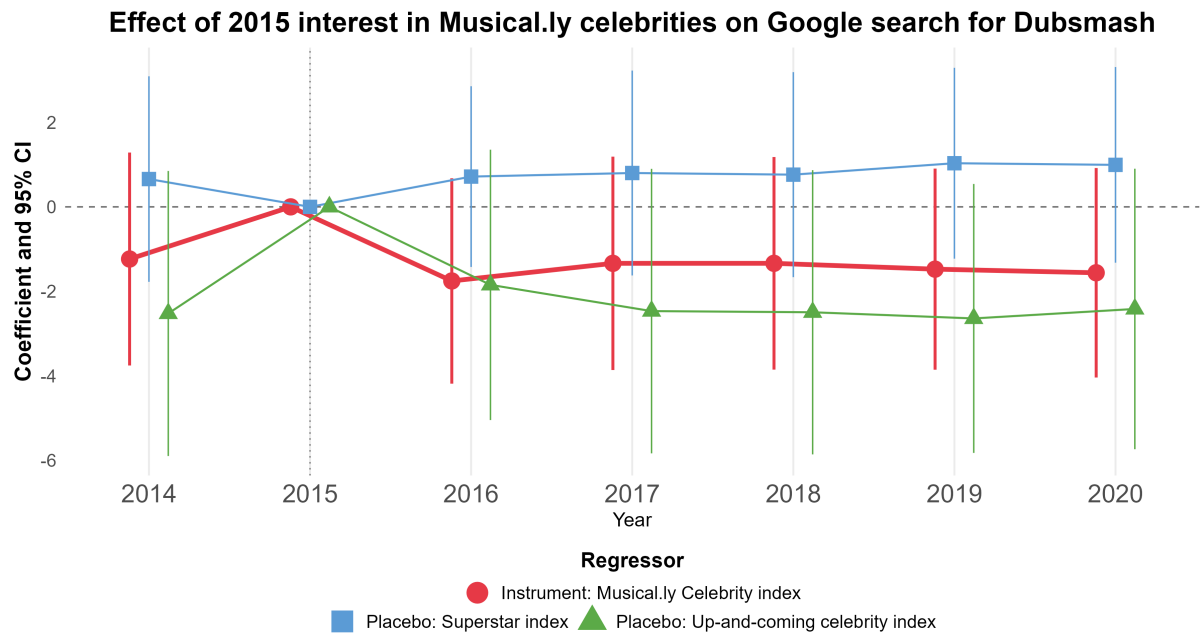


Figure C2: The Musical.ly Celebrity index and Google search interest in Dubsmash

Note: This figure plots point estimates and 95% confidence intervals from a regression of DMA-by-year Google search interest in Dubsmash on the three 2015 celebrity indices, each interacted with year dummies, with DMA and year fixed effects (205 DMAs). 2015 is the omitted reference year. Each year's DMA-level Google Trends series is normalized within year, so to make coefficients comparable across years each DMA's interest is rescaled by national annual search for Dubsmash. Dots are point estimates and bars are 95% confidence intervals, based on standard errors clustered by DMA. The Musical.ly Celebrity index is in red, the superstar and up-and-coming placebo indices in blue and green.

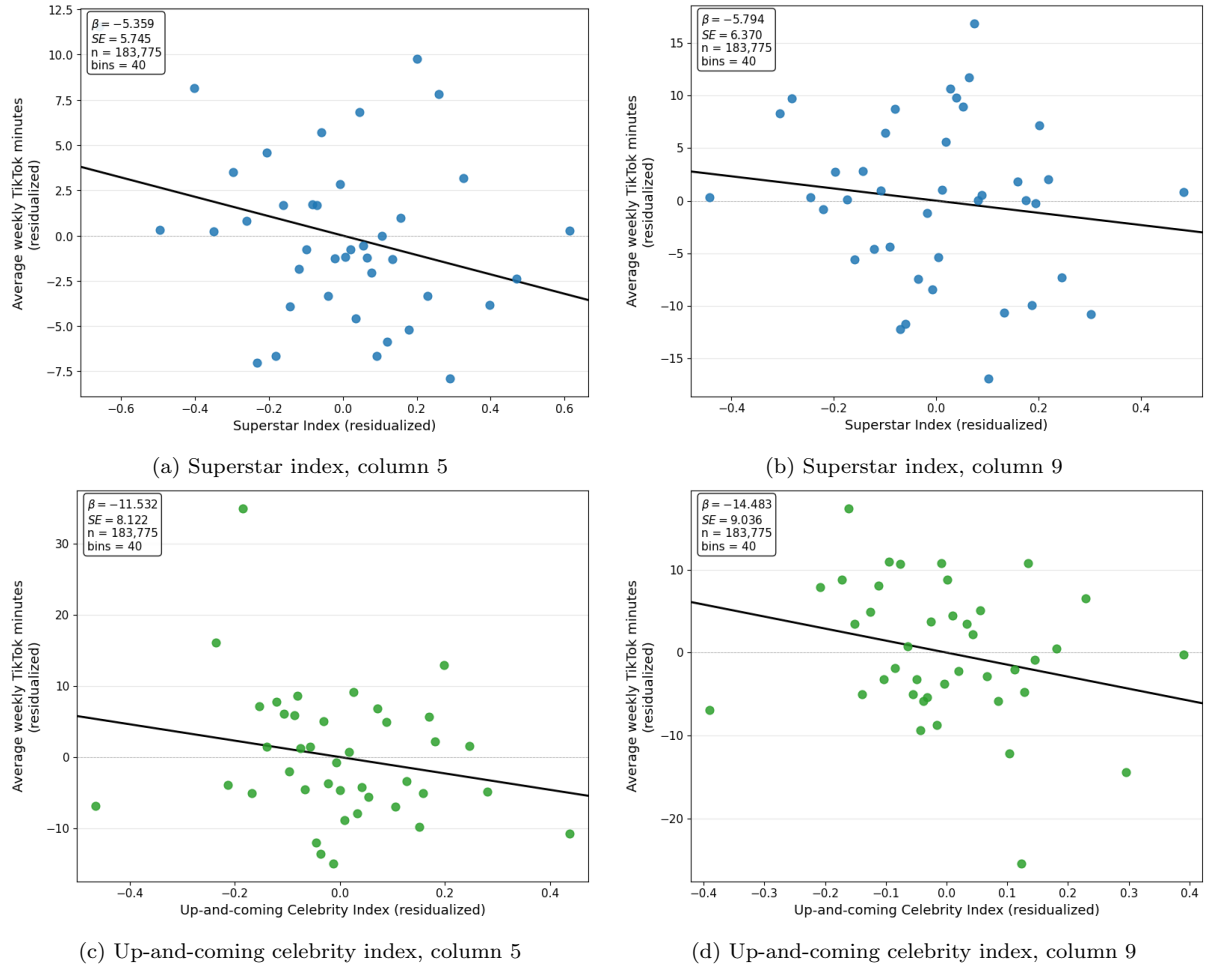


Figure C3: Binned residual scatter plots of average weekly TikTok minutes and the placebo celebrity indices

Note: This figure repeats Figure 5 for the two placebo indices, plotting residualized average weekly TikTok minutes against the residualized superstar index (top row) and up-and-coming celebrity index (bottom row), across device-years. The left column residualizes on the four DMA balance controls, namely the DMA shares of women, Hispanic residents, residents in poverty and high-school graduates, together with the individual controls; the right column residualizes on the full set of controls. *Individual controls* are the device user's gender, age in 5-year bins, device type (iPhone, iPad, or Android), indicators for Black and for Hispanic users, household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, an indicator for children in the household, and the race and ethnicity indicators interacted with the age bins. *DMA controls* are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates; indicators for deciles of DMA total population; the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older; the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races; and the DMA shares of adults with at least a high-school diploma, a bachelor's degree, and a master's degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. The data are binned into 40 equal-interval bins on the x-axis and dots represent average values within each bin. The line is an OLS fit on the unbinned data; its slope reproduces the corresponding coefficient.

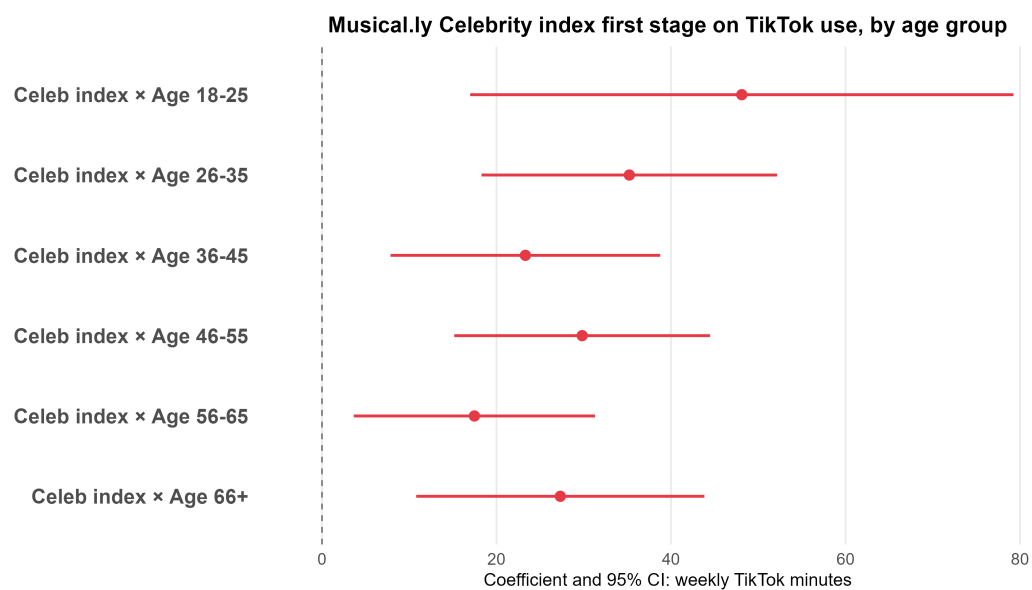


Figure C4: First stage by age group

Note: This figure plots point estimates and 95% confidence intervals from a single regression of average weekly TikTok minutes on the Musical.ly Celebrity index interacted with indicators for the device user's age group, with the placebo indices and the full set of controls entering analogously. Standard errors are clustered by DMA.

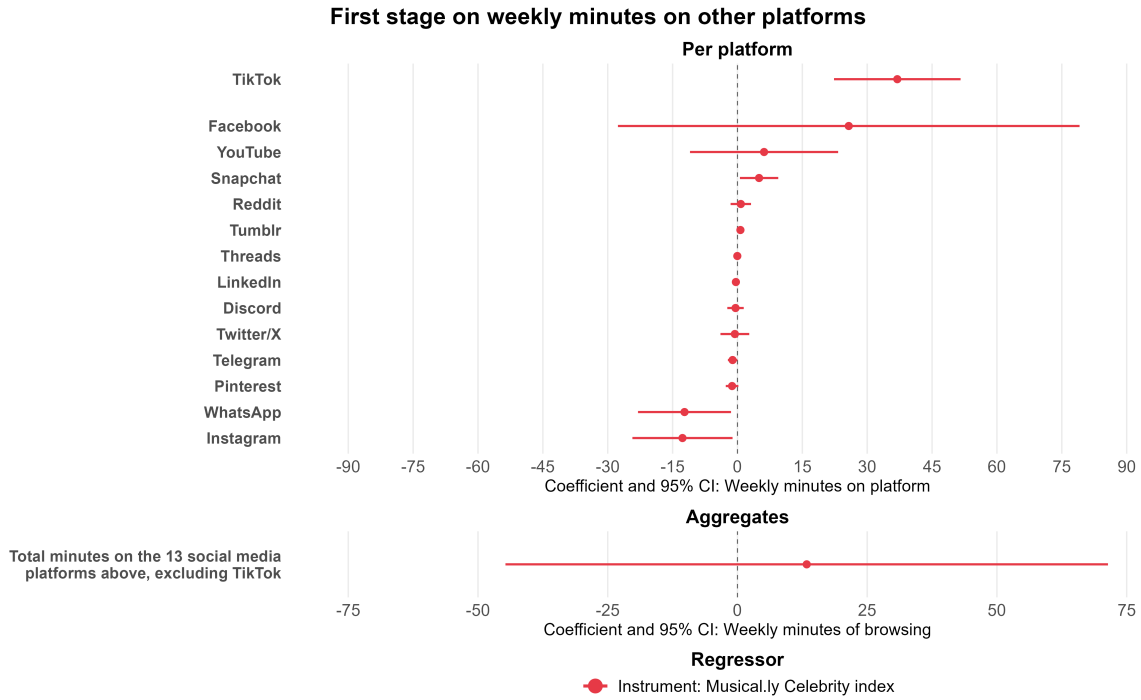


Figure C5: First stage on other social media platforms

Note: The unit of observation is a device. Each estimate comes from a separate regression of a device's average weekly minutes on the indicated platform, averaged over the weeks the device is observed, on the three 2015 celebrity indices; the figure plots the coefficient on the Musical.ly Celebrity index with 95% confidence intervals. All regressions include the full set of controls. *Individual controls* are the device user's gender, age in 5-year bins, device type (iPhone, iPad, or Android), indicators for Black and for Hispanic users, household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, an indicator for children in the household, and the race and ethnicity indicators interacted with the age bins. *DMA controls* are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates; indicators for deciles of DMA total population; the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older; the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races; and the DMA shares of adults with at least a high-school diploma, a bachelor's degree, and a master's degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. The bottom panel reports the same coefficient for total minutes across the thirteen platforms above, excluding TikTok. Standard errors are clustered by DMA.

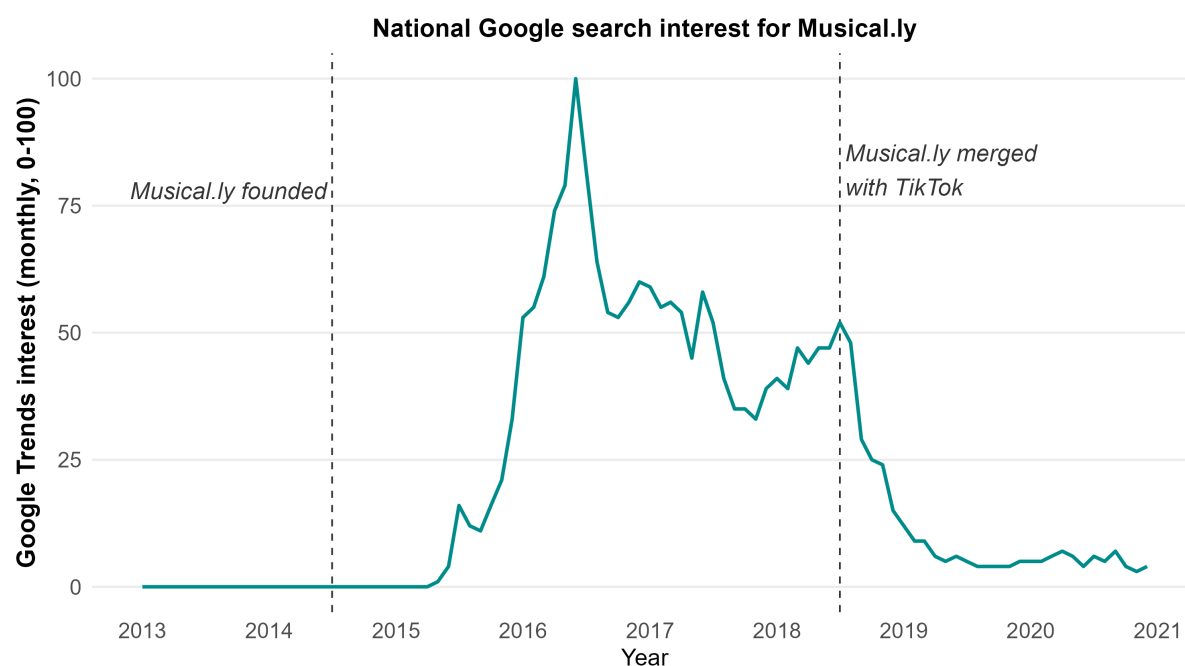


Figure C6: National Google search interest in Musical.ly, 2013–2020

Note: This figure plots monthly national Google Trends search interest in Musical.ly (0–100 scale) from 2013 to 2020. Interest is negligible through early 2015, rises sharply over 2015–2016, peaks in 2016, and declines after the 2018 merger with TikTok. This national series is used to rescale the within-year DMA-level interest underlying Figure 4.

Table C1: First stage: median weekly TikTok minutes and share of days with use

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A									
Dependent variable	Median weekly minutes on TikTok								
Musical.ly Celebrity index	29.483*** (7.559)	27.103*** (6.255)	27.468*** (6.204)	28.472*** (6.124)	28.942*** (6.139)	29.906*** (6.098)	27.872*** (6.228)	30.504*** (6.286)	35.271*** (6.982)
Superstar index	-1.505 (7.132)	-1.231 (5.644)	-2.548 (5.459)	-4.773 (5.349)	-4.989 (5.301)	-1.143 (5.244)	-0.776 (5.415)	-3.451 (5.730)	-5.198 (5.864)
Up-and-coming celebrity index	-4.884 (10.689)	-8.585 (8.069)	-10.677 (7.966)	-10.538 (7.766)	-10.005 (7.694)	-14.894* (7.942)	-12.912 (8.284)	-12.785 (8.153)	-12.404 (8.780)
Mean dependent variable	105.38	105.38	105.38	105.38	105.38	105.38	105.38	105.38	105.38
R ²	0.00	0.08	0.08	0.08	0.08	0.08	0.08	0.09	0.09
Cluster-robust F-statistic	15.21	18.78	19.60	21.61	22.23	24.05	20.03	23.55	25.52
Panel B									
Dependent variable	Share of days with TikTok use								
Musical.ly Celebrity index	0.023*** (0.008)	0.020*** (0.006)	0.022*** (0.006)	0.022*** (0.006)	0.023*** (0.006)	0.024*** (0.006)	0.022*** (0.006)	0.024*** (0.006)	0.024*** (0.006)
Superstar index	0.002 (0.007)	0.002 (0.006)	-0.002 (0.005)	-0.003 (0.005)	-0.003 (0.005)	0.002 (0.005)	0.001 (0.005)	-0.000 (0.005)	-0.000 (0.005)
Up-and-coming celebrity index	-0.008 (0.010)	-0.010 (0.008)	-0.011 (0.007)	-0.011 (0.007)	-0.011 (0.007)	-0.018** (0.008)	-0.015** (0.007)	-0.016** (0.007)	-0.017** (0.007)
Mean dependent variable	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15
R ²	0.00	0.12	0.13	0.13	0.14	0.14	0.14	0.14	0.14
Cluster-robust F-statistic	8.91	10.80	13.01	13.81	14.17	15.27	14.12	15.55	15.58
Number of devices	183,775	183,775	183,775	183,775	183,775	183,775	183,775	183,775	183,775
Number of DMAs	205	205	205	205	205	205	205	205	205
DMA controls for balance	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Gender, 5y age buckets, device type		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Race and ethnicity			Yes	Yes	Yes	Yes	Yes	Yes	Yes
HH income, size, children				Yes	Yes	Yes	Yes	Yes	Yes
Race and ethnicity specific age buckets					Yes	Yes	Yes	Yes	Yes
DMA population deciles						Yes	Yes	Yes	Yes
DMA age composition							Yes	Yes	Yes
DMA race composition								Yes	Yes
DMA socio-economic									Yes

Notes: The unit of observation is a device, and both panels use the same estimation sample. In Panel A the dependent variable is the median of a device's weekly TikTok minutes across the weeks it is observed; in Panel B it is the share of observed days on which the device registers any TikTok use. Each column regresses the dependent variable on the three celebrity indices. Standard errors clustered by DMA. The DMA balance controls, included in every column, are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates. The remaining controls are added cumulatively across columns and are defined as follows. *Gender, 5y age buckets, device type* is the device user's gender, age in 5-year bins, and device type (iPhone, iPad, or Android). *Race and ethnicity* is an indicator for Black users and an indicator for Hispanic users. *HH income, size, children* is household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, and an indicator for children in the household. *Race and ethnicity specific age buckets* interacts the race and ethnicity indicators with the age bins. *DMA population deciles* is a set of indicators for deciles of DMA total population. *DMA age composition* is the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older. *DMA race composition* is the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races. *DMA socio-economic* is the DMA shares of adults with at least a high-school diploma, a bachelor's degree, and a master's degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C2: First stage: TikTok use above thresholds of the use distribution

	(1)	(2)	(3)	(4)	(5)
Dependent variable	Dummy = 1 if average weekly TikTok use above:				
Threshold	> 0	>20th percentile	>40th percentile	>60th percentile	>80th percentile
Musical.ly Celebrity index	0.028*** (0.010)	0.027** (0.011)	0.026*** (0.010)	0.028*** (0.008)	0.029*** (0.006)
Superstar index	0.002 (0.009)	-0.001 (0.009)	-0.004 (0.008)	0.001 (0.008)	-0.005 (0.006)
Up-and-coming celebrity index	-0.015 (0.010)	-0.024** (0.012)	-0.025** (0.011)	-0.034*** (0.009)	-0.018** (0.008)
Mean dependent variable	0.654	0.523	0.392	0.262	0.131
Number of devices	183,775	183,775	183,775	183,775	183,775
Number of DMAs	205	205	205	205	205
R ²	0.06	0.10	0.10	0.10	0.10
Cluster-robust F-statistic	7.73	6.53	7.30	10.68	20.80
Individual controls	Yes	Yes	Yes	Yes	Yes
DMA controls	Yes	Yes	Yes	Yes	Yes

Notes: The unit of observation is a device. The dependent variable in each column is an indicator equal to one if the device's average weekly TikTok use, averaged over the weeks it is observed, exceeds the stated threshold; percentiles are defined among devices with positive use. Column (1) is therefore the extensive margin. Each column regresses the dependent variable on the three celebrity indices. Standard errors clustered by DMA. All columns include the full set of controls. *Individual controls* are the device user's gender, age in 5-year bins, device type (iPhone, iPad, or Android), indicators for Black and for Hispanic users, household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, an indicator for children in the household, and the race and ethnicity indicators interacted with the age bins. *DMA controls* are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates; indicators for deciles of DMA total population; the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older; the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races; and the DMA shares of adults with at least a high-school diploma, a bachelor's degree, and a master's degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C3: First stage on other platforms and aggregate social media use

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Dependent variable	Average weekly minutes on:													
Platform	Facebook	Instagram	Twitter/X	Snapchat	YouTube	Reddit	Pinterest	LinkedIn	Discord	Threads	Tumblr	WhatsApp	Telegram	Social media excl. TikTok
Musical.ly Celebrity index	25.735 (27.232)	-12.715** (5.908)	-0.603 (1.698)	4.999** (2.262)	6.153 (8.739)	0.776 (1.206)	-1.246* (0.744)	-0.364** (0.180)	-0.449 (0.969)	-0.036 (0.096)	0.684 (0.463)	-12.244** (5.487)	-1.127** (0.556)	13.346 (29.620)
Superstar index	-70.184*** (21.990)	28.985*** (5.828)	2.555* (1.319)	-2.798* (1.654)	2.876 (6.466)	0.080 (0.740)	0.149 (0.543)	0.535*** (0.175)	-1.426** (0.702)	0.205** (0.081)	-0.141 (0.356)	10.772** (5.237)	2.108*** (0.368)	-29.936 (21.610)
Up-and-coming celebrity index	3.236 (28.038)	1.923 (6.530)	-1.469 (2.026)	-2.617 (2.111)	9.101 (7.856)	-0.196 (1.063)	-0.995 (0.674)	0.162 (0.162)	2.543*** (0.882)	-0.089 (0.095)	-0.058 (0.453)	19.036** (7.783)	-0.292 (0.550)	28.546 (28.677)
Mean dependent variable	766.49	73.99	16.44	21.46	331.16	8.14	7.89	1.48	6.22	0.45	1.74	24.14	2.73	1253.57
Number of devices	182,501	182,501	182,501	182,501	182,501	182,501	182,501	182,501	182,501	182,501	182,501	182,501	182,501	183,775
Number of DMAs	205	205	205	205	205	205	205	205	205	205	205	205	205	205
R ²	0.15	0.08	0.01	0.10	0.05	0.01	0.03	0.01	0.02	0.00	0.00	0.03	0.00	0.10
Cluster-robust F-statistic	0.89	4.63	0.13	4.89	0.50	0.41	2.81	4.11	0.21	0.14	2.19	4.98	4.12	0.20
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
DMA controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The unit of observation is a device. Each column regresses a device's average weekly minutes on the indicated platform, averaged over the weeks the device is observed, on the three celebrity indices. Column (14) is the sum across the thirteen platforms in columns (1)–(13). Standard errors clustered by DMA. All columns include the full set of controls. *Individual controls* are the device user's gender, age in 5-year bins, device type (iPhone, iPad, or Android), indicators for Black and for Hispanic users, household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, an indicator for children in the household, and the race and ethnicity indicators interacted with the age bins. *DMA controls* are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates; indicators for deciles of DMA total population; the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older; the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races; and the DMA shares of adults with at least a high-school diploma, a bachelor's degree, and a master's degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Mobile news consumption

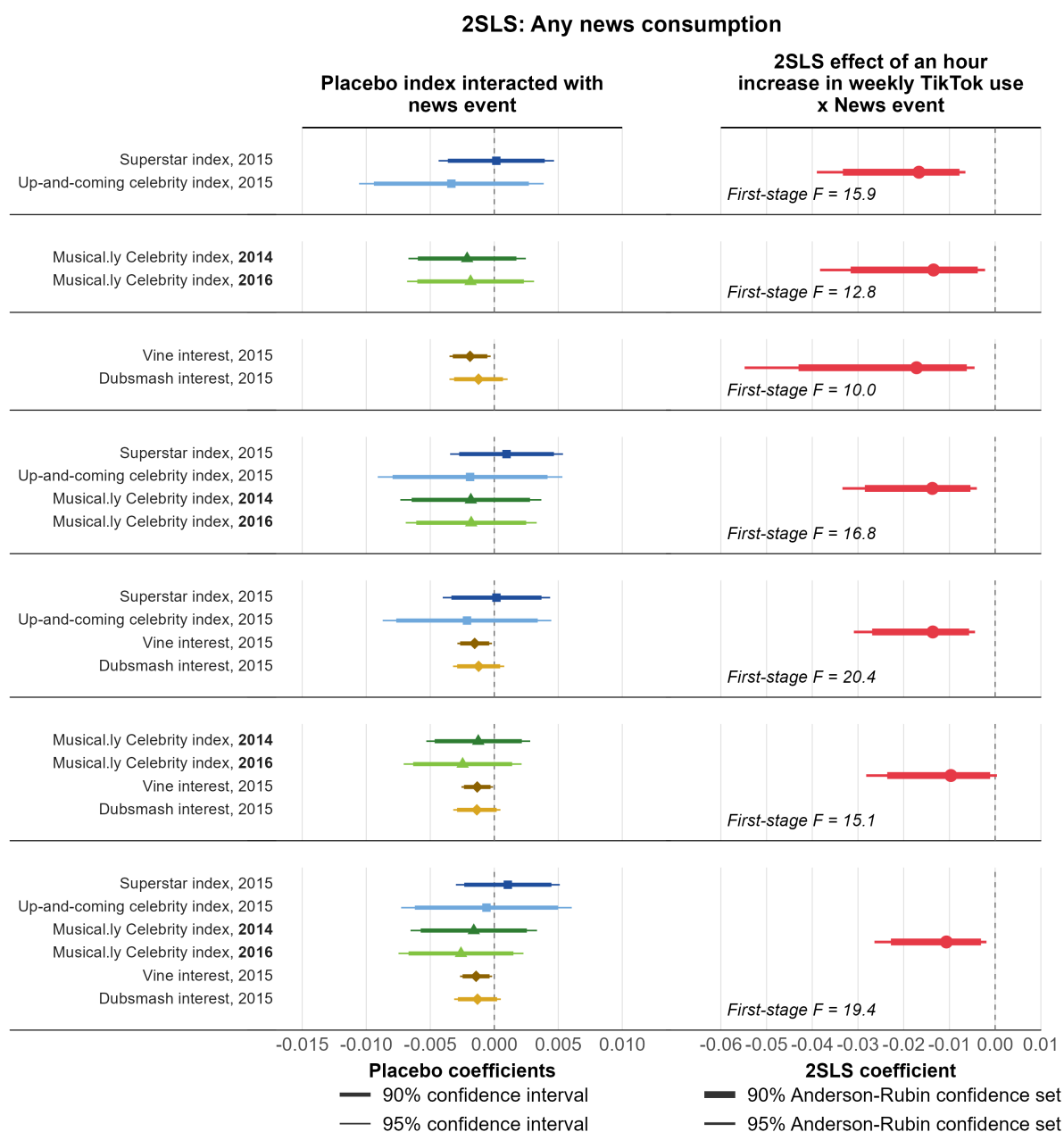


Figure C7: Alternative placebo sets: extensive margin

Note: The unit of observation is a device-week. The upper panel plots two-stage least squares estimates of the effect of an additional weekly hour of TikTok use interacted with the news event indicator, with 90% and 95% Anderson–Rubin confidence sets, which are robust to weak identification. The dependent variable is an indicator for any news consumption in the week. The lower panel plots the coefficients on the placebo indices included in the same specification, with 90% and 95% confidence intervals. Each position on the horizontal axis is a separate specification, defined by which placebo indices enter: the 2015 superstar and up-and-coming celebrity indices, the Musical.ly Celebrity index measured in 2014 and in 2016, and 2015 search interest in Vine and Dubsmash, alone and in combination. The dot matrix beneath records which indices enter each specification, and the cluster-robust first-stage F-statistic is reported below it. All specifications include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator, as reported in Table C13. Standard errors are clustered by DMA.

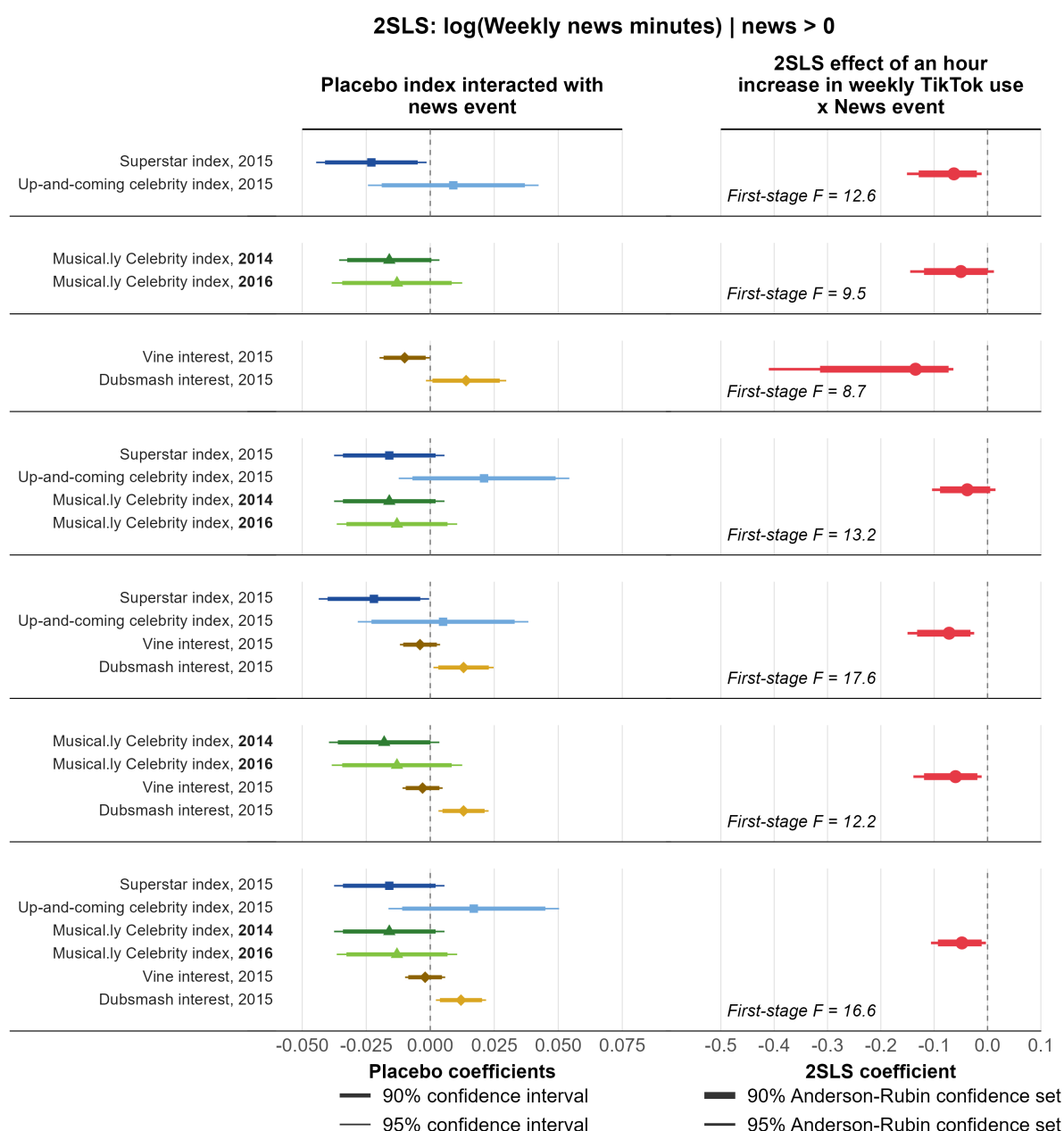


Figure C8: Alternative placebo sets: intensive margin

Note: The unit of observation is a device-week. The upper panel plots two-stage least squares estimates of the effect of an additional weekly hour of TikTok use interacted with the news event indicator, with 90% and 95% Anderson-Rubin confidence sets, which are robust to weak identification. The dependent variable is the logarithm of weekly news minutes among device-weeks with positive news consumption. The lower panel plots the coefficients on the placebo indices included in the same specification, with 90% and 95% confidence intervals. Each position on the horizontal axis is a separate specification, defined by which placebo indices enter: the 2015 superstar and up-and-coming celebrity indices, the Musical.ly Celebrity index measured in 2014 and in 2016, and 2015 search interest in Vine and Dubsmash, alone and in combination. The dot matrix beneath records which indices enter each specification, and the cluster-robust first-stage F-statistic is reported below it. All specifications include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator, as reported in Table C13. Standard errors are clustered by DMA.

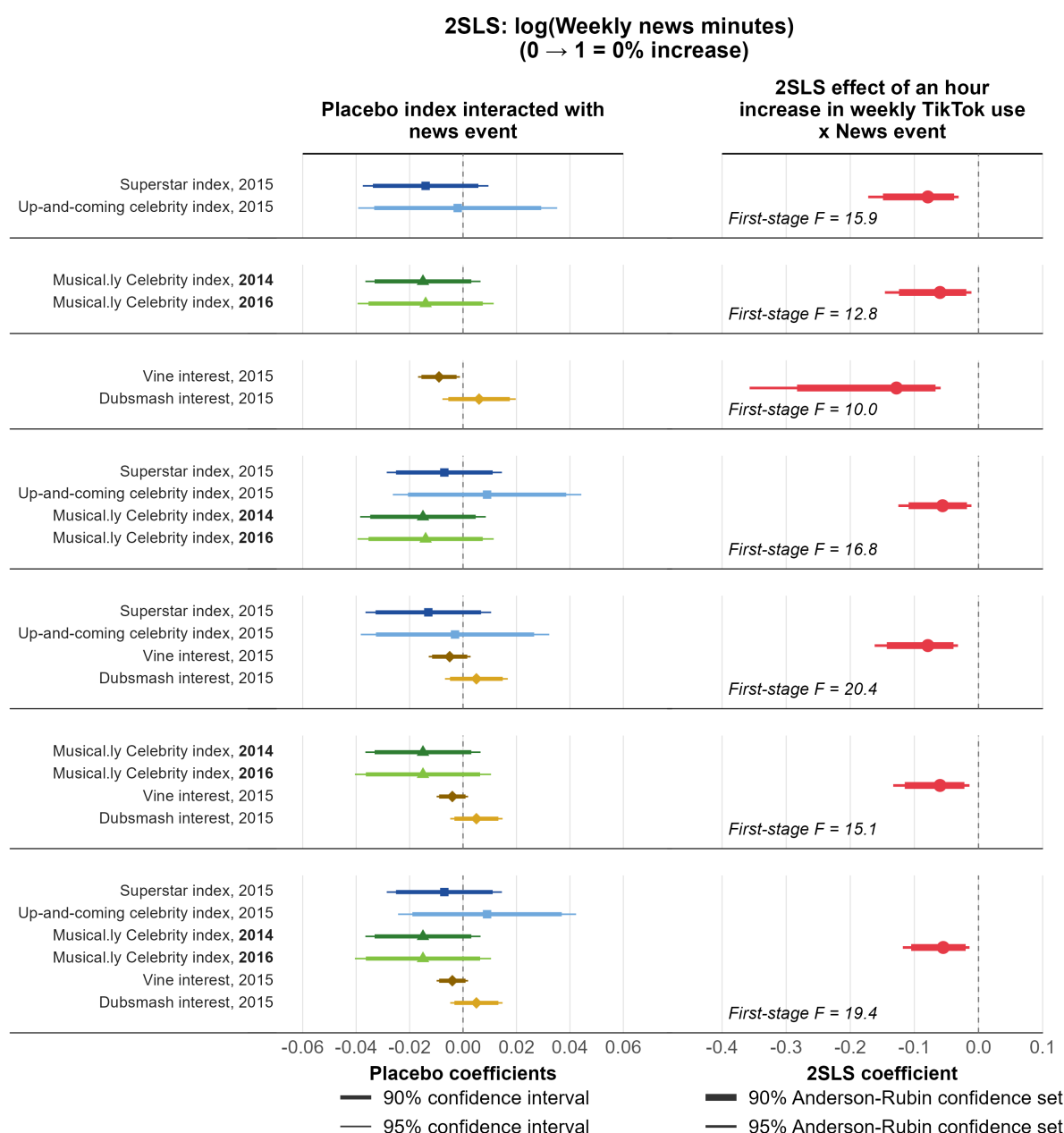


Figure C9: Alternative placebo sets: combined margin, extensive margin valued at zero

Note: The unit of observation is a device-week. The upper panel plots two-stage least squares estimates of the effect of an additional weekly hour of TikTok use interacted with the news event indicator, with 90% and 95% Anderson-Rubin confidence sets, which are robust to weak identification. The dependent variable is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued at zero. The lower panel plots the coefficients on the placebo indices included in the same specification, with 90% and 95% confidence intervals. Each position on the horizontal axis is a separate specification, defined by which placebo indices enter: the 2015 superstar and up-and-coming celebrity indices, the Musical.ly Celebrity index measured in 2014 and in 2016, and 2015 search interest in Vine and Dubsmash, alone and in combination. The dot matrix beneath records which indices enter each specification, and the cluster-robust first-stage F-statistic is reported below it. All specifications include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator, as reported in Table C13. Standard errors are clustered by DMA.

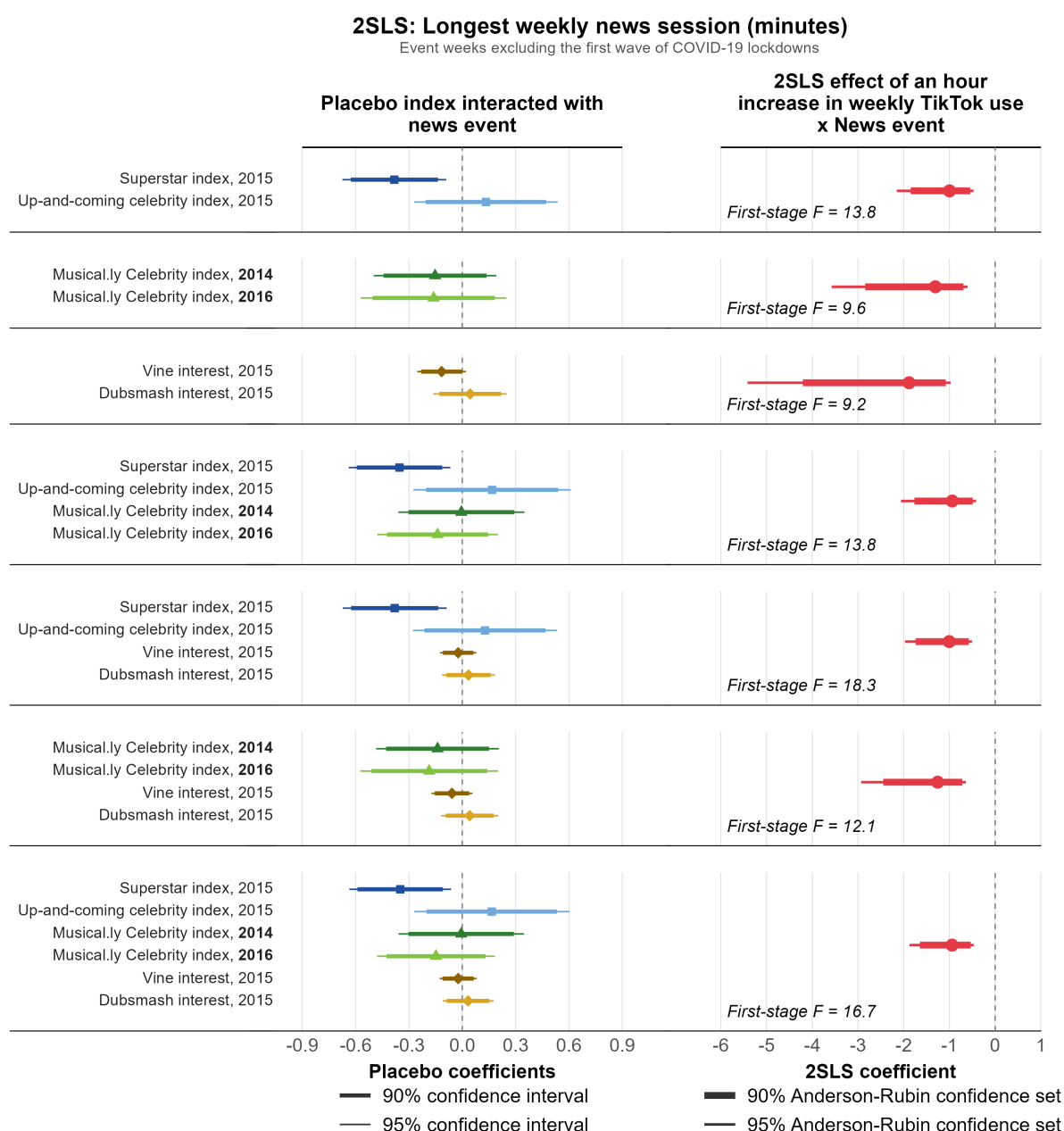


Figure C10: Alternative placebo sets: length of the longest news session

Note: The unit of observation is a device-week, and the sample is device-weeks with positive news consumption. The upper panel plots two-stage least squares estimates of the effect of an additional weekly hour of TikTok use interacted with the news event indicator, with 90% and 95% Anderson–Rubin confidence sets, which are robust to weak identification. The dependent variable is the length of the longest news session within the device-week, in minutes. Event weeks exclude the first wave of COVID-19 stay-at-home orders, as in Panel B of Table 4. The lower panel plots the coefficients on the placebo indices included in the same specification, with 90% and 95% confidence intervals. Each position on the horizontal axis is a separate specification, defined by which placebo indices enter: the 2015 superstar and up-and-coming celebrity indices, the Musical.ly Celebrity index measured in 2014 and in 2016, and 2015 search interest in Vine and Dubsmash, alone and in combination. The dot matrix beneath records which indices enter each specification, and the cluster-robust first-stage F-statistic is reported below it. All specifications include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator, as reported in the notes to Table 4. Standard errors are clustered by DMA.

Table C4: Cross-sectional 2SLS: adding controls sequentially

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A									
Dependent variable	Any news consumption, all devices								
Avg. weekly TikTok hours	0.011 (0.009)	0.011 (0.009)	0.009 (0.008)	0.007 (0.008)	0.007 (0.008)	0.006 (0.008)	0.010 (0.009)	0.007 (0.007)	0.006 (0.008)
90% AR CI	[-0.003, 0.031]	[-0.004, 0.027]	[-0.006, 0.024]	[-0.007, 0.021]	[-0.007, 0.021]	[-0.008, 0.019]	[-0.004, 0.026]	[-0.006, 0.020]	[-0.007, 0.019]
95% AR CI	[-0.005, 0.037]	[-0.007, 0.031]	[-0.009, 0.028]	[-0.010, 0.025]	[-0.010, 0.024]	[-0.011, 0.023]	[-0.008, 0.030]	[-0.009, 0.023]	[-0.010, 0.022]
Superstar index	-0.018*** (0.004)	-0.016*** (0.004)	-0.015*** (0.004)	-0.013*** (0.004)	-0.013*** (0.004)	-0.012*** (0.004)	-0.014*** (0.005)	-0.015*** (0.005)	-0.013*** (0.004)
Up-and-coming celebrity index	-0.005 (0.007)	-0.001 (0.006)	-0.002 (0.006)	-0.002 (0.006)	-0.003 (0.006)	-0.004 (0.006)	-0.003 (0.006)	0.000 (0.005)	-0.000 (0.005)
First-stage F-statistic	14.71	17.69	18.97	20.33	20.85	22.70	19.34	22.57	24.51
Panel B									
Dependent variable	log(Weekly news minutes), devices with news > 0								
Avg. weekly TikTok hours	0.053 (0.103)	0.086 (0.097)	0.070 (0.095)	0.090 (0.093)	0.087 (0.091)	0.090 (0.087)	0.131 (0.098)	0.105 (0.086)	0.088 (0.077)
90% AR CI	[-0.109, 0.273]	[-0.073, 0.276]	[-0.089, 0.248]	[-0.062, 0.266]	[-0.064, 0.258]	[-0.053, 0.252]	[-0.026, 0.331]	[-0.035, 0.272]	[-0.041, 0.230]
95% AR CI	[-0.142, 0.340]	[-0.107, 0.326]	[-0.125, 0.292]	[-0.095, 0.310]	[-0.095, 0.301]	[-0.083, 0.292]	[-0.059, 0.387]	[-0.064, 0.315]	[-0.068, 0.265]
Superstar index	-0.070* (0.037)	-0.079*** (0.030)	-0.063* (0.032)	-0.074** (0.031)	-0.072** (0.031)	-0.057 (0.036)	-0.068* (0.040)	-0.080** (0.032)	-0.082** (0.033)
Up-and-coming celebrity index	-0.007 (0.059)	0.046 (0.048)	0.045 (0.047)	0.053 (0.046)	0.051 (0.046)	0.031 (0.051)	0.021 (0.048)	0.055 (0.041)	0.049 (0.042)
First-stage F-statistic	14.29	17.68	18.88	20.25	20.91	21.63	15.62	19.34	22.40
Panel C									
Dependent variable	log(Weekly news minutes), all devices (0 → 1 = 100% increase)								
Avg. weekly TikTok hours	0.058 (0.114)	0.088 (0.105)	0.068 (0.103)	0.084 (0.101)	0.081 (0.099)	0.076 (0.094)	0.119 (0.097)	0.094 (0.083)	0.084 (0.077)
90% AR CI	[-0.120, 0.299]	[-0.087, 0.288]	[-0.110, 0.257]	[-0.085, 0.270]	[-0.086, 0.264]	[-0.081, 0.248]	[-0.040, 0.304]	[-0.045, 0.248]	[-0.046, 0.222]
95% AR CI	[-0.157, 0.371]	[-0.125, 0.341]	[-0.150, 0.303]	[-0.122, 0.316]	[-0.123, 0.309]	[-0.115, 0.289]	[-0.073, 0.352]	[-0.074, 0.285]	[-0.074, 0.254]
Superstar index	-0.120*** (0.046)	-0.117*** (0.036)	-0.097** (0.039)	-0.105*** (0.037)	-0.103*** (0.037)	-0.083* (0.042)	-0.093** (0.044)	-0.112*** (0.035)	-0.110*** (0.036)
Up-and-coming celebrity index	-0.018 (0.074)	0.041 (0.058)	0.036 (0.057)	0.043 (0.056)	0.042 (0.056)	0.015 (0.061)	0.005 (0.057)	0.051 (0.047)	0.047 (0.047)
First-stage F-statistic	14.71	17.69	18.97	20.33	20.85	22.70	19.34	22.57	24.51
Panel D									
Dependent variable	log(Weekly total mobile minutes), all devices								
Avg. weekly TikTok hours	0.207*** (0.068)	0.103** (0.045)	0.109** (0.044)	0.096** (0.041)	0.094** (0.040)	0.098*** (0.037)	0.105** (0.043)	0.078** (0.037)	0.055* (0.031)
90% AR CI	[0.112, 0.367]	[0.029, 0.189]	[0.037, 0.193]	[0.027, 0.172]	[0.027, 0.168]	[0.038, 0.168]	[0.039, 0.191]	[0.016, 0.147]	[0.001, 0.111]
95% AR CI	[0.095, 0.421]	[0.013, 0.211]	[0.022, 0.215]	[0.012, 0.191]	[0.012, 0.186]	[0.025, 0.186]	[0.025, 0.215]	[0.003, 0.164]	[-0.010, 0.123]
Superstar index	-0.078** (0.032)	-0.040** (0.016)	-0.044*** (0.016)	-0.035** (0.015)	-0.034** (0.015)	-0.032* (0.016)	-0.037** (0.019)	-0.028 (0.019)	-0.021 (0.016)
Up-and-coming celebrity index	-0.053 (0.058)	-0.026 (0.030)	-0.018 (0.028)	-0.023 (0.027)	-0.024 (0.027)	-0.037 (0.028)	-0.034 (0.028)	-0.036 (0.026)	-0.041* (0.023)
First-stage F-statistic	14.72	17.74	19.04	20.39	20.93	22.75	19.39	22.63	24.59
DMA controls for balance	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Gender, 5y age buckets, device type		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Race and ethnicity			Yes	Yes	Yes	Yes	Yes	Yes	Yes
HH income, size, children				Yes	Yes	Yes	Yes	Yes	Yes
Race and ethnicity specific age buckets					Yes	Yes	Yes	Yes	Yes
DMA population deciles						Yes	Yes	Yes	Yes
DMA age composition							Yes	Yes	Yes
DMA race composition								Yes	Yes
DMA socio-economic									Yes

Notes: The unit of observation is a device, and there are 205 media markets in every regression. Each cell reports the two-stage least squares estimate of equation (2) for the dependent variable stated in the panel heading. Average weekly TikTok hours is instrumented with the Musical.ly Celebrity index; the superstar and up-and-coming placebo indices enter as exogenous controls and their coefficients are reported. In Panel A the dependent variable is an indicator equal to one if the device consumes any news, with a mean of 0.943 across 183,775 devices. In Panel B it is the logarithm of average weekly news minutes among the 173,273 devices with positive news consumption, with a mean of 2.582. In Panel C it is average weekly news minutes over all 183,775 devices, transformed as $\log(x)$ for $x \geq 1$ and -1 for $x < 1$, so that devices with little or no news consumption are retained and the move from below one minute to one minute is valued as a doubling; its mean is 2.409. In Panel D it is the logarithm of average weekly total mobile minutes, with a mean of 7.687 across 183,772 devices. All quantities are averaged over the weeks the device is observed. Anderson-Rubin confidence sets are robust to weak identification. The DMA balance controls, included in every column, are the DMA shares of women, Hispanic residents, residents in poverty, and high-school graduates. The remaining controls are added cumulatively across columns and are defined as follows. *Gender, 5y age buckets, device type* is the device user's gender, age in 5-year bins, and device type (iPhone, iPad, or Android). *Race and ethnicity* is an indicator for Black users and an indicator for Hispanic users. *HH income, size, children* is household-income buckets, indicators for household size of 1, 2, 3, 4, and 5+, and an indicator for children in the household. *Race and ethnicity specific age buckets* interacts the race and ethnicity indicators with the age bins. *DMA population deciles* is a set of indicators for deciles of DMA total population. *DMA age composition* is the DMA population shares aged 0–19, 20–39, 40–59 and 60 or older. *DMA race composition* is the DMA population shares that are white, Black, Asian, Native American, Pacific Islander and of two or more races. *DMA socio-economic* is the DMA shares of adults with at least a high-school diploma, a bachelor's degree, and a master's degree, log median household income, the unemployment rate and the Gini index. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C5: News consumption during news event weeks, valuing the extensive margin at zero

	(1)	(2)	(3)
Specification	OLS	Reduced form	2SLS
Dependent variable	log(Weekly news minutes) (0 → 1 = 0% increase)		
Avg. weekly TikTok hours × News event	0.003*** (0.000)		-0.079*** (0.030)
90% AR CI			[-0.149, -0.038]
95% AR CI			[-0.172, -0.031]
Musical.ly Celebrity index × News event		-0.038*** (0.012)	
Superstar index × News event	-0.016 (0.011)	-0.003 (0.011)	-0.014 (0.012)
Up-and-coming celebrity index × News event	0.006 (0.017)	0.012 (0.016)	-0.002 (0.019)
Mean dependent variable	2.34	2.34	2.34
Observations	5,660,705	5,660,705	5,660,705
Number of devices	176,321	176,321	176,321
Number of weeks	235	235	235
Number of DMAs	205	205	205
R ²	0.665	0.665	
Within-R ²	0.0002	0.0001	
Cluster-robust F-statistic			15.95
Week & device FEs	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes

Notes: The unit of observation is a device-week. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. The dependent variable is the logarithm of weekly news minutes, with weeks of less than one minute coded as zero and the move from zero to one minute valued at zero rather than as a doubling; column (3) of Table 3 values the same move as a doubling. Column (1) reports ordinary least squares estimates. Column (2) reports the reduced form, in which the Musical.ly Celebrity index interacted with the event indicator enters the outcome equation directly in place of TikTok use. Column (3) instruments average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musical.ly Celebrity index interacted with the same indicator; the corresponding first stage is column (1) of Table C6. Anderson–Rubin confidence sets are robust to weak identification. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C6: First stage for news event weeks

	(1)	(2)
Dependent variable	Avg. weekly TikTok hours $\times$ News event	
Sample	All device-weeks	News > 0
Musical.ly Celebrity index $\times$ News event	0.484*** (0.121)	0.468*** (0.132)
Superstar index $\times$ News event	-0.134* (0.080)	-0.138 (0.085)
Up-and-coming celebrity index $\times$ News event	-0.177 (0.120)	-0.177 (0.123)
Mean dependent variable	0.28	0.27
Observations	5,660,705	3,852,220
Number of devices	176,321	160,833
Number of weeks	235	235
Number of DMAs	205	205
R ²	0.256	0.261
Within-R ²	0.0658	0.0534
Cluster-robust F-statistic	15.95	12.64
Week & device FEs	Yes	Yes
Controls $\times$ News event	Yes	Yes

Notes: The unit of observation is a device-week. This table reports equation (5), the first stage underlying Panel C of Table 3. The dependent variable is average weekly TikTok hours in the four weeks before the event, interacted with the event indicator. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. Column (1) uses all device-weeks and is the first stage for columns (1) and (3) of Table 3; column (2) restricts to device-weeks with positive news consumption and is the first stage for column (2). Both columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user's gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor's and a master's degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C7: Selection on mobile use: conditioning on total mobile time and the news share

	(1)	(2)
Dependent variable	log(Weekly news minutes) (0 → 1 = 100% increase)	News share of mobile time (news min. / total min.)
Panel A: 2SLS		
Avg. weekly TikTok hours × News event	-0.116*** (0.038)	-0.0020* (0.0011)
90% AR CI	[-0.209, -0.065]	[-0.0047, -0.0004]
95% AR CI	[-0.241, -0.056]	[-0.0054, -0.0001]
Total weekly mobile hours × News event	0.012*** (0.001)	
Superstar index × News event	-0.012 (0.014)	-0.0003 (0.0005)
Up-and-coming celebrity index × News event	-0.001 (0.022)	0.0003 (0.0007)
Cluster-robust F-statistic	15.32	16.82
Panel B: OLS		
Avg. weekly TikTok hours × News event	-0.0046*** (0.0003)	-0.00011*** (0.00001)
Total weekly mobile hours × News event	0.0089*** (0.0001)	
R ²	0.652	0.596
Within-R ²	0.0070	0.0001
Panel C: First stage		
Dependent variable	Avg. weekly TikTok hours × News event	
Musical.ly Celebrity index × News event	0.466*** (0.119)	0.507*** (0.124)
Total weekly mobile hours × News event	0.024*** (0.001)	
Superstar index × News event	-0.122 (0.081)	-0.143* (0.080)
Up-and-coming celebrity index × News event	-0.165 (0.120)	-0.183 (0.117)
Cluster-robust F-statistic	15.32	16.82
R ²	0.272	0.256
Within-R ²	0.0869	0.0659
Mean dependent variable	1.99	0.04
Observations	5,660,705	5,656,737
Number of devices	176,321	176,300
Week & device FEs, controls × News event	Yes	Yes

Notes: The unit of observation is a device-week, 235 weeks and 205 media markets. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. Column (1) repeats column (3) of Table 3, adding total weekly mobile hours in the four weeks before the event, interacted with the event indicator, as a control. Column (2) instead replaces the outcome with news minutes as a share of total mobile minutes in the week, and so does not condition on total mobile time separately. Panel A reports two-stage least squares estimates instrumenting average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musical.ly Celebrity index interacted with the same indicator; Panel B reports ordinary least squares estimates; and Panel C reports the corresponding first stage, equation (5), whose dependent variable has a mean of 0.28 in both columns. Anderson–Rubin confidence sets are robust to weak identification. Both columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C8: Alternative thresholds for defining news event weeks

	(1)	(2)	(3)	(4)
Specification	First stage	2SLS	2SLS	2SLS
Margin		Extensive	Intensive	Combined
Dependent variable	Avg. weekly TikTok hrs × News event	Any news consumption	log(Weekly news minutes) news > 0	log(Weekly news minutes) (0 → 1 = 100% increase)
Panel B: Events more than 1.5 standard deviations above the rolling average				
Avg. weekly TikTok hours × News event		-0.023* (0.013)	-0.051 (0.040)	-0.105* (0.055)
90% AR CI		[-0.064, -0.007]	[-0.159, 0.012]	[-0.267, -0.030]
95% AR CI		[-0.083, -0.004]	[-0.204, 0.028]	[-0.339, -0.017]
Musically Celebrity index × News event	0.402*** (0.126)			
Superstar index × News event	-0.044 (0.076)	0.001 (0.003)	-0.027** (0.013)	-0.011 (0.018)
Up-and-coming celebrity index × News event	-0.239** (0.114)	-0.002 (0.005)	0.021 (0.020)	0.007 (0.027)
Mean dependent variable	0.25	0.69	3.43	2.07
Number of events	19	19	19	19
Observations	2,841,545	2,841,545	1,963,436	2,841,545
Number of devices	134,755	134,755	120,242	134,755
Number of weeks	113	113	113	113
Cluster-robust F-statistic	10.10	10.10	8.67	10.10
Panel A: Events more than 0.75 standard deviations above the rolling average				
Avg. weekly TikTok hours × News event		-0.011** (0.005)	-0.048** (0.022)	-0.070** (0.029)
90% AR CI		[-0.026, -0.003]	[-0.113, -0.018]	[-0.151, -0.031]
95% AR CI		[-0.031, -0.002]	[-0.134, -0.011]	[-0.178, -0.024]
Musically Celebrity index × News event	0.496*** (0.125)			
Superstar index × News event	-0.143* (0.079)	0.0001 (0.0019)	-0.018** (0.009)	-0.012 (0.011)
Up-and-coming celebrity index × News event	-0.216* (0.119)	-0.002 (0.003)	0.012 (0.014)	0.003 (0.017)
Mean dependent variable	0.33	0.68	3.38	1.99
Number of events	58	58	58	58
Observations	6,941,070	6,941,070	4,723,820	6,941,070
Number of devices	181,196	181,196	167,047	181,196
Number of weeks	287	287	287	287
Cluster-robust F-statistic	15.84	15.84	12.56	15.84
Week & device FEs	Yes	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes	Yes

Notes: The unit of observation is a device-week, and all panels cover 205 media markets. The panels repeat Table 3 with news event weeks defined by alternative thresholds: a week is a news event if national news consumption exceeds its surrounding eight-week rolling average by more than 1.5 standard deviations in Panel A and by more than 0.75 standard deviations in Panel B. The baseline threshold in Table 3 is one standard deviation. In column (2) the dependent variable is an indicator for any news consumption in the week; in column (3) it is the logarithm of weekly news minutes among device-weeks with positive news consumption; and in column (4) it is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued as a doubling. Column (1) reports the first stage of equation (5) on the full sample, and columns (2)–(4) instrument average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musically Celebrity index interacted with the same indicator. Column (3) is estimated on device-weeks with positive news consumption, so its sample and first-stage F differ. Anderson–Rubin confidence sets are robust to weak identification. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user's gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor's and a master's degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C9: News event weeks defined as a percentage rise above the rolling average

	(1)	(2)	(3)	(4)
Specification	First stage	2SLS	2SLS	2SLS
Margin		Extensive	Intensive	Combined
Dependent variable	Avg. weekly TikTok hrs × News event	Any news consumption	log(Weekly news minutes) news > 0	log(Weekly news minutes) (0 → 1 = 100% increase)
Avg. weekly TikTok hours × News event		-0.019** (0.008)	-0.061* (0.032)	-0.091** (0.036)
90% AR CI		[-0.039, -0.009]	[-0.142, -0.015]	[-0.181, -0.041]
95% AR CI		[-0.046, -0.007]	[-0.171, -0.006]	[-0.211, -0.032]
Musically Celebrity index × News event	0.524*** (0.131)			
Superstar index × News event	-0.134 (0.085)	0.002 (0.003)	-0.026** (0.013)	-0.011 (0.015)
Up-and-coming celebrity index × News event	-0.228* (0.126)	-0.006 (0.004)	0.012 (0.019)	-0.011 (0.023)
Mean dependent variable	0.27	0.68	3.38	1.98
Number of events	30	30	30	30
Observations	4,575,385	4,575,385	3,095,589	4,575,385
Number of devices	163,991	163,991	148,184	163,991
Number of weeks	191	191	191	191
Cluster-robust F-statistic	16.02	16.02	11.68	16.02
Week & device FEs	Yes	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes	Yes

Notes: The unit of observation is a device-week, and all columns cover 205 media markets. This table repeats Table 3 with news events defined as the 30 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than 5%, rather than by more than one standard deviation. In column (2) the dependent variable is an indicator for any news consumption in the week; in column (3) it is the logarithm of weekly news minutes among device-weeks with positive news consumption; and in column (4) it is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued as a doubling. Column (1) reports the first stage of equation (5) on the full sample, and columns (2)–(4) instrument average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musically Celebrity index interacted with the same indicator. Column (3) is estimated on device-weeks with positive news consumption, so its sample and first-stage F differ. Anderson–Rubin confidence sets are robust to weak identification. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C10: News events excluding the first wave of COVID-19 lockdowns

	(1)	(2)	(3)	(4)
Specification	First stage	2SLS	2SLS	2SLS
Margin		Extensive	Intensive	Combined
Dependent variable	Avg. weekly TikTok hrs × News event	Any news consumption	log(Weekly news minutes) news > 0	log(Weekly news minutes) (0 → 1 = 100% increase)
Avg. weekly TikTok hours × News event		-0.014** (0.006)	-0.089*** (0.032)	-0.105*** (0.036)
90% AR CI		[-0.028, -0.006]	[-0.182, -0.050]	[-0.198, -0.059]
95% AR CI		[-0.032, -0.005]	[-0.214, -0.043]	[-0.227, -0.051]
Musical.ly Celebrity index × News event	0.577*** (0.131)			
Superstar index × News event	-0.165* (0.090)	0.0004 (0.0025)	-0.030** (0.014)	-0.018 (0.016)
Up-and-coming celebrity index × News event	-0.201 (0.130)	-0.004 (0.004)	0.005 (0.021)	-0.009 (0.026)
Mean dependent variable	0.28	0.67	3.36	1.94
Number of events	35	35	35	35
Observations	5,131,152	5,131,152	3,453,415	5,131,152
Number of devices	174,592	174,592	158,277	174,592
Number of weeks	220	220	220	220
Cluster-robust F-statistic	19.40	19.40	14.35	19.40
Week & device FEs	Yes	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes	Yes

Notes: The unit of observation is a device-week, and all columns cover 205 media markets. This table repeats Table 3 dropping the event weeks affected by the first wave of COVID-19 lockdowns: the event weeks falling inside the period of state and territorial stay-at-home orders, which ran from March through May 2020 (Moreland, 2020), together with 1 June 2020, whose ± 4 -week control window reaches back into that period, leaving 35 events. News events are otherwise defined exactly as in Table 3, as the weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. Forty-two states and territories issued mandatory stay-at-home orders over this period. Mobile device use rose sharply while they were in force, so these estimates ask whether the baseline result is driven by it. In column (2) the dependent variable is an indicator for any news consumption in the week; in column (3) it is the logarithm of weekly news minutes among device-weeks with positive news consumption; and in column (4) it is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued as a doubling. Column (1) reports the first stage of equation (5) on the full sample, and columns (2)–(4) instrument average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musical.ly Celebrity index interacted with the same indicator. Column (3) is estimated on device-weeks with positive news consumption, so its sample and first-stage F differ. Anderson–Rubin confidence sets are robust to weak identification. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user's gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor's and a master's degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C11: News events defined by news pressure

	(1)	(2)	(3)	(4)
Specification	First stage	2SLS	2SLS	2SLS
Margin		Extensive	Intensive	Combined
Dependent variable	Avg. weekly TikTok hrs × Top 10% news pressure	Any news consumption	log(Weekly news minutes) news > 0	log(Weekly news minutes) (0 → 1 = 100% increase)
Avg. weekly TikTok hours × Top 10% news pressure		-0.013* (0.007)	0.004 (0.027)	-0.042 (0.030)
90% AR CI		[-0.033, -0.004]	[-0.049, 0.064]	[-0.126, 0.004]
95% AR CI		[-0.041, -0.002]	[-0.064, 0.083]	[-0.157, 0.015]
Musically Celebrity index × Top 10% news pressure	0.466*** (0.137)			
Superstar index × Top 10% news pressure	-0.103 (0.085)	0.0002 (0.0020)	-0.012 (0.009)	-0.004 (0.009)
Up-and-coming celebrity index × Top 10% news pressure	-0.237* (0.126)	0.0003 (0.0031)	0.022 (0.014)	0.012 (0.013)
Mean dependent variable	0.39	0.69	3.41	2.05
Number of events	33	33	33	33
Observations	3,700,695	3,700,695	2,548,675	3,700,695
Number of devices	142,460	142,460	129,086	142,460
Number of weeks	141	141	141	141
Cluster-robust F-statistic	11.54	11.54	9.68	11.54
Week & device FEs	Yes	Yes	Yes	Yes
Controls × Top 10% news pressure	Yes	Yes	Yes	Yes

Notes: The unit of observation is a device-week, and all columns cover 205 media markets. This table repeats Table 3 with news events defined as the 33 weeks in the top decile of the news-pressure distribution, rather than as the weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. News pressure is the number of minutes the NBC, ABC and CBS evening newscasts devote to their top three stories in a week (Eisensee & Strömberg, 2007; Durante & Zhuravskaya, 2018). In column (2) the dependent variable is an indicator for any news consumption in the week; in column (3) it is the logarithm of weekly news minutes among device-weeks with positive news consumption; and in column (4) it is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued as a doubling. Column (1) reports the first stage of equation (5) on the full sample, and columns (2)–(4) instrument average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musically Celebrity index interacted with the same indicator. Column (3) is estimated on device-weeks with positive news consumption, so its sample and first-stage F differ. Anderson–Rubin confidence sets are robust to weak identification. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C12: Event-week first stage under alternative placebo sets

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	Avg. weekly TikTok hours $\times$ News event						
Musical.ly Celebrity index 2015 $\times$ News event	0.484*** (0.121)	0.455*** (0.127)	0.298*** (0.094)	0.565*** (0.138)	0.525*** (0.116)	0.520*** (0.134)	0.621*** (0.141)
Superstar index 2015 $\times$ News event	-0.134* (0.080)			-0.128 (0.080)	-0.141* (0.078)		-0.130* (0.078)
Up-and-coming celebrity index 2015 $\times$ News event	-0.177 (0.120)			-0.133 (0.120)	-0.160 (0.116)		-0.106 (0.119)
Musical.ly Celebrity index 2014 $\times$ News event		-0.248** (0.118)		-0.164 (0.118)		-0.249** (0.114)	-0.173 (0.114)
Musical.ly Celebrity index 2016 $\times$ News event		0.035 (0.104)		0.075 (0.103)		0.001 (0.099)	0.040 (0.099)
Vine interest 2015 $\times$ News event			-0.090*** (0.031)		-0.088*** (0.029)	-0.094*** (0.031)	-0.090*** (0.030)
Dubsmash interest 2015 $\times$ News event			0.013 (0.038)		0.014 (0.037)	0.010 (0.036)	0.012 (0.037)
Cluster-robust F-statistic	15.95	12.83	10.03	16.80	20.45	15.08	19.43
R ²	0.256	0.255	0.256	0.256	0.256	0.256	0.256
Within-R ²	0.0658	0.0657	0.0658	0.0658	0.0658	0.0658	0.0658
Mean dependent variable	0.28	0.28	0.28	0.28	0.28	0.28	0.28
Observations	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705
Number of devices	176,321	176,321	176,321	176,321	176,321	176,321	176,321
Number of weeks	235	235	235	235	235	235	235
Number of DMAs	205	205	205	205	205	205	205
Interest in similar celebrities in 2015	Yes			Yes	Yes		Yes
Interest in Musical.ly celebrities in 2014+2016		Yes		Yes		Yes	Yes
Interest in contemporaneous short-form video platforms			Yes		Yes	Yes	Yes

Notes: The unit of observation is a device-week. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. The dependent variable is average weekly TikTok hours in the four weeks before the event, interacted with the event indicator. Placebo indices enter in pairs: interest in similar celebrities in 2015 is the 2015 superstar and up-and-coming celebrity indices; interest in Musical.ly celebrities in 2014+2016 is the Musical.ly Celebrity index measured in 2014 and in 2016; interest in contemporaneous short-form video platforms is 2015 search interest in Vine and Dubsmash. Each column includes a different combination of these pairs, and every index is interacted with the event indicator. The 2014 Musical.ly Celebrity index excludes Silentó and “Watch Me Whip / Nae Nae”. All columns are estimated on the same sample. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C13: TikTok use and news consumption during news events: alternative placebos (combined)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	log(Weekly news minutes) (0 → 1 = 100% increase)						
Avg. weekly TikTok hours × News event	-0.098*** (0.036)	-0.077** (0.034)	-0.150** (0.059)	-0.071** (0.030)	-0.097*** (0.034)	-0.074** (0.031)	-0.070** (0.028)
90% AR CI	[-0.182, -0.049]	[-0.157, -0.028]	[-0.335, -0.080]	[-0.138, -0.027]	[-0.173, -0.049]	[-0.144, -0.029]	[-0.130, -0.028]
95% AR CI	[-0.210, -0.041]	[-0.184, -0.018]	[-0.422, -0.070]	[-0.158, -0.019]	[-0.196, -0.041]	[-0.166, -0.020]	[-0.148, -0.021]
Superstar index 2015 × News event	-0.015 (0.014)			-0.008 (0.013)	-0.014 (0.014)		-0.007 (0.013)
Up-and-coming celebrity index 2015 × News event	-0.004 (0.022)			0.009 (0.021)	-0.004 (0.022)		0.010 (0.021)
Musical.ly Celebrity index 2014 × News event		-0.018 (0.013)		-0.017 (0.014)		-0.017 (0.013)	-0.017 (0.014)
Musical.ly Celebrity index 2016 × News event		-0.015 (0.016)		-0.015 (0.016)		-0.017 (0.015)	-0.017 (0.016)
Vine interest 2015 × News event			-0.011** (0.005)		-0.006 (0.004)	-0.005 (0.004)	-0.005 (0.004)
Dubsmash interest 2015 × News event			0.006 (0.008)		0.005 (0.007)	0.005 (0.006)	0.004 (0.006)
Cluster-robust F-statistic	15.95	12.83	10.03	16.80	20.45	15.08	19.43
Mean dependent variable	1.99	1.99	1.99	1.99	1.99	1.99	1.99
Observations	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705
Number of devices	176,321	176,321	176,321	176,321	176,321	176,321	176,321
Number of weeks	235	235	235	235	235	235	235
Number of DMAs	205	205	205	205	205	205	205
Week & device FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Interest in similar celebrities in 2015	Yes			Yes	Yes		Yes
Interest in Musical.ly celebrities in 2014+2016		Yes		Yes		Yes	Yes
Interest in contemporaneous short-form video platforms			Yes		Yes	Yes	Yes

Notes: The unit of observation is a device-week. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. The dependent variable is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued as a doubling. Each column instruments average weekly TikTok hours in the four weeks before the event, interacted with the event indicator, with the Musical.ly Celebrity index measured in 2015 and interacted with the same indicator. Placebo indices enter in pairs: interest in similar celebrities in 2015 is the 2015 superstar and up-and-coming celebrity indices; interest in Musical.ly celebrities in 2014+2016 is the Musical.ly Celebrity index measured in 2014 and in 2016; interest in contemporaneous short-form video platforms is 2015 search interest in Vine and Dubsmash. Each column includes a different combination of these pairs, and every index is interacted with the event indicator. The 2014 Musical.ly Celebrity index excludes Silentó and “Watch Me Whip / Nae Nae”. All columns are estimated on the same sample. Anderson-Rubin confidence sets are robust to weak identification. The corresponding first stage is reported in Table C12. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C14: TikTok use and news consumption during news events: reduced form (extensive margin)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	Any news consumption						
Musical.ly Celebrity index 2015 $\times$ News event	-0.008*** (0.002)	-0.006** (0.003)	-0.005*** (0.002)	-0.008*** (0.003)	-0.007*** (0.002)	-0.005* (0.003)	-0.007** (0.003)
Superstar index 2015 $\times$ News event	0.002 (0.002)			0.003 (0.002)	0.002 (0.002)		0.002 (0.002)
Up-and-coming celebrity index 2015 $\times$ News event	-0.0004 (0.0030)			-0.0001 (0.0031)	0.0001 (0.0029)		0.001 (0.003)
Musical.ly Celebrity index 2014 $\times$ News event		0.001 (0.002)		0.0004 (0.0024)		0.001 (0.002)	0.0002 (0.0024)
Musical.ly Celebrity index 2016 $\times$ News event		-0.002 (0.002)		-0.003 (0.002)		-0.002 (0.002)	-0.003 (0.002)
Vine interest 2015 $\times$ News event			-0.0003 (0.0006)		-0.0003 (0.0006)	-0.0004 (0.0006)	-0.0005 (0.0006)
Dubsmash interest 2015 $\times$ News event			-0.001* (0.001)		-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)
R ²	0.480	0.480	0.480	0.480	0.480	0.480	0.480
Within-R ²	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Mean dependent variable	0.68	0.68	0.68	0.68	0.68	0.68	0.68
Observations	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705
Number of devices	176,321	176,321	176,321	176,321	176,321	176,321	176,321
Number of weeks	235	235	235	235	235	235	235
Number of DMAs	205	205	205	205	205	205	205
Week & device FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls $\times$ News event	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>p-values for the equality of coefficients with the 2015 Musical.ly Celebrity index</i>							
Superstar index 2015	0.002***			0.003***	0.006***		0.010**
Up-and-coming celebrity index 2015	0.075*			0.075*	0.092*		0.097*
Musical.ly Celebrity index 2014		0.098*		0.064*		0.171	0.125
Musical.ly Celebrity index 2016		0.251		0.136		0.470	0.302
Vine interest 2015			0.021**		0.006***	0.118	0.036**
Dubsmash interest 2015			0.090*		0.031**	0.221	0.084*
<i>p-values for equality of coefficient for categories of alternative indices with the 2015 Musical.ly Celebrity index</i>							
Interest in similar celebrities in 2015	0.008***			0.012**	0.019**		0.030**
Interest in Musical.ly celebrities in 2014+2016		0.251		0.162		0.384	0.307
Interest in contemporaneous short-form video platforms			0.044**		0.010**	0.201	0.073*
All indices in the column	0.008***	0.251	0.044**	0.048**	0.026**	0.393	0.085*
Interest in similar celebrities in 2015	Yes			Yes	Yes		Yes
Interest in Musical.ly celebrities in 2014+2016		Yes		Yes		Yes	Yes
Interest in contemporaneous short-form video platforms			Yes		Yes	Yes	Yes

Notes: The unit of observation is a device-week. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. The dependent variable is an indicator for any news consumption in the week. Each column reports the reduced form, in which the Musical.ly Celebrity index interacted with the event indicator enters the outcome equation directly in place of TikTok use. Placebo indices enter in pairs: interest in similar celebrities in 2015 is the 2015 superstar and up-and-coming celebrity indices; interest in Musical.ly celebrities in 2014+2016 is the Musical.ly Celebrity index measured in 2014 and in 2016; interest in contemporaneous short-form video platforms is 2015 search interest in Vine and Dubsmash. Each column includes a different combination of these pairs, and every index is interacted with the event indicator. The 2014 Musical.ly Celebrity index excludes Silentó and “Watch Me Whip / Nae Nae”. All columns are estimated on the same sample. The lower block reports Wald tests of equality between the reduced-form coefficient on the 2015 Musical.ly Celebrity index and the placebo coefficients, using the same DMA-clustered covariance matrix as the reported standard errors. All indices are standardized and interacted with the same event indicator, so their coefficients share units. The first six rows test equality against one placebo at a time, the next three against both members of a pair jointly, and the last against every index in the column. A low *p*-value indicates that the 2015 index and the placebo do not move the outcome alike. Cells are empty where the index does not enter the column. Stars on the *p*-values mark rejection of equality at the 10%, 5% and 1% levels. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table C15: TikTok use and news consumption during news events: reduced form (intensive margin)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	log(Weekly news minutes) news > 0						
Musical.ly Celebrity index 2015 × News event	-0.030** (0.013)	-0.021* (0.013)	-0.039*** (0.009)	-0.021 (0.014)	-0.037*** (0.013)	-0.029** (0.013)	-0.029** (0.014)
Superstar index 2015 × News event	-0.014 (0.012)			-0.011 (0.011)	-0.012 (0.011)		-0.009 (0.011)
Up-and-coming celebrity index 2015 × News event	0.020 (0.016)			0.026* (0.016)	0.016 (0.015)		0.022 (0.015)
Musical.ly Celebrity index 2014 × News event		-0.004 (0.012)		-0.010 (0.012)		-0.004 (0.012)	-0.008 (0.012)
Musical.ly Celebrity index 2016 × News event		-0.016 (0.011)		-0.017 (0.011)		-0.015 (0.010)	-0.016 (0.011)
Vine interest 2015 × News event			0.004 (0.003)		0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
Dubsmash interest 2015 × News event			0.012** (0.005)		0.012** (0.005)	0.012** (0.005)	0.011** (0.005)
R ²	0.575	0.575	0.575	0.575	0.575	0.575	0.575
Within-R ²	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Mean dependent variable	3.38	3.38	3.38	3.38	3.38	3.38	3.38
Observations	3,852,220	3,852,220	3,852,220	3,852,220	3,852,220	3,852,220	3,852,220
Number of devices	160,833	160,833	160,833	160,833	160,833	160,833	160,833
Number of weeks	235	235	235	235	235	235	235
Number of DMAs	205	205	205	205	205	205	205
Week & device FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>p-values for the equality of coefficients with the 2015 Musical.ly Celebrity index</i>							
Superstar index 2015	0.448			0.636	0.201		0.331
Up-and-coming celebrity index 2015	0.019**			0.029**	0.009***		0.014**
Musical.ly Celebrity index 2014		0.453		0.629		0.237	0.359
Musical.ly Celebrity index 2016		0.786		0.831		0.379	0.432
Vine interest 2015			<0.001***		0.002***	0.017**	0.029**
Dubsmash interest 2015			<0.001***		<0.001***	0.004***	0.010**
<i>p-values for equality of coefficient for categories of alternative indices with the 2015 Musical.ly Celebrity index</i>							
Interest in similar celebrities in 2015	0.065*			0.088*	0.030**		0.048**
Interest in Musical.ly celebrities in 2014+2016		0.748		0.888		0.479	0.629
Interest in contemporaneous short-form video platforms			<0.001***		0.003***	0.013**	0.029**
All indices in the column	0.065*	0.748	<0.001***	0.218	<0.001***	<0.001***	0.002***
Interest in similar celebrities in 2015	Yes			Yes	Yes		Yes
Interest in Musical.ly celebrities in 2014+2016		Yes		Yes		Yes	Yes
Interest in contemporaneous short-form video platforms			Yes		Yes	Yes	Yes

Notes: The unit of observation is a device-week. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. The dependent variable is the logarithm of weekly news minutes among device-weeks with positive news consumption. Each column reports the reduced form, in which the Musical.ly Celebrity index interacted with the event indicator enters the outcome equation directly in place of TikTok use. Placebo indices enter in pairs: interest in similar celebrities in 2015 is the 2015 superstar and up-and-coming celebrity indices; interest in Musical.ly celebrities in 2014+2016 is the Musical.ly Celebrity index measured in 2014 and in 2016; interest in contemporaneous short-form video platforms is 2015 search interest in Vine and Dubsmash. Each column includes a different combination of these pairs, and every index is interacted with the event indicator. The 2014 Musical.ly Celebrity index excludes Silentó and “Watch Me Whip / Nae Nae”. All columns are estimated on the same sample. The lower block reports Wald tests of equality between the reduced-form coefficient on the 2015 Musical.ly Celebrity index and the placebo coefficients, using the same DMA-clustered covariance matrix as the reported standard errors. All indices are standardized and interacted with the same event indicator, so their coefficients share units. The first six rows test equality against one placebo at a time, the next three against both members of a pair jointly, and the last against every index in the column. A low *p*-value indicates that the 2015 index and the placebo do not move the outcome alike. Cells are empty where the index does not enter the column. Stars on the *p*-values mark rejection of equality at the 10%, 5% and 1% levels. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table C16: TikTok use and news consumption during news events: reduced form (combined, extensive margin valued as a doubling)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	log(Weekly news minutes) (0 → 1 = 100% increase)						
Musical.ly Celebrity index 2015 × News event	-0.047*** (0.014)	-0.035** (0.014)	-0.045*** (0.010)	-0.040*** (0.015)	-0.051*** (0.014)	-0.038*** (0.014)	-0.043*** (0.016)
Superstar index 2015 × News event	-0.002 (0.013)			0.001 (0.013)	-0.001 (0.013)		0.002 (0.013)
Up-and-coming celebrity index 2015 × News event	0.013 (0.019)			0.019 (0.019)	0.012 (0.019)		0.017 (0.019)
Musical.ly Celebrity index 2014 × News event		0.002 (0.013)		-0.005 (0.013)		0.002 (0.013)	-0.005 (0.013)
Musical.ly Celebrity index 2016 × News event		-0.018 (0.013)		-0.020 (0.013)		-0.017 (0.013)	-0.020 (0.013)
Vine interest 2015 × News event			0.002 (0.003)		0.002 (0.003)	0.002 (0.003)	0.001 (0.003)
Dubsmash interest 2015 × News event			0.004 (0.005)		0.004 (0.005)	0.004 (0.005)	0.004 (0.005)
R ²	0.649	0.649	0.649	0.649	0.649	0.649	0.649
Within-R ²	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Mean dependent variable	1.99	1.99	1.99	1.99	1.99	1.99	1.99
Observations	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705
Number of devices	176,321	176,321	176,321	176,321	176,321	176,321	176,321
Number of weeks	235	235	235	235	235	235	235
Number of DMAs	205	205	205	205	205	205	205
Week & device FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>p-values for the equality of coefficients with the 2015 Musical.ly Celebrity index</i>							
Superstar index 2015	0.039**			0.056*	0.024**		0.039**
Up-and-coming celebrity index 2015	0.020**			0.025**	0.018**		0.023**
Musical.ly Celebrity index 2014		0.135		0.148		0.109	0.119
Musical.ly Celebrity index 2016		0.391		0.311		0.291	0.240
Vine interest 2015			<0.001***		<0.001***	0.011**	0.006***
Dubsmash interest 2015			<0.001***		<0.001***	0.009***	0.007***
<i>p-values for equality of coefficient for categories of alternative indices with the 2015 Musical.ly Celebrity index</i>							
Interest in similar celebrities in 2015	0.022**			0.039**	0.017**		0.031**
Interest in Musical.ly celebrities in 2014+2016		0.326		0.343		0.271	0.283
Interest in contemporaneous short-form video platforms			<0.001***		0.001***	0.031**	0.022**
All indices in the column	0.022**	0.326	<0.001***	0.134	0.004***	0.008***	0.032**
Interest in similar celebrities in 2015	Yes			Yes	Yes		Yes
Interest in Musical.ly celebrities in 2014+2016		Yes		Yes		Yes	Yes
Interest in contemporaneous short-form video platforms			Yes		Yes	Yes	Yes

Notes: The unit of observation is a device-week. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. The dependent variable is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued as a doubling. Each column reports the reduced form, in which the Musical.ly Celebrity index interacted with the event indicator enters the outcome equation directly in place of TikTok use. Placebo indices enter in pairs: interest in similar celebrities in 2015 is the 2015 superstar and up-and-coming celebrity indices; interest in Musical.ly celebrities in 2014+2016 is the Musical.ly Celebrity index measured in 2014 and in 2016; interest in contemporaneous short-form video platforms is 2015 search interest in Vine and Dubsmash. Each column includes a different combination of these pairs, and every index is interacted with the event indicator. The 2014 Musical.ly Celebrity index excludes Silentó and “Watch Me Whip / Nae Nae”. All columns are estimated on the same sample. The lower block reports Wald tests of equality between the reduced-form coefficient on the 2015 Musical.ly Celebrity index and the placebo coefficients, using the same DMA-clustered covariance matrix as the reported standard errors. All indices are standardized and interacted with the same event indicator, so their coefficients share units. The first six rows test equality against one placebo at a time, the next three against both members of a pair jointly, and the last against every index in the column. A low *p*-value indicates that the 2015 index and the placebo do not move the outcome alike. Cells are empty where the index does not enter the column. Stars on the *p*-values mark rejection of equality at the 10%, 5% and 1% levels. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table C17: TikTok use and news consumption during news events: reduced form and 2SLS (combined, extensive margin valued at zero)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	log(Weekly news minutes) (0 → 1 = 0% increase)						
Musical.ly Celebrity index 2015 × News event	-0.038*** (0.012)	-0.027** (0.012)	-0.038*** (0.009)	-0.031** (0.013)	-0.042*** (0.012)	-0.031** (0.012)	-0.034*** (0.013)
Superstar index 2015 × News event	-0.003 (0.011)			-0.0002 (0.0109)	-0.002 (0.011)		0.001 (0.011)
Up-and-coming celebrity index 2015 × News event	0.012 (0.016)			0.017 (0.016)	0.010 (0.016)		0.015 (0.016)
Musical.ly Celebrity index 2014 × News event		-0.0002 (0.0114)		-0.006 (0.011)		-0.0000 (0.0114)	-0.005 (0.011)
Musical.ly Celebrity index 2016 × News event		-0.016 (0.011)		-0.018 (0.011)		-0.015 (0.011)	-0.017 (0.011)
Vine interest 2015 × News event			0.002 (0.003)		0.002 (0.003)	0.001 (0.003)	0.001 (0.003)
Dubsmash interest 2015 × News event			0.005 (0.005)		0.004 (0.005)	0.004 (0.005)	0.004 (0.005)
R ²	0.665	0.665	0.665	0.665	0.665	0.665	0.665
Within-R ²	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Mean dependent variable	2.34	2.34	2.34	2.34	2.34	2.34	2.34
Observations	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705	5,660,705
Number of devices	176,321	176,321	176,321	176,321	176,321	176,321	176,321
Number of weeks	235	235	235	235	235	235	235
Number of DMAs	205	205	205	205	205	205	205
Week & device FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls × News event	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>p-values for the equality of coefficients with the 2015 Musical.ly Celebrity index</i>							
Superstar index 2015	0.066*			0.098*	0.041**		0.067*
Up-and-coming celebrity index 2015	0.024**			0.031**	0.020**		0.027**
Musical.ly Celebrity index 2014		0.188		0.214		0.144	0.166
Musical.ly Celebrity index 2016		0.495		0.419		0.359	0.314
Vine interest 2015			<0.001***		<0.001***	0.015**	0.011**
Dubsmash interest 2015			<0.001***		0.001***	0.010**	0.010**
<i>p-values for equality of coefficient for categories of alternative indices with the 2015 Musical.ly Celebrity index</i>							
Interest in similar celebrities in 2015	0.035**			0.061*	0.025**		0.047**
Interest in Musical.ly celebrities in 2014+2016		0.420		0.459		0.342	0.372
Interest in contemporaneous short-form video platforms			<0.001***		0.002***	0.037**	0.034**
All indices in the column	0.035**	0.420	<0.001***	0.181	0.005***	0.007***	0.040**
Interest in similar celebrities in 2015	Yes			Yes	Yes		Yes
Interest in Musical.ly celebrities in 2014+2016		Yes		Yes		Yes	Yes
Interest in contemporaneous short-form video platforms			Yes		Yes	Yes	Yes

Notes: The unit of observation is a device-week. *News event* equals one in the 40 weeks in which national news consumption exceeds its surrounding eight-week rolling average by more than one standard deviation. The dependent variable is the logarithm of weekly news minutes across all device-weeks, where weeks with less than one minute of news are coded as zero and the move from zero to one minute is valued at zero. Each column reports the reduced form, in which the Musical.ly Celebrity index interacted with the event indicator enters the outcome equation directly in place of TikTok use. Placebo indices enter in pairs: interest in similar celebrities in 2015 is the 2015 superstar and up-and-coming celebrity indices; interest in Musical.ly celebrities in 2014+2016 is the Musical.ly Celebrity index measured in 2014 and in 2016; interest in contemporaneous short-form video platforms is 2015 search interest in Vine and Dubsmash. Each column includes a different combination of these pairs, and every index is interacted with the event indicator. The 2014 Musical.ly Celebrity index excludes Silentó and “Watch Me Whip / Nae Nae”. All columns are estimated on the same sample. The lower block reports Wald tests of equality between the reduced-form coefficient on the 2015 Musical.ly Celebrity index and the placebo coefficients, using the same DMA-clustered covariance matrix as the reported standard errors. All indices are standardized and interacted with the same event indicator, so their coefficients share units. The first six rows test equality against one placebo at a time, the next three against both members of a pair jointly, and the last against every index in the column. A low *p*-value indicates that the 2015 index and the placebo do not move the outcome alike. Cells are empty where the index does not enter the column. Stars on the *p*-values mark rejection of equality at the 10%, 5% and 1% levels. All columns include device and week fixed effects and the individual-level and DMA-level controls interacted with the event indicator: the device user’s gender, age in 5-year bins, device type, indicators for Black and for Hispanic users, those two indicators interacted with the age bins, household-income buckets, household-size indicators, an indicator for children in the household, and, at the DMA level, the 2015 shares of women, Hispanic residents, residents in poverty and high-school graduates; the shares white, Black, Asian, Native American, Pacific Islander and of two or more races; the shares aged 0–19, 20–39 and 60 or older; log median household income, the Gini coefficient, the unemployment rate and the shares holding a bachelor’s and a master’s degree; and indicators for DMA population decile. Variable sources are listed in Table A1. Standard errors clustered by DMA. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

C.3 Survey evidence

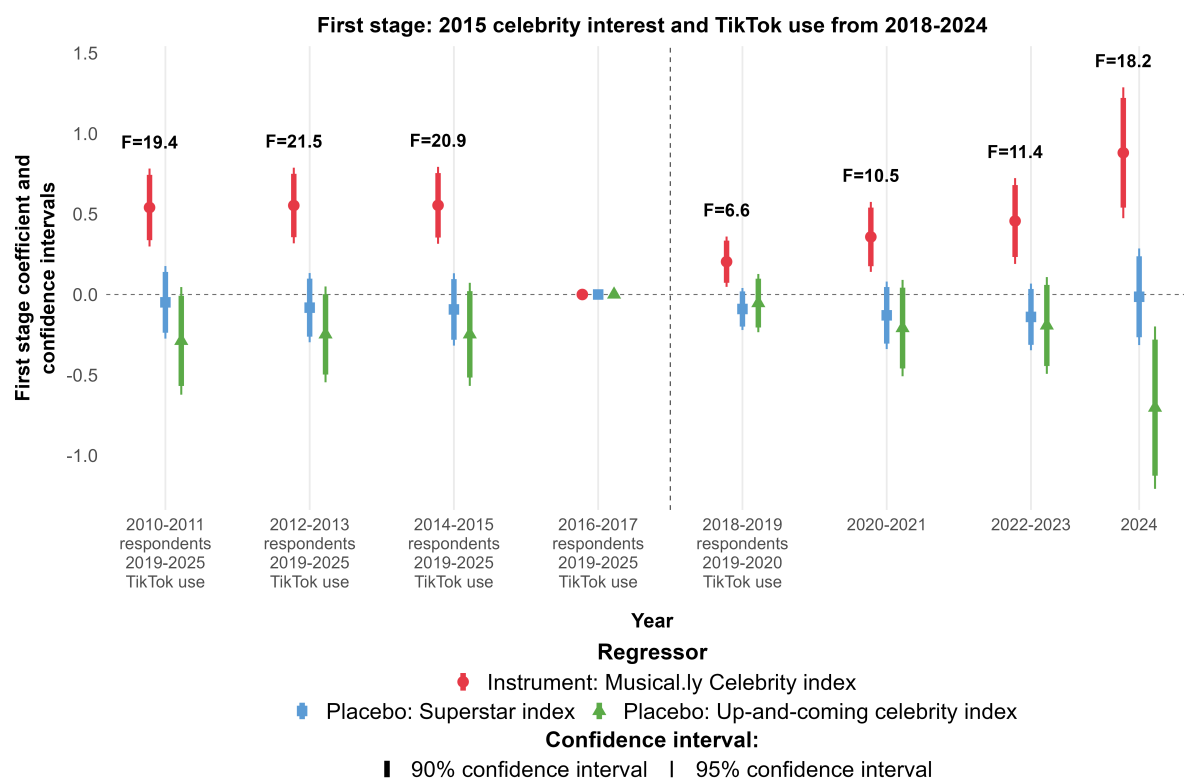


Figure C11: First stage among CES respondents by survey period, all periods

Note: This figure reports survey respondent-level regressions. It plots the coefficients on the three 2015 celebrity indices from a regression of media-market-level average demographically adjusted weekly TikTok hours on each index interacted with survey-period indicators, with 2016–2017 as the omitted reference period. Respondents surveyed before 2018 are assigned their market’s average TikTok use over the whole panel, since no contemporaneous measure exists. Cluster-robust first-stage F-statistics are reported above each period. Thick bars are 90% and thin bars 95% confidence intervals. The specification includes DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with the period indicators, and individual controls for age, gender, race and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA.

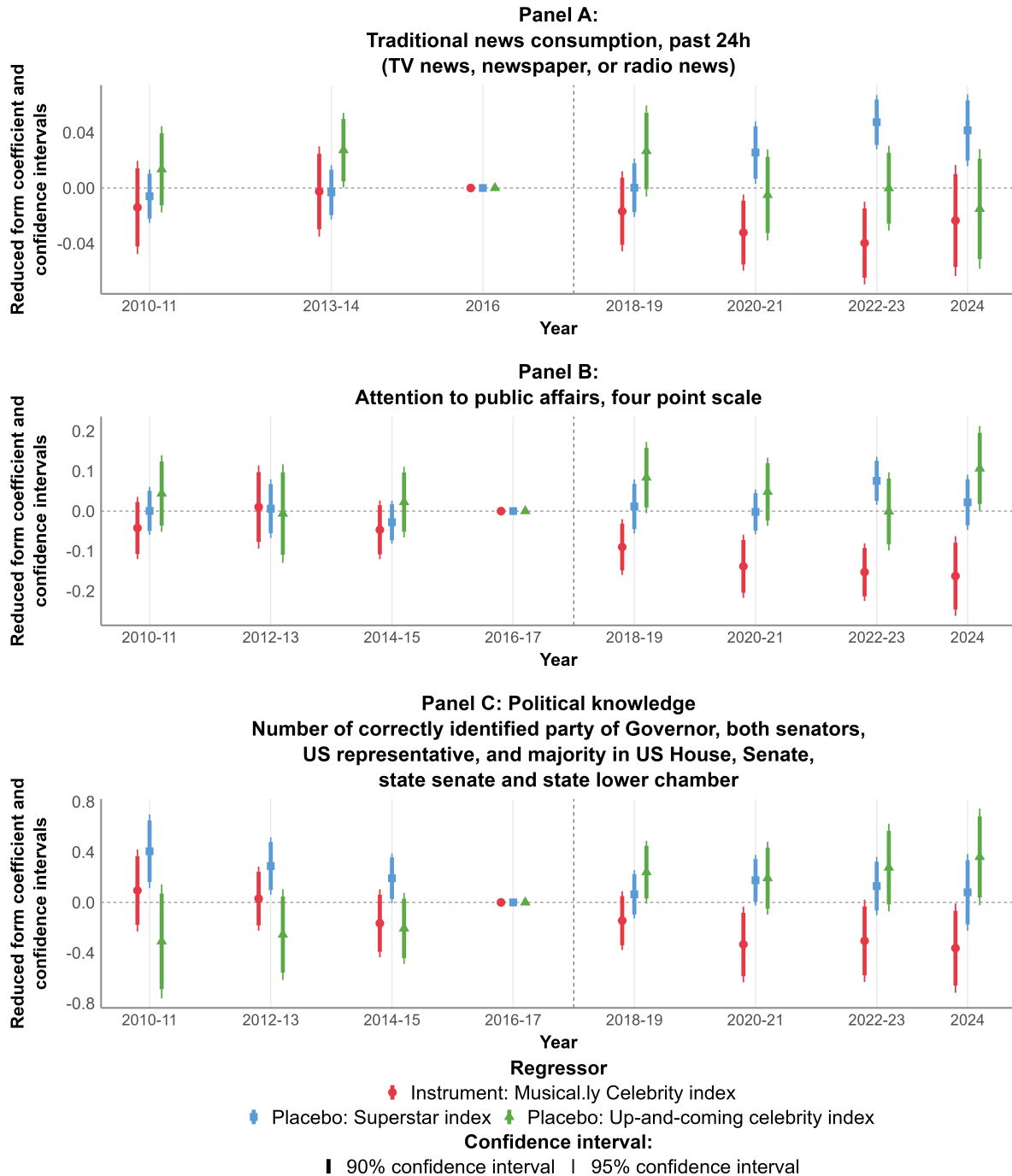


Figure C12: Reduced form among CES respondents: News consumption and political information, by survey period

Note: This figure reports survey respondent-level regressions. It plots the reduced-form coefficients on the three 2015 celebrity indices, each interacted with survey-period indicators, with 2016–2017 as the omitted reference. In Panel A the outcome is an indicator for having watched television news, read a newspaper or listened to radio news in the past 24 hours. In Panel B the outcome is how closely the respondent reports following government and public affairs, on a four-point scale. In Panel C the outcome is the number of correct answers to eight factual questions: the party of the respondent’s governor, of both U.S. senators and of their U.S. representative, and the majority party in the U.S. House, the U.S. Senate, the respondent’s state senate and their state’s lower chamber. Because the traditional media items are not fielded in 2012, 2015 or 2017, Panel A pools survey years into 2010–2011, 2013–2014 and 2016 before the merger and uses the same post-2018 periods as the other panels. Thick bars are 90% and thin bars 95% confidence intervals. The specification includes DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with the period indicators, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA.

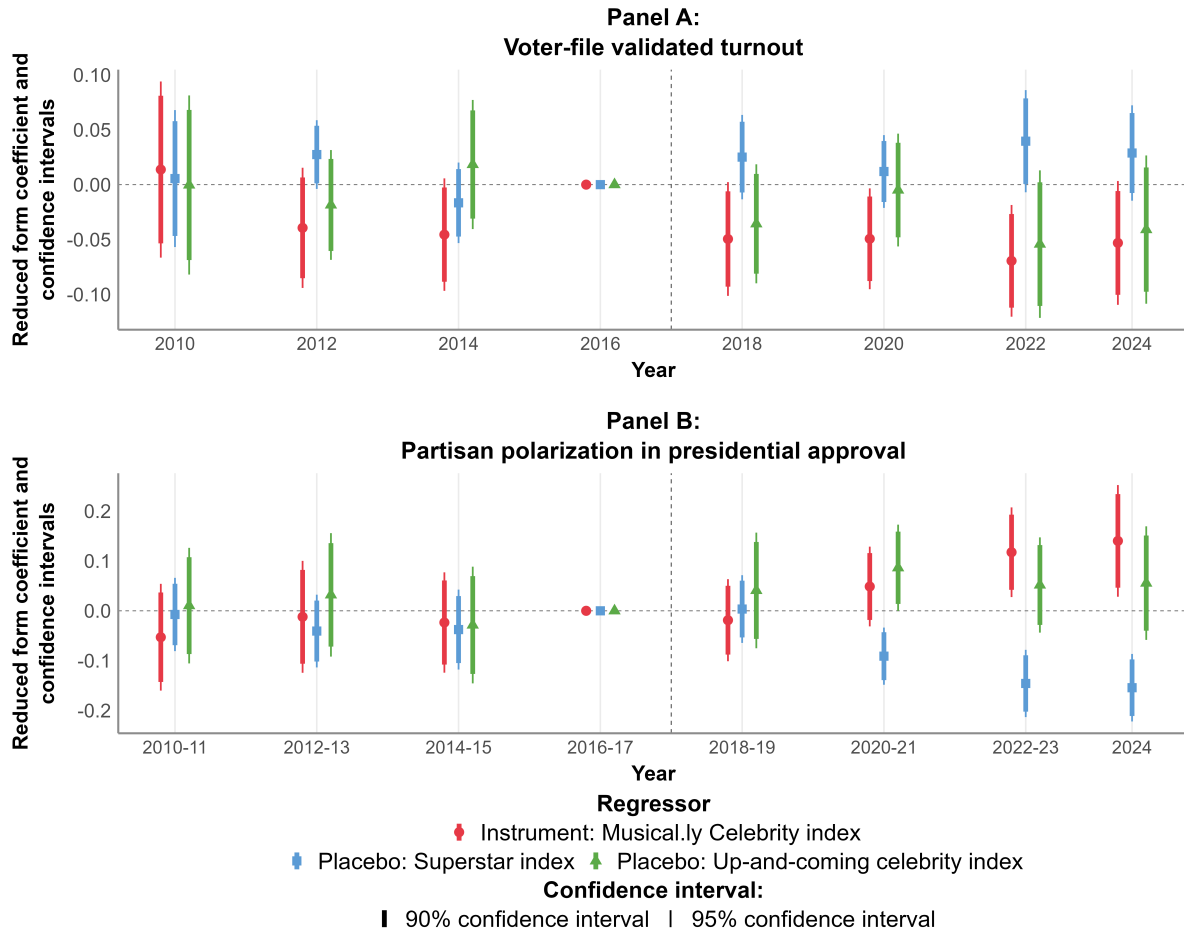


Figure C13: Reduced form among CES respondents: Turnout and partisan polarization in presidential approval, by survey period

Note: This figure reports survey respondent-level regressions. It plots the reduced-form coefficients on the three 2015 celebrity indices, each interacted with survey-period indicators, with 2016 as the omitted reference in Panel A and 2016–2017 in Panel B. In Panel A the outcome is voter-file validated turnout, which is available only in federal-election years. In Panel B the outcome is presidential approval signed by partisan alignment, among respondents who describe themselves as Democrats or Republicans. Thick bars are 90% and thin bars 95% confidence intervals. The specification includes DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with the period indicators, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA.

Table C18: Ordinary least squares and reduced-form estimates

	(1)	(2)	(3)	(4)	(5)
		Political information		Political behavior and attitudes	
Dependent variable	Traditional news, past 24h	Attention to public affairs	Political knowledge	Validated turnout	Partisan polarization in presidential approval
Panel A: OLS					
Avg. weekly TikTok hours $\times$ Post	-0.003 (0.004)	-0.013 (0.009)	-0.015 (0.036)	-0.002 (0.005)	0.001 (0.009)
Superstar index $\times$ Post	0.035*** (0.006)	0.008 (0.019)	-0.129 (0.085)	0.011 (0.012)	-0.085*** (0.018)
Up-and-coming celebrity index $\times$ Post	-0.031*** (0.011)	-0.013 (0.031)	0.323** (0.137)	-0.035* (0.020)	0.092*** (0.027)
R ²	0.074	0.122	0.175	0.134	0.048
Panel B: Reduced form					
Musical.ly Celebrity index $\times$ Post	-0.029*** (0.008)	-0.118*** (0.022)	-0.316*** (0.111)	-0.031* (0.017)	0.118*** (0.030)
Superstar index $\times$ Post	0.042*** (0.006)	0.034** (0.017)	-0.058 (0.077)	0.017 (0.013)	-0.111*** (0.017)
Up-and-coming celebrity index $\times$ Post	-0.022** (0.010)	0.022 (0.030)	0.411*** (0.137)	-0.026 (0.019)	0.058** (0.026)
R ²	0.074	0.122	0.176	0.134	0.049
Mean dependent variable	0.805	3.184	4.973	0.538	0.997
Number of respondents	512,216	580,025	574,436	463,193	368,331
Number of survey years	12	15	15	8	15
Number of DMAs	205	205	204	205	205
DMA & survey-year FEs	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes
DMA balance controls $\times$ Post	Yes	Yes	Yes	Yes	Yes

Notes: This table reports survey respondent-level regressions. Survey years 2010–2024 are pooled and *Post* equals one after 2018, the year of the Musical.ly–TikTok merger. Panel A regresses each outcome directly on media-market-level average demographically adjusted weekly TikTok hours interacted with *Post*, without instrumenting. Panel B reports the reduced form, regressing each outcome on the Musical.ly Celebrity index interacted with *Post*. The placebo indices interacted with *Post* enter both panels and their coefficients are reported. The outcomes are as in Table 5: in column (1), an indicator for having watched television news, read a newspaper or listened to radio news in the past 24 hours; in column (2), how closely the respondent reports following government and public affairs, coded 1–4; in column (3), the number of correct answers to eight factual questions: the party of the respondent’s governor, of both U.S. senators and of their U.S. representative, and the majority party in the U.S. House, the U.S. Senate, the respondent’s state senate and their state’s lower chamber; in column (4), voter-file validated turnout, available only in federal-election years; and in column (5), presidential approval signed by partisan alignment, among respondents who describe themselves as Democrats or Republicans. All columns include DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with *Post*, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C19: Traditional news consumption by medium

	(1)	(2)	(3)
Dependent variable	TV news	Newspaper	Radio news
Avg. weekly TikTok hours $\times$ Post	-0.041* (0.021)	-0.089*** (0.029)	0.008 (0.018)
95% AR CI	[-0.090, -0.001]	[-0.169, -0.041]	[-0.028, 0.050]
Superstar index $\times$ Post	0.026*** (0.008)	0.030*** (0.010)	-0.005 (0.009)
Up-and-coming celebrity index $\times$ Post	-0.016 (0.013)	-0.038** (0.016)	0.015 (0.014)
Cluster-robust F-statistic	28.44	28.44	28.44
Mean dependent variable	0.645	0.404	0.334
Number of respondents	512,216	512,216	512,216
Number of survey years	12	12	12
Number of DMAs	205	205	205
DMA & survey-year FEs	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
DMA balance controls $\times$ Post	Yes	Yes	Yes

Notes: This table reports survey respondent-level regressions. Survey years 2010–2024 are pooled and *Post* equals one after 2018, the year of the Musical.ly–TikTok merger. Each column reports two-stage least squares estimates. Media-market-level average demographically adjusted weekly TikTok hours interacted with *Post* is instrumented with the Musical.ly Celebrity index interacted with *Post*, with the superstar and up-and-coming placebo indices entering as exogenous controls. The cluster-robust F-statistic is computed on each column’s own sample. Anderson–Rubin confidence sets are robust to weak identification. The dependent variable is an indicator for having used the medium in the past 24 hours: television news in column (1), a newspaper in column (2) and radio news in column (3). Column (2) of Table 5 takes the maximum of the three. The three items come from a single check-all battery, so a respondent who answered it and did not select a medium is coded zero; the battery’s social-media and blog items are not used, the latter because the CES carries it only through 2019. The battery is not fielded in 2012, 2015 or 2017, so these columns cover twelve of the fifteen survey years. All columns include DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with *Post*, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C20: Components of the political knowledge score

	(1)	(2)
Dependent variable	State and district	National
Avg. weekly TikTok hours $\times$ Post	-0.533** (0.229)	-0.100** (0.050)
95% AR CI	[-1.129, -0.135]	[-0.221, -0.008]
Superstar index $\times$ Post	-0.122 (0.095)	0.006 (0.019)
Up-and-coming celebrity index $\times$ Post	0.205 (0.145)	0.004 (0.032)
Cluster-robust F-statistic	27.46	27.46
Mean dependent variable	3.764	1.209
Number of respondents	574,436	574,436
Number of survey years	15	15
Number of DMAs	204	204
DMA & survey-year FEs	Yes	Yes
Individual controls	Yes	Yes
DMA balance controls $\times$ Post	Yes	Yes

Notes: This table reports survey respondent-level regressions. Survey years 2010–2024 are pooled and *Post* equals one after 2018, the year of the Musical.ly–TikTok merger. Each column reports two-stage least squares estimates. Media-market-level average demographically adjusted weekly TikTok hours interacted with *Post* is instrumented with the Musical.ly Celebrity index interacted with *Post*, with the superstar and up-and-coming placebo indices entering as exogenous controls. The cluster-robust F-statistic is computed on each column’s own sample. Anderson–Rubin confidence sets are robust to weak identification. The columns split the eight-item political-knowledge score into its two components: column (1) covers the party of the governor, both U.S. senators and the respondent’s U.S. representative, and the majority party in the state upper and lower chambers; column (2) covers the majority party in the U.S. House and Senate. All columns include DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with *Post*, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C21: Presidential approval by partisan alignment and definition of partisanship

	(1)	(2)	(3)	(4)	(5)	(6)
Definition of partisanship	Self-described partisans			Leaners included		
Dependent variable	Partisan polarization	Approval: copartisans	Approval: out-partisans	Partisan polarization	Approval: copartisans	Approval: out-partisans
Avg. weekly TikTok hours $\times$ Post	0.223*** (0.076)	0.198* (0.101)	-0.085 (0.052)	0.141** (0.061)	0.106 (0.084)	-0.037 (0.042)
95% AR CI	[0.100, 0.432]	[0.017, 0.453]	[-0.219, 0.005]	[0.037, 0.303]	[-0.053, 0.307]	[-0.143, 0.038]
Superstar index $\times$ Post	-0.088*** (0.029)	-0.100*** (0.035)	0.028 (0.019)	-0.072*** (0.024)	-0.070** (0.029)	0.022 (0.017)
Up-and-coming celebrity index $\times$ Post	0.132*** (0.041)	0.144*** (0.050)	-0.043 (0.032)	0.107*** (0.036)	0.098** (0.043)	-0.038 (0.029)
Cluster-robust F-statistic	27.73	25.22	28.14	28.83	25.89	30.30
Mean dependent variable	3.497	3.279	1.261	3.478	3.232	1.255
Number of respondents	368,331	201,037	167,294	483,597	256,496	227,101
Number of survey years	15	15	15	15	15	15
Number of DMAs	205	205	205	205	205	205
DMA & survey-year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
DMA balance controls $\times$ Post	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Each column is an individual-level 2SLS regression of a presidential-approval outcome on average weekly TikTok hours $\times$ Post, instrumented by the 2015 Musical.ly index $\times$ Post, with the Superstar and Upcoming indices $\times$ Post as exogenous controls. The two blocks differ only in how partisanship is defined. Partisanship comes from the seven-point party identification item, which is built from a branching sequence in which respondents who do not name a party are asked which they feel closer to. Columns (1)–(3) use only respondents who describe themselves as Democrats (codes 1–2) or Republicans (6–7), so codes 3 (Lean Democrat), 4 (Independent) and 5 (Lean Republican) are all excluded; this is the definition used in the main table and in the event studies. Columns (4)–(6) also count as partisans those who name no party but say they feel closer to one, collapsing the item to Democrat (1–3) and Republican (5–7) so that only pure independents are dropped. Everything else is held fixed, so differences between the blocks are attributable to the leaners. Within each block: “Partisan polarization” is the 1–4 presidential job-approval item recoded so that higher values indicate an evaluation more in line with the respondent’s own party — 4 where a copartisan strongly approves or an out-partisan strongly disapproves, down to 1 where a copartisan strongly disapproves or an out-partisan strongly approves. The copartisan and out-partisan columns report the job-approval item itself on the same 1–4 scale (4 = strongly approves) within the two subsamples, so a positive coefficient among copartisans and a negative one among out-partisans both correspond to more partisan-consistent approval. “Not sure” responses are treated as missing rather than as a neutral midpoint. “First-stage F” is the cluster-robust Wald statistic for the excluded instrument on each column’s own sample. The bracketed “95% AR CI” rows report the weak-instrument-robust Anderson–Rubin 95% confidence set. No R^2 is reported (it is not meaningful for 2SLS). “Average weekly TikTok hours $\times$ Post” is demographically adjusted DMA-level TikTok use matched to the respondent’s own survey years: 2019 respondents receive the device-weighted average of 2019 and 2020, 2020 and 2021 respondents the device-weighted average of those two years, 2022 and 2023 respondents the average of those two, and 2024 respondents the 2024 value; pre-period respondents receive zero. All columns include DMA and survey-year fixed effects, the four 2015 DMA balance controls $\times$ Post, and individual controls for age (3 bins), gender, race (5 categories) and Hispanic origin. Survey years 2010–2024 are pooled (Post = 1 after 2018). Survey-weighted; SEs clustered by DMA. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C22: Ideological polarization: where respondents place the parties and themselves

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sample	Perception of parties			Distance from self to a party		The respondent's own position		
	All respondents	Self-described partisans				All respondents		
	Distance between the two parties	Opposing party's perceived ideological extremity	Own party's perceived ideological extremity	Distance from self to the opposing party	Distance from self to own party	Own ideological extremity (7pt)	Own ideological extremity (5pt)	Strength of party identification
Avg. weekly TikTok hours × Post	0.134 (0.090)	0.081* (0.047)	0.036 (0.060)	0.159 (0.099)	0.014 (0.053)	-0.040 (0.060)	-0.016 (0.046)	-0.058 (0.054)
95% AR CI	[-0.049, 0.334]	[-0.014, 0.186]	[-0.096, 0.160]	[-0.041, 0.381]	[-0.096, 0.129]	[-0.182, 0.072]	[-0.126, 0.073]	[-0.189, 0.041]
Superstar index × Post	-0.141*** (0.030)	-0.049*** (0.019)	-0.069*** (0.020)	-0.082* (0.045)	0.050*** (0.017)	-0.014 (0.025)	-0.013 (0.016)	-0.047** (0.024)
Up-and-coming celebrity index × Post	0.152*** (0.053)	0.108*** (0.028)	0.100*** (0.037)	0.149** (0.072)	-0.052* (0.030)	0.004 (0.044)	0.006 (0.028)	0.009 (0.037)
Cluster-robust F-statistic	28.09	27.55	27.76	27.58	27.80	28.51	27.70	28.22
Mean dependent variable	3.694	2.498	1.538	4.036	1.124	1.457	0.848	1.884
Number of respondents	516,646	339,840	346,713	336,397	343,171	563,610	552,680	577,445
Number of survey years	15	15	15	15	15	15	15	15
Number of DMAs	205	205	205	205	205	205	205	205
DMA & survey-year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
DMA balance controls × Post	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports survey respondent-level regressions. Survey years 2010-2024 are pooled and *Post* equals one after 2018, the year of the Musical.ly-TikTok merger. Each column reports two-stage least squares estimates. Media-market-level average demographically adjusted weekly TikTok hours interacted with *Post* is instrumented with the Musical.ly Celebrity index interacted with *Post*, with the superstar and up-and-coming placebo indices entering as exogenous controls. The cluster-robust F-statistic is computed on each column's own sample. Anderson-Rubin confidence sets are robust to weak identification. The ideological items place the respondent, the Democratic Party and the Republican Party on one 7-point scale running from 1 "Very Liberal" through 4 "Middle of the Road" to 7 "Very Conservative"; "Not sure" is treated as missing rather than as the midpoint. The columns are ordered by whose position is being measured, and the *Sample* row records which of them require the respondent to be classified as a Democrat or a Republican. Columns (1)-(3) concern where the respondent places the two parties. Column (1) is how far apart they see them: the gap between where they put the Republicans and where they put the Democrats on the 7-point ideological scale from 1 "Very Liberal" to 7 "Very Conservative." Columns (2) and (3) ask instead how far from "Middle of the Road" each party is placed: the opposing party in column (2) and the respondent's own party in column (3), on a scale from 0 (middle) to 3 (very conservative/liberal). Columns (4) and (5) use the same ideology scale, measuring the gap between respondent's placement of themselves and the opposing party, and the gap between the respondent and their own party. Columns (2)-(5) all require knowing which party is the respondent's own, so they are estimated on self-described partisans. Columns (6)-(8) concern the respondent's own position and need no classification. The first two are how far from the center of the scale respondents place themselves, on the 7-point item and then on the 5-point one. The 5-point item is asked separately. In column (8) the outcome is the strength of respondent's party identification on a scale from 0 (independent) to 3 (strong democrat/republican). All columns include DMA and survey-year fixed effects, the four 2015 DMA balance controls interacted with *Post*, and individual controls for age in three bins, gender, race in five categories and Hispanic origin. Respondents are weighted to represent the adult U.S. population and standard errors are clustered by DMA. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.